WORK IN PROGRESS - ONGOING - NON-EXHAUSTIVE - NOT THE FINAL PRODUCT

Allocating AI-assisted educational publishing for greater access

A living research paper on populations plausibly helped, educational needs, language and register access, open materials, evidence, and production effort

Working snapshot 0.6.0 | 7 September 2026

This paper asks how AI-assisted educational publishing can create more usable access worldwide. It compares populations plausibly helped, educational needs, delivery, evidence, and production effort without ranking the worth of communities.

Every language and exact variety remains eligible. Where comparable research is absent, the same disclosed population baseline remains visible; missing evidence cannot lower or delete a community.

The worldwide screen now contains 775 non-English language-and-script edition profiles, with a real 100-row reach order and a distinct 100-row fixed-package workload proxy. Simplified Standard Written Chinese is first in both; Indonesian, Bangla, Hindi, Ukrainian, signed languages, displaced communities, accessibility formats, and adult-register English remain explicit.

Academic-register access is a governing thesis. Evidence improves the audience, register, package, delivery, fidelity, cost estimate, and scale; it cannot be used to infer innate capacity or withhold access.

This checkpoint adds an audited fixed 100,000-source-token package comparison to the 53-study effect register, 18 planning scenarios, replaceable ten-program hedge, nonordinal 100-profile inventory, and normalized model covering 1,069 interventions while retaining 10,625 possible outrankers.

The worldwide study remains ongoing and non-exhaustive. Reach positions and fixed-package proxy positions answer named quantitative questions; neither is yet a final target-specific funding order.

Living reader: https://kokunoyumeto.github.io/allocating-ai-translation-compute-for-educational-access/

Contents

This table describes the current working paper. Later versions may add evidence or revise conclusions.

Allocating AI-assisted educational publishing for greater access

Working manuscript. Empirical synthesis in progress; not a completed publication or funding ranking.

Abstract

AI-assisted translation can expand educational access when it moves complete material into a language, register and format that people can actually use. This working paper asks how a fixed production-compute budget can create the largest additional functional educational understanding. It does not rank the worth of languages or communities. The analysis first estimates who may be reached without deleting under-researched communities, then compares concrete language × stage × subject × package × format interventions on compatible educational endpoints. Under the disclosed common missing-data assumption, the largest displayed non-English reach opportunities begin with Simplified Standard Written Chinese, Hindi, Spanish, Arabic, French, Urdu, Bangla, Portuguese, Indonesian and Russian. Simplified Standard Written Chinese is the first proposed writing profile for the Mandarin row, not a claim that all Sinitic communities use one edition. English is also an access target: ordinary English use does not establish comfortable independent use of academic English, so adult register-to-register translations remain a separate, currently unmeasured global opportunity.

The paper compares language, subject, educational stage and delivery format from early learning through research. It retains every language when evidence is missing, separates population plausibly helped from learning effect and production cost, and gives no automatic coverage credit to a colonial, national, prestige or geopolitically coercive language. Initial source comparisons identify substantial foundational and secondary opportunities, research-stage language burdens, accessibility needs, and distinct routes for signed, displaced, local-language and independent learners. The new register-translation method preserves every proposition, definition, quantifier, proof dependency, number, citation and uncertainty marker while changing how the content is expressed for a specified adult audience. The worldwide synthesis is ongoing and non-exhaustive; missing evidence is a research obligation, never evidence of low need. The current allocation work has two levels: a 775-edition fixed-package budget calculation, and a nine-route PISA-context workload sensitivity. The latter is not an observed effect or a funding order. Together they show how a stated workload budget can be allocated under explicit scenarios while leaving the unknown educational effect visible.

Current checkpoint: worldwide reach and a fixed-package compute proxy

The public comparison now contains 775 non-English language-and-script edition profiles rather than a short convenience sample. The first table below shows the first ten of a real 100-row potential-reach table. A separate 100-row table then divides the same reach numerator by the estimated token-event workload for one identical 100,000-source-token text package and one nominal quality target. Stable profile identifiers are never positions.

The point seed remains 89.57%, the population-weighted share who did not report English across 17 India Census 2011 C-17 mother-tongue rows: 1,044,314,726 of 1,165,872,176 people. The sensitivity range is 70% to 99%. This is a transparent preservation assumption, not an observed worldwide rate of academic-English access. It keeps an under-researched community visible. Comparable target research replaces it only when audience, educational task, register threshold, stage and material endpoint match.

Reach position	Exact edition profile	Population point	Plausibly helped at common point
1	Simplified Standard Written Chinese profile	1.286 billion	1.152 billion
2	Hindi, Devanagari-script profile	580.3 million	519.8 million
3	Spanish, Latin-script profile	507.6 million	454.7 million
4	Arabic, Arabic-script profile	378.8 million	339.3 million
5	French, Latin-script profile	333.0 million	298.2 million
6	Urdu, Arabic-script profile	313.1 million	280.4 million
7	Bangla, Bangla-script profile	279.9 million	250.7 million
8	Portuguese, Latin-script profile	249.5 million	223.5 million
9	Indonesian, Latin-script profile	248.5 million	222.6 million
10	Russian, Cyrillic-script profile	201.2 million	180.2 million

The Indonesian point retains the direct 2022 BPS estimate of people aged five and older able to understand and produce comprehensible Indonesian; it does not equate oral ability with academic reading or package effect. The other rows use CLDR 48 interface-fluent L1/L2 estimates resolved to likely scripts. CLDR is a locale-planning source, not a complete census of independently usable academic editions. Multilingual, macrolanguage/member and multiscript rows overlap and cannot be summed into a world total.

Under the fixed-package workload proxy, the first ten are Simplified Standard Written Chinese, Hindi, Spanish, Arabic, French, Urdu, Portuguese, Indonesian, Bangla and Russian. Indonesian moves from reach position 9 to proxy position 8; Chinese remains first. These positions do not yet measure learning effect, delivery, physical compute, price, shared terminology, accessibility work or overlapping beneficiaries. Fresh-equivalent token events are workload scenarios, not hardware operations, energy, time, money or quality.

Ukrainian is present at reach position 62 and fixed-package proxy position 62 under the pinned CLDR layer: 24.1 million interface-fluent users and 21.6 million at the common point. That source omits part of the displaced diaspora, so this is not a final addressable-audience estimate. Russian knowledge receives no automatic coverage credit. Any overlap deduction requires the same material to be technically sufficient, linguistically usable, socially and politically acceptable, accessible, safe and actually available. The same rule applies worldwide to colonial, state, prestige and regional lingua-franca routes.

The complete two 100-row tables, all 775 rows, exact source and workload bindings, formulas and limitations appear in the appendix Worldwide reach and common-package allocation checkpoint and in the accompanying CSV and JSON files. They restore the quantitative allocation chain; they do not replace the target-specific educational-effect and physical-compute research still needed.

Current checkpoint: a nine-route PISA-context workload sensitivity

The catalogue and the worldwide reach screen are not, by themselves, the allocation answer. The allocation question remains active and is answered only for explicitly declared scenario units below. The decision unit is a concrete intervention: one language or register, one educational stage and endpoint, one package and format, one delivery route and one time horizon. The first checked subset in which a common context measure is available contains nine secondary quantitative-catch-up routes. For route i, the current conditional coefficient is N_i/C_i, where N_i is a source-defined prevalence or context count and C_i is the audited token-event workload. It is not yet a measure of people helped. Actual additional access per workload is $(N_i/C_i)q_i$, where q_i is the still-unmeasured average net probability that a person in that context gains the specified endpoint because of the intervention.

Route	Context count N	Point token-event workload C	Conditional coefficient per million token events if q=1	q relative to Indonesian required to lead
Indonesian	3,095,344	1,342,749	2,305,230	1.00×
Brazilian Portuguese	1,660,603	1,331,473	1,247,193	1.85×
Filipino	1,498,419	1,574,846	951,470	2.42×
Thai	412,820	1,677,044	246,160	9.36×
Vietnamese	264,513	1,455,161	181,776	12.68×
Standard Malay, Malaysia	230,285	1,574,846	146,227	15.76×
Japanese	122,172	1,533,971	79,644	28.94×
Central Khmer	111,241	2,318,470	47,980	48.05×
Ukrainian	70,228	1,576,517	44,546	51.75×

This is a conditional comparison, not a claim that Indonesian currently has the largest real effect. For example, Brazilian Portuguese exceeds Indonesian if its target-specific q is more than about 1.85 times as large; Filipino does so above about 2.42 times. Those ratios are reversal-threshold sensitivities, not value-of-information estimates or a measurement-priority ranking: no prior, measurement likelihood, decision utility or measurement cost has been specified. Thai, Vietnamese, Standard Malay, Japanese, Central Khmer and Ukrainian remain possible outrankers if their target-specific net yields are sufficiently higher. The thresholds show exactly how much the unknown effect would have to differ; they do not rank people or entitlement to educational material.

The effect term is being bound through six observable stages: exact-audience fit, reach, acquisition, practical usability, active use and endpoint gain, minus any endpoint loss. This factorization is a measurement plan, not an independence assumption; a directly comparable intent-to-treat estimate would replace the product. Every component is currently recorded as unmeasured rather than zero. Fifteen routes with numeric context remain in the frontier; six use different endpoints and are retained without being forced into this cross-ranking. All 1,022 candidate routes remain possible outrankers.

Token-event workload is still not physical compute. Provider accelerator work cannot be recovered from cached-input, fresh-input, output-token, elapsed-time or price counters. The final denominator will therefore use a declared reference replay: accelerator-seconds per accepted package plus full-system joules, under cold-isolated and stated operational scenarios. Until that replay exists, the physical-compute field is explicitly unidentified.

Current checkpoint: academic English is also a translation target

The English source population is 1.5 billion in the same rounded source, but the share unable to use academic English for a specified task is not measured globally. The honest expression is therefore 1.5 billion multiplied by q for the relevant academic-register mismatch. English-first-language identity, conversational English, years of schooling, or conformity to British or American prestige norms cannot set that mismatch to zero.

The governing premise is a sufficiently plausible access thesis, not a go/no-go hypothesis: inherited command of academic register is one institutionally distributed advantage, and any credible nonzero educational benefit makes a source-faithful adult register edition eligible. Research determines how the translation should be designed, checked, measured and revised. It may not turn correlations among schooling, family education, income, test performance, language exposure or attainment into claims of innate capability or reasons to withhold translation.

The method is full-content diaphasic intralingual translation: translation between registers within English for a declared adult audience. It is not abridgment, remediation, child-directed explanation, or replacement of the original. The target may be longer. It may unpack dense grammar, define terms, split overloaded sentences, make logical relations explicit, resolve references, reorder with tracked dependencies, and add clearly labeled examples or background. It may not omit, generalize, silently strengthen or weaken, or substitute intuition for a proof.

There is no single universal Plain English. The current design distinguishes at least international/ELF-friendly adult English, educated non-specialist English, profession-adjacent novice English, localized adult Englishes, an academic-convention companion, and separately commissioned accessibility editions. Every version preserves the same protected content while making different lexical, grammatical, pragmatic and navigation choices.

The evidence is supportive but appropriately bounded. Parent education and socioeconomic position are associated with unequal educational trajectories; qualitative work documents tacit institutional discourse and hidden rules; academic-language knowledge predicts reading comprehension in several school samples; experiments show that jargon and document design can change processing or understanding; and one randomized university intervention found that making background-linked challenges and navigation strategies explicit improved first-generation outcomes. No identified study tests the complete chain from parent education through full-content register translation to long-run attainment. That gap defines what to measure; it does not veto the intervention.

Ordinary text simplification is not an adequate production objective. One document-level dataset reports deletion as 44% of sampled operations, and factuality studies show omissions and unsupported additions that common metrics miss. This project therefore requires bidirectional source-target claim alignment and zero critical errors for definitions, propositions, quantifiers, conditions, numbers, formulas, proof dependencies, cross-references, attribution and epistemic status. Readability and similarity scores are alarms, not proof of fidelity.

The full evidence synthesis, 42-source register, six target profiles, 13 protected content classes and 20-step production workflow are published in ACADEMIC_REGISTER_TRANSLATION_RESEARCH_v1.md (source file: `ACADEMIC_REGISTER_TRANSLATION_RESEARCH_v1.md`). The anti-hegemony model and deterministic counterfactual tests are published in HEGEMONY_BIAS_AUDIT_v1.md (source file: `HEGEMONY_BIAS_AUDIT_v1.md`).

1. Research question

Which educational publications could create the greatest additional usable access per standardized unit of AI production effort? The relevant decision is not simply which language has the most speakers. It is which material, in which language or shared register, would help which learners do something educationally valuable that their existing options do not adequately support.

Education includes preschool and school learning, vocational development, university study and research. A mathematical research text can be an important educational resource even when few school learners need it. Equally, a technically advanced translation does not address missing early arithmetic. The analysis therefore keeps stages and learning tasks explicit instead of assigning each language one universal need score.

Throughout the paper, the stage codes are P for preschool/early childhood, F for foundational/primary, S for secondary, V for vocational/practical, U for university or taught tertiary, and R for postgraduate/research. They identify different educational uses; they are not grades, scores or a priority order.

Autonomous learners are a central audience. The availability of teachers, uninterrupted internet, expensive software or institutional subscriptions is not assumed. Downloadable, phone-readable material with prerequisites, exercises, answers and usable navigation can offer a different route from classroom instruction. Classroom use, independent study and research/reference use are analyzed as distinct, potentially overlapping endpoints.

The unit is an intervention: a specified population and language/variety/script profile, a defined educational package, a format and a time horizon. A multilingual learner may have several viable routes. Those alternatives are compared without counting the learner repeatedly in a combined portfolio.

2. What counts as additional access?

For a specified educational task, additional access is the difference between what learners can use with the intervention and what they could use otherwise. A file's existence is not proof that everyone has access to it. Conversely, failure to find a title on one website is not proof that an entire language lacks educational resources.

Open licensing is important because it enables lawful translation, adaptation and redistribution without charging each learner or institution. Price, independent-download access, device compatibility and permission to adapt are separate attributes. A free-to-view video, a reusable textbook and a complete offline course are different resources. Existing paid or institutionally restricted material may serve some people while leaving substantial unmet need among others.

The programme used as a case study does not change the estimated educational needs of a language community. Its local files, completed translations, recent publication counts or sunk production effort are not inputs to global need, eligibility or priority. The research concerns the world's educational opportunities, not a schedule of what one project should do next.

3. Evidence and comparison method

3.1 Population and inclusion

The candidate universe starts with the frozen Glottolog language-level register and overlays exact varieties, scripts, signing communities, displaced populations, accessibility modes and possible interlanguages. Historical and unresolved entries remain traceable with their appropriate interpretation. Registry membership is not itself proof of a current speaker population, and a macro-language is not automatically an intelligible edition for each constituent community.

Population evidence preserves the construct a source actually measures: mother tongue, home language, additional-language use, literacy or technical reading. These cannot be substituted without explanation. Historical counts retain their dates; cross-year projections or cross-source combinations are explicit inferences. National totals may constrain a resident subset when the relationship is defensible, but do not bound an entire worldwide diaspora.

Missing evidence does not remove a community or assign it zero need. A related educational system can motivate a hypothesis and, where its mechanism is supported, a bounded inference. It does not automatically supply the same population, need rate or rank. The analysis records both expected candidates and unexpected possible contenders.

3.2 Need, benefit and cost

Let B describe a specified population with a stated educational need. Let q describe its average net change in usable access, relative to independent existing options, for a specified task and time horizon. With access normalized between zero and one, q lies between -1 and 1. The simplest common-endpoint net benefit is Bq. Both components may vary across subgroups; the general analysis sums over mutually exclusive learner strata.

Net change is not the same as the gross number newly gaining access. For binary access, q equals the share gaining access minus the share losing it. A nonnegative scenario explicitly assumes no access losses; their absence is not established by missing evidence. An optional additional resource can motivate that assumption for availability when existing routes remain intact, but it does not establish a nonnegative effect on learning. Being below a proficiency benchmark also does not mean a learner had no usable resources beforehand.

Benefit per standardized production cost is Bq/C. A large B can coexist with modest benefit per person. A smaller population can justify a high-value intervention because its unmet need, language fit or practical delivery is different. The analysis reports these components separately.

When q is not measured, useful conditional comparisons remain possible. For positive population and cost terms and $q_B>0$, intervention A exceeds B on a matched endpoint when q_A/q_B exceeds $(B_B/B_A)(C_A/C_B)$. Otherwise the analysis compares $B_A q_A C_B$ with $B_B q_B C_A$ directly, preserving signed and zero effects. This identifies what would have to be true for one option to be more efficient. It does not disguise an assumed common yield as an empirical estimate.

Observed data, evidence-based transport and illustrative scenarios are labelled separately. Common priors cannot produce a high-confidence rank. Uncertain comparisons remain visible alongside the information that could change them. The final Top 10 and Top 100 are intended as transparent portfolios with possible alternatives, not an eligibility boundary for everyone outside them.

3.3 Hypotheses and anomaly investigation

Before the next ordering, we record twelve hypotheses and one hundred expected candidate profiles. They predict large additional-language opportunities, school-to-academic-language transitions, stage-specific shortages, useful interlanguage designs, independent-learning routes and displacement-related needs. Initial stage annotations never exclude other stages.

The initial high-reach expectation includes Indonesian, Hindi, Bangla, Simplified Written Chinese and Urdu, with Telugu, Tamil, Marathi, Hausa and Kiswahili as further strong investigations. This is not a predicted exact order or a guaranteed final ten. The complete 100-profile forecast (source file: `HYPOTHESES_AND_EXPECTED_CANDIDATES_v1.md`) includes regional, research, signed-language and shared-register candidates; its sections are not priority bands. Profiles outside that forecast remain eligible. The forecast was recorded after earlier analyses, so it is prospective only for the next analysis, not an original preregistration.

Expected pattern	A result that would require investigation	What could legitimately change the expectation?
Large additional-language audiences make Indonesian and other common educational languages important contenders.	The model counts only mother-tongue speakers, or removes a language because a local project translated material.	Source-specific literacy, educational-stage need and independently usable provision can change a package's expected benefit.
A need found in one Slavic community suggests checking comparable needs elsewhere, including Ukrainian.	Russian is compared while Ukrainian or other relevant communities disappear.	Different population sizes, interruption of schooling, publication markets or usable alternatives can produce different results. They must be investigated, not assumed.
School-to-academic-language transitions create opportunities beyond early schooling.	Regional-language learners vanish because their country has an official academic lingua franca.	Evidence of comfortable subject access or an equally usable existing package can weaken the proposed intervention.
Established scientific languages can still have specific educational gaps.	School provision removes Japanese, Chinese, English or another language from every stage.	An accessible equivalent can weaken a specific proposed package; it does not answer every other subject and stage.
Formula-rich interlanguages may retain enough benefit to justify their lower shared-production cost.	Every design is assigned either zero comprehension or the entire language-family population.	Findings on exact mathematical tasks, scripts, reader preparation and design cost can support or weaken the advantage.
Written-language comparisons miss some signing, displacement and access-format needs.	An entire modality or displaced community is absent without an investigated reason.	Actual language routes and task-specific provision can distinguish useful from ineffective packages.

An unexpected absence prompts a trace through identity, population, educational need, provision, overlap, cost and implementation. If a rule caused the absence, it is checked across the entire retained candidate-stage universe. For example, a population-data join that excludes one language must be tested against every initially unmeasured language, not patched only for the noticed case. Evidence can justify unequal outcomes; it cannot justify equating missing measurement with no need.

The expectation-versus-investigation table (source file: `EXPECTED_CANDIDATE_COVERAGE_v1.md`) exposes the work still missing. In this bounded calculation, 63 of the 100 expected profiles have at least one proposed route and 37 do not; separate shared-register analyses are not counted in those figures. These are investigation counts, not rankings or proportions of educational need covered. All 100 retain preschool, foundation, secondary, vocational, university and research stages. Preschool is an explicit column, not an invisible part of foundation. Source references, proposed designs, identity qualifications and unresolved population estimates are represented separately; none supplies an implicit low rank.

3.4 Standardized production scenarios

The currently reusable production model measures token traffic, not physical computing operations, electricity, subscription consumption or money. It combines a declared workflow with recorded tokenization measurements from 1,012 aligned general-domain sentences. Target counts use `o200k_base`; the fixed source reference uses `cl100k_base`. That mixed reference is explicit and is not a measurement of a contemporary model's actual mathematical-translation workflow.

For the base text scenario, traffic is approximately `T = S(7.585897 + 4.73R)`, where S is the source-reference token count and R is the target working-token count divided by the source-reference count. The coefficients describe stipulated production and checking work, not observed learning benefit. Extra model work remains an explicit additional term; rendering, speech and signed-video production require separate resources.

Edition benchmark	Base-scenario token traffic for 100,000 source-reference tokens
Indonesian	1.343 million
Bangla	1.559 million
Urdu	1.531 million
Hindi	1.494 million
Simplified Chinese	1.342 million
Japanese	1.534 million
Ukrainian	1.577 million
Russian	1.423 million
Hausa	1.484 million
Kiswahili	1.453 million

These scenario costs can inform conditional benefit thresholds; they cannot determine a language's need. An unmeasured language retains a symbolic cost or an explicitly labelled predictive scenario, not a penalty or exclusion. General-domain tokenization is also an imperfect transport to mathematics, where shared formulas may change relative expansion. The exact source bindings, coefficients and reconstruction checks are retained in `structured/STANDARDIZED_COMPUTE_REUSE_20260904.json`; no local programme completion or sunk-compute adjustment enters this model.

3.5 Complete courses and smaller additions must be compared on the same outcome

A short companion is not automatically more efficient than a complete course. For the same community and educational outcome, let a smaller package have net additional access `G_s` and positive cost `C_s`, and a fuller sequence have `G_f` and positive cost `C_f`. The smaller package has more access per cost only when `G_s × C_f > G_f × C_s`. If both packages address the same potential population and have positive yields, this becomes `q_s/q_f > C_s/C_f`. Negative or zero effects require the cross-product form rather than that ratio.

The source scale is no longer represented solely by an arbitrary 100,000-token canon. A bounded measurement of the exact OpenStax Prealgebra 2e source at commit `38cae454e644abf9f0a623e876994553881597c9` finds 75 unique CNXML modules across 11 chapters. NFC-normalized text extracted from the learning-content elements contains 364,270 `cl100k_base` tokens. A declared six-chapter quantitative sequence-whole numbers, integers, fractions, percents, linear equations and graphs-contains 173,958 tokens, or 47.8% of that measured full source. The fractions chapter contains 41,336 tokens, or 11.3%. These are source representations, not teaching-time, difficulty or learning-effect measurements. The archive excludes media, and text extraction omits mathematical and document structure; tokenizing the complete XML instead gives 2,811,119 tokens because markup and metadata are included (OpenStax, 2026a).

Under the declared text-workflow coefficient and the general-domain Indonesian expansion measurement, the corresponding traffic scenarios before unmeasured extra work are approximately 4.89 million, 2.34 million and 0.56 million token events. They are neither actual model usage nor money or physical compute. The same measured English source scale is carried through all ten currently measured language profiles rather than charging each a different source canon.

At zero common setup cost and equal per-source-token workflow, the six-chapter package needs more than 47.8% of the full course's net benefit on their common endpoint to be more efficient. If shared setup costs both alternatives an additional 10% of the full course's variable cost, the threshold is 52.5%; if setup equals the full variable cost, it is 73.9%. The corresponding fractions-only thresholds are 11.3%, 19.4% and 55.7%. These figures demonstrate why a small-content shortcut can lose its apparent advantage once terminology, formats and validation infrastructure are substantial. They do not estimate either package's educational effect.

The comparison must also preserve outcomes that exist only in the full course. A fractions unit and a complete prealgebra sequence can be compared on fractions access, while the latter's additional algebra, decimal, geometry and graph outcomes are reported separately or assigned explicit weights. Counting every stage as a new person would double-count overlapping learners. Conversely, once common infrastructure has been produced, the next decision compares the marginal benefit and cost of the next unit; sunk production never reduces a community's educational need.

4. Initial findings

4.1 Indonesian access includes a large additional-language audience

BPS's 2022 tables report 248,501,794 Indonesian-capable residents aged five or above, compared with 71,971,184 whose first-acquired language was Indonesian. These are different constructs. With the reported national marginals treated as fixed, approximately 176.53-181.71 million residents can use Indonesian without having acquired it first. This is a conditional joint-event bound, not a measure of advanced reading. It nevertheless demonstrates why an L1-only audience would omit a substantial potential route (BPS, 2022a, 2022b).

Regional-language needs do not disappear because Indonesian is widely used. Javanese, Sundanese and other routes may offer distinct benefits for particular ages, tasks or readers. They must be compared as overlapping interventions rather than assigned either complete national-language coverage or complete separation.

4.2 Secondary mathematics, reading and science give matched need measures

The complete PISA 2022 comparison covers 81 jurisdictions and uses the same represented-student population frame for mathematics, reading and science. Among Indonesia's represented cohort, approximately 3.10 million students are below Level 2 in mathematics, 2.83 million in reading and 2.50 million in science. The corresponding shares are 81.7%, 74.5% and 65.8%. The Philippines has similarly large observed shares-84.0%, 76.3% and 77.2%-while Japan's are 12.0%, 13.8% and 8.0%. These are educational-system observations, not language-specific translation-benefit counts (OECD, 2023a, 2023b).

The three counts overlap within students and cannot be added. Because the published marginal tables do not identify each student's joint status, the derivative publishes sharp bounds rather than assuming independence. For Indonesia, between approximately 0.84 million and 2.50 million represented students could be below Level 2 in all three subjects; between 3.10 million and the full 3.79-million represented cohort could be below Level 2 in at least one. These bounds describe assessment status, not the audience or effect of a translated package.

The matched comparison supports substantial Indonesian secondary-foundation investigation and shows why the package cannot be chosen from a mathematics score alone. Reading prerequisites may affect whether a learner can use mathematics, science, finance or computing material. It does not show that every assessed learner needs a translation, that Japanese education has no gaps, or that university, research, minority-language and accessibility needs are unimportant. Independent provision, exact language fit, delivery and the package's actual contribution remain consequential.

A bounded inspection of current official Indonesian supply changes the content hypothesis without lowering the measured need. The SIBI interface displayed 642 Kurikulum Merdeka PDF books overall and 115 at SMP/MTS level. Its subject filters displayed five mathematics, six science, six Indonesian-language and six informatics books at that level. The acquired Grade VII sample contains a student book and a separate teacher guide for mathematics, science and Indonesian language. All six PDFs are readable and visually substantive; the three teacher guides contain answer-key markers, while the corresponding student books do not contain that phrase in extractable text. This is evidence of existing public supply and teacher-facing feedback-not evidence of adoption, learner possession, independent usability, open adaptation rights or educational effect (Kementerian Pendidikan Dasar dan Menengah, n.d.).

The resulting first secondary hypothesis is not another undifferentiated textbook. It is three separable but interoperable learner companions-mathematics, reading and scientific language, and science-sharing a prerequisite diagnostic, route map, worked explanations, learner-facing answers and error feedback, remediation, cumulative review and download-once navigation. Science additionally needs low-equipment activities, safety guidance and offline substitutes for linked resources. That bounded companion must be compared with a complete independently usable lower-secondary foundation sequence at the same outcome. Neither alternative has a measured effect, complete production cost or rank; source length alone cannot choose between them.

Early-grade language support is a different intervention. Indonesian law establishes Indonesian as the national educational medium while allowing regional languages at the initial stage when needed. A ministry policy brief describes 718 local languages, reports that approximately 73% of people aged five or above use a local language at home, and presents several readiness-based pathways into Indonesian. Those national statements are not exact-language learner denominators. A defensible portfolio therefore specifies each local language, variety, script or orthography, location and stage rather than treating “regional languages” as one production target (Badan Pemeriksa Keuangan, 2003; Pusat Standar dan Kebijakan Pendidikan, 2023).

Delivery remains part of the intervention. BPS reports national 2024 marginals of 72.78% having accessed the internet, 68.65% owning a mobile phone and 18.52% of households owning a computer. These figures support phone-readable semantic HTML or EPUB, tagged PDF, printable units, local sharing and resumable offline progress. They do not imply universal phone ownership or continuous affordable access, and they are not student-by-language cross-tabs that can be multiplied into a beneficiary estimate (Badan Pusat Statistik, 2025).

Coverage can change the comparison materially. Under an explicit subset assumption and treating source population estimates as fixed, allowing all unrepresented learners' proficiency to remain unknown gives a Cambodia age-15 need range of approximately 111,000-333,000, compared with 122,000-210,000 in Japan. The represented point estimates alone therefore cannot settle that comparison. The complete calculation retains sampling cautions and source inconsistencies rather than repairing them silently.

The Ukrainian data require particular care. The same sample represents 165,591.78 students in both an 18-region and a national population frame. The observed regional estimates below Level 2 are approximately 70,228 in mathematics, 67,803 in reading and 56,326 in science. The corresponding conditional national envelopes are about 70,000-303,000, 68,000-301,000 and 56,000-289,000. They do not extend participating-region percentages to all Ukraine or add the national frame as a second population. These are 2022 observations, not a current wartime educational census.

4.3 Different regional needs suggest different packages

Bangladesh's MICS 2025 indicators report foundational reading for 50.2% and numeracy for 39.2% of children aged 7-14. Pakistan's HIES 2024-25 reports 28% of children aged 5-16 out of school. These findings support investigating foundational catch-up, school re-entry and secondary progression, while preserving the different survey populations and constructs (UNICEF Bangladesh, 2026; Pakistan Bureau of Statistics, 2026).

For Bangladesh, combining the age-specific numeracy rate with projected age groups and explicit within-group age assumptions gives approximately 15.3-15.7 million children aged 7-14 not meeting the numeracy criterion. This is a range across three cross-source scenarios, not a confidence interval or direct census count. Without a within-age-bin composition assumption, the fixed-source sensitivity range is much wider, about 10.0-19.5 million. The calculation does not multiply a child learning rate by an all-age language population. It motivates a concrete package of number concepts, comparison, addition, patterns and worked practice, while existing national textbooks remain part of the provision baseline (UNICEF Data, n.d.; BBS & UNICEF Bangladesh, 2026).

Actual content inspection narrows that proposal. The sampled 2026 NCTB Grade 2 book already covers all four numerical domains with visual explanations, worked answers and practice; selected Bangla Khan Academy lessons also provide explanations and exercises. For this task set, a portable diagnostic/feedback companion is a stronger provisional addition than an undifferentiated replacement book. It would link to existing examples and supply entry checks, prerequisite routing, self-check keys and short explanations of wrong answers. The sample does not establish complete national supply or universal autonomous usability, and it does not lower the population need estimate. Other stages and subject packages remain separate questions (National Curriculum and Textbook Board, 2026; Khan Academy, n.d.).

The language choices differ. Bangladesh has a large broad Bangla-language route, with exact-variety questions still important. Pakistan's census records several substantial mother-tongue communities: Urdu cannot stand for all Pakistanis. Urdu editions and Punjabi, Sindhi, Pashto, Saraiki and other editions require explicit capability and overlap comparisons.

Pakistan's 2023 census provides a matched historical count: 25,373,350 out-of-school children among 71,270,068 children aged 5-16 in its detailed-characteristics population. These are published school-status counts and summed single-year age counts, respectively. The newer HIES rate of 28% would imply 19.96 million only if that older cohort size and relevant frames were held constant; this is a scenario, not a measured decline between the two sources. Headcount-only populations missing detailed characteristics, AJK, Gilgit-Baltistan and diaspora communities remain outside this particular estimate, not outside the research (Pakistan Bureau of Statistics, 2023a, 2023c, 2026).

The same census makes a stronger language-specific inference possible. Rural Sindh has 25,623,864 residents, 23,605,495 with Sindhi as mother tongue, and 5,117,164 out-of-school children aged 5-16. Even if every non-Sindhi resident were in that last group, at least 3,098,795 out-of-school children must be Sindhi-mother-tongue children. Combining ten disjoint geographic cells gives a national detailed-frame interval of 3,098,795-8,757,513 for that intersection. This is a logical bound on the recorded margins, not an estimate of comfortable readers or beneficiaries. It supports investigating a distinct Sindhi foundational-learning route without assuming that a province is a language or that Urdu serves everyone (Pakistan Bureau of Statistics, 2023b, 2023c).

Considering several editions requires joint population accounting. Urdu, Punjabi, Sindhi, Pashto and Saraiki mother-tongue categories together contain between 9,929,282 and 25,373,350 of the recorded out-of-school children, conditional on the ten geographic cells' margins. Adding their separately maximized upper bounds would instead yield an impossible 54,141,695. The integrated calculation retains the shared constraints and feasible endpoint allocations. It also adds a Western/Pakistani Punjabi written-profile challenger outside the original forecast, while preserving uncertainty about how the census category maps to an exact edition. None of these membership bounds is a measured package-benefit count.

Existing distance education is a material countercheck. Bangladesh Open University already lists Bangla financial-management and business-statistics content; Pakistan's Virtual University lists extensive mathematics and related subjects. The useful additional package may therefore be a prerequisite companion, complete worked practice, accessible offline material or a specific missing topic-not another generic introductory volume. Catalogue presence alone does not establish which learners can use the material or what adaptation rights apply.

The Bangladesh comparison has now been extended across all six stages. The 2024 official report records 9,063,422 students in classes 6-10, 1,619,349 technical and vocational enrolments, and 1,069,579 university students. These are stage-scale context counts, not Bangla-only populations, unmet-need totals or expected beneficiaries. The accompanying source freeze contains 299 NCTB secondary, Dakhil, technical and higher-secondary catalog rows, of which 297 expose at least one link. It also contains 394 Bangladesh Open University course cells-326 unique program-title keys-including 65 cells in explicitly Bangla-medium BBA and MBA programs. Two sampled BOU mathematics units visibly contain Bangla explanations, but their legacy font encoding defeats ordinary Unicode extraction; a sampled computing unit is in English. The finding is therefore substantial but heterogeneous supply, not either universal adequacy or universal absence (Bangladesh Bureau of Educational Information and Statistics [BANBEIS], 2026; Bangladesh Open University, n.d.; National Curriculum and Textbook Board, 2026).

Delivery evidence changes the endpoint. The national 2024-25 ICT survey reports household computer access of 9.0%, home internet access of 55.1%, individual internet use of 53.4%, mobile-phone use of 88.4% and mobile-phone ownership of 65.5%. Rural individual internet use is 43.5%, compared with 84.2% in city-corporation areas. Among people who do not use the internet, 49.4% report not knowing how to use it and 18.6% report that service cost is too high. These are differently conditioned, sometimes multiple-response indicators, not additive barriers. They support phone-readable, low-bandwidth, downloadable, resumable and printable editions with simple onboarding; they rule out treating continuous connectivity or personal smartphone ownership as universal (Bangladesh Bureau of Statistics, 2026).

Institutional access is also stage-dependent. Among the 20,631 secondary institutions in the BANBEIS table, 98.94% report electricity, 87.90% a computer facility, 80.98% internet and 69.52% multimedia. Junior-secondary institutions report only 49.12% internet and 24.15% multimedia, compared with 84.70% and 74.75% for secondary schools. Institution-level presence does not prove reliable personal learner access, but it makes a single delivery assumption indefensible. An offline learner edition, a printable unit and a classroom projection asset can be complementary outputs rather than rivals (BANBEIS, 2026, Tables 3.5.1-3.5.3).

The package implications differ by stage. At preschool level, the present evidence is still missing and the route remains visible rather than being assigned zero need. At foundational level, a twelve-item diagnostic/feedback companion can be tested against a complete accessible sequence on the same numerical tasks. At secondary level, a coherent mathematics, reading/academic-language and science pathway should be compared with curriculum-linked feedback and transition companions. Vocational work must be tied to a named curriculum, occupation or qualification; the official total spans 9,310 institutions and changed scope in 2024. At university level, full Bangla open textbooks should be compared with bilingual Bangla-English terminology, prerequisite and worked-practice bridges. At research level, full advanced translations and smaller bilingual navigation or terminology companions remain separate candidates. Readiness for a bounded test is not an educational-need or funding rank.

The university bridge is more than a generic analogy. A qualitative study of seven senior university teachers reports a preference for Bangla and recommends a bilingual curriculum. A second qualitative study interviewed eight teachers in four Dhaka universities and identifies English-medium learning and mother-tongue textbook mechanisms. A 2025 open-access study surveyed 301 first-year engineering students and conducted focus groups with ten students; it reports moderate difficulties with English-medium lectures, assignments, examinations and communicating subject knowledge. The last article is internally inconsistent about whether participants came from one or two universities, and none of the three estimates national prevalence or the effect of a translated package. Together they justify a bounded Bangla-English engineering-transition test, not a national effect claim (Karim et al., 2023; Sultana & Fang, 2024; Islam et al., 2025).

Finally, the national Bangla route is not the whole country. The universal register contains 50 Glottolog language-level nodes linked to Bangladesh. Bengali (`ben`, `beng1280`) is the national written route considered here, but it cannot be counted as automatic coverage for Sylheti (`syl`), Chittagonian (`ctg`), Rohingya (`rhg`), signing communities or the other country-linked nodes. Glottolog currently links Indian Sign Language (`ins`, `indi1237`) to Bangladesh, India and Pakistan; that catalog identity does not establish that one standardized signed route serves every Bangladeshi signing community. Each remains a separate evidence obligation.

Regional investigations also identify distinct transition mechanisms: regional-language schooling to English technical education in India; Kiswahili-supported entry to English-medium secondary subjects in Tanzania; and Hausa-French learning routes in Niger. Their population magnitudes, independent provision and package effects must be estimated separately. A familiar mechanism is a research lead, not permission to copy one country's rate into another.

India illustrates how a hypothesis should change when confronted with evidence. Among higher-secondary students in the 2017-18 NSS survey's home-language categories, the English-medium shares were 19.5% for Hindi, 37.4% for Tamil, 74.5% for Telugu and 28.2% for Marathi. Thus a universal story of switching to English only upon entering university is particularly weak for the Telugu group. These historical cross-sections are neither English-proficiency measurements nor the same learners followed into university (Ministry of Statistics and Programme Implementation, 2020, Statement 4.17).

A more precise candidate is an optional bilingual prerequisite companion. Inspection of actual Hindi and Telugu NPTEL pages finds substantial existing explanations, worked examples, answers and glossaries. Both inspected mathematics books open with Rolle's theorem. A provisional eight-page entry insert can supply prerequisite orientation, six transfer tasks with explanatory answers, two compact foundation bridges and precise navigation to the existing lectures. This is a design scenario, not a demonstrated optimum or proof that similar support is unavailable elsewhere. Tamil and Marathi content remains uninspected in this comparison. English-medium learners remain eligible for local-language support; regional-language learners are not presumed incapable of English (NPTEL, n.d.).

Tanzania offers a different, source-bound transition example. Uwezo's 2017 Standard 7 reading margins imply that 39%-53% of the assessed cohort could pass the basic Kiswahili task but not the English task. Adding its subtraction-test margin gives a 19%-53% interval for pupils passing both Kiswahili and subtraction while not passing English, without assuming independence. This supports investigating paired subject explanations and English task practice, with separate foundational support. The tests concern basic competencies among assessed ages 6-16, not current technical-reading ability or a guaranteed translation effect (Uwezo Tanzania, 2019). A separately labeled transport scenario scales these bounds to historical administrative grade enrollment; it does not turn them into a current population estimate.

4.4 A high-consequence omission gate changes missing evidence into a research obligation

A source-bound omission audit now tests the prospective portfolio against all 22 scheduled-language groups in the Census of India 2011 C-16/C-17 frame. The frame contains one group above 100 million, seven at or above 50 million and thirteen at or above 10 million. These are historical census-group populations, not exact modern varieties, current learner counts or benefit estimates. The audit retains every group and gives every missing route an explicit possible-outranker state (Office of the Registrar General & Census Commissioner, India, 2011a, 2011b).

Published Census language group	2011 population	Prior internal scenario stages	Unbound stages in that scenario set	NPTEL catalogue title labels
Hindi	528,347,193	university	preschool, foundational, secondary, vocational, research	292
Bengali	97,237,669	foundational, vocational	preschool, secondary, university, research	181
Marathi	83,026,680	university	preschool, foundational, secondary, vocational, research	199

Published Census language group	2011 population	Prior internal scenario stages	Unbound stages in that scenario set	NPTEL catalogue title labels
Telugu	81,127,740	university	preschool, foundational, secondary, vocational, research	209
Tamil	69,026,881	university	preschool, foundational, secondary, vocational, research	248
Gujarati	55,492,554	none	all six stages	212
Urdu	50,772,631	foundational	preschool, secondary, vocational, university, research	not inspected in the current 12-page census

Table note. The scenario-stage columns are an internal research inventory, not worldwide educational coverage. Prior project work is a case study and has zero effect on a population's educational need or funding priority. The catalogue column records unique title labels extracted from one NPTEL language page. It does not measure complete courses, open licensing, accessibility, uptake, quality or adequate provision. Existing material therefore does not reduce the language group's educational need; it only changes the additionality question for a particular proposed package (NPTEL, n.d.).

The immediate finding is not a new rank. It is that Gujarati cannot disappear because no matched intervention route has yet been assembled, and that one university-stage hypothesis for Hindi, Marathi, Telugu or Tamil does not amount to access across the educational life course. The same audit retains Kannada, Odia, Malayalam and the Punjabi-linked profile as route-free possible outrankers below the 50-million alarm line, and retains Assamese, Bodo, Dogri, Kashmiri, Konkani, Manipuri, Sanskrit and Sindhi outside the frozen 100 without treating that forecast boundary as a funding boundary.

The language-group labels themselves require care. The published Hindi group cannot be substituted for one exact Hindi variety, and the Punjabi group cannot be substituted for an exact Eastern Punjabi readership without resolving mother-tongue composition, script and community. Reporting English among up to two subsidiary languages is not academic proficiency; not reporting English or Hindi in those slots is not proof that neither is understood. These measures expose overlap questions, not ready-made access multipliers.

This is the first national high-consequence omission gate, not a sufficient global audit. The same operation must be repeated for every country and territory, large official and minoritized language, displaced and signed-language population, and then for communities that national tables fail to enumerate. Population magnitude sounds an alarm; it does not supply a need score.

4.5 Existing Chinese and Japanese provision tests the package hypotheses

Existing materials can weaken a proposed intervention without resolving a community's broader needs. A Chinese probability/statistics textbook already supplies Python explanations and advertises maintained source code. A teacher-qualified support offer does not establish that all supplementary material is restricted (Xu, 2023). The University of Tokyo provides public Japanese programming lessons, scientific-computing topics and exercise-answer routes (University of Tokyo, n.d.).

These are concrete alternatives, not evidence that all learners are served. They require a new package to identify its additional contribution: a different task, prerequisite sequence, price, format or otherwise unavailable explanation. They also weaken any automatic inference that Chinese or Japanese opportunity must move wholesale from school to research. The remaining comparison is package-specific at every stage.

4.5.1 One Chinese score would collapse six different allocation questions

The new China and Simplified Written Chinese checkpoint (source file: `CHN_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) makes that package-specific rule operational from preschool through research. The Ministry of Education reports 280.4043 million formal learners and 18.7010 million full-time teachers in 2025. Its stage counts include 32.2552 million preschool children, 101.7829 million primary learners, 55.0934 million junior-secondary learners, 30.3950 million ordinary senior-secondary learners, 11.1877 million secondary-vocational learners, 39.5396 million ordinary and vocational undergraduate/associate learners, and 4.2995 million postgraduate learners (Ministry of Education of the People's Republic of China, 2026). These are administrative contact populations, not unmet-need counts or automatic beneficiaries.

Domestic supply is also large. The national Smart Education platform reports more than 130,000 K-12 resources, more than 12,500 vocational courses, about 145,000 higher-education courses and more than 178 million users (Ministry of Education of the People's Republic of China, 2025). That is strong counterevidence to a blanket claim that Chinese educational material is scarce. It does not establish item-level open licences, accessible equivalence, answer completeness, autonomous-study coherence, resumable offline use or the additional effect of a proposed package. A national score would therefore obscure the actual comparisons: early-language and numeracy routes at P; foundational feedback and mobility continuity at F; subject/prerequisite bridges at S; qualification- or task-specific practical learning at V; exact open course alternatives at U; and proof, source-search and academic-language bridges at R.

The language boundary is equally consequential. A conditional calculation gives about 1.052-1.121 billion age-15+ users of normative Chinese characters under explicit matching-population assumptions, but it is only a written-standard proxy. It is not an exact Simplified-Chinese readership, academic-comprehension, need or benefit interval. The Ministry's 80.72% Putonghua prevalence is a separate construct and is not multiplied into that proxy (Ministry of Education of the People's Republic of China, 2021). Independently, the universal register retains 452 Glottolog language-level nodes linked to `CN`, including nineteen bookkeeping nodes and five sign-language nodes. Registry linkage can cover historical, regional, cross-border, Hong Kong, Macau, Taiwan and multinational associations; none is automatically a present mainland audience or covered by `zh-Hans-CN`.

Cross-cutting constraints also remain separate. The official 2020 analysis reports 138 million migration-affected children under its definitions; the 2025 migrant-worker report counts 301.15 million workers, 65.4% with junior-secondary education or less. Disability sources report more than 85 million disabled people, over 17 million people with visual disability and about 27.8 million with hearing disability, alongside adapted-reading shortages (National Bureau of Statistics of China, UNICEF, & UNFPA, 2023; National Press and Publication Administration, 2026). These overlapping measures motivate continuity, practical-learning, semantic HTML/MathML, screen-reader, large-print, braille/tactile, audio, caption and exact signed-language comparisons. They are never added or substituted for language audiences.

The resulting finding is deliberately asymmetric with neither blanket inclusion nor blanket exclusion: Chinese packages are high-consequence candidates, while no national rank is assigned. All nine measured OpenStax sources remain available as complete-versus-bounded comparison baselines; Open Logic and the Stacks Project provide separate university/research baselines. Exact audience, incremental educational effect, physical compute, currency cost, dominance and rank remain null until a named language-stage-subject-format package is compared with current usable provision at a common endpoint. The constructor passes 57 source and structural checks, an independently implemented readback passes 82, the manuscript/state integration passes 77, and two complete builds are byte-identical.

4.5.2 India is not one Hindi target

The India cross-stage checkpoint (source file: `IND_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) applies the same six-stage comparison without substituting either a country or a Census category for an exact language. It carries all 518 Glottolog language-level nodes whose registry association includes `IN`, including 27 bookkeeping nodes and four signed-language nodes. It also retains all 22 scheduled-language groups in the Census frame. Registry association does not establish current residence, use, vitality, need or comprehension of Hindi or English; the scheduled-language groups are historical administrative categories that can aggregate multiple mother tongues.

That distinction is especially important for the 2011 Census `HINDI` group. Its published total is 528,347,193, but the source decomposes it into 57 named or residual rows. The largest source-labelled components include Hindi (322,230,097), Bhojpuri (50,579,447), Rajasthani (25,806,344), Chhattisgarhi (16,245,190), Magadhi/Magahi (12,706,825), Haryanvi (9,806,519), Khortha/Khotta (8,038,735) and Marwari (7,831,749). Some labels map cleanly to a Glottolog language, some remain unresolved, and some are residual buckets. The group total therefore cannot be used as the addressable audience for Standard Hindi in Devanagari. Reporting English among at most two subsidiary languages is likewise a reporting variable, not a test of comfortable academic comprehension (Office of the Registrar General & Census Commissioner, India, 2011a, 2011b).

The stage evidence identifies different questions rather than one national deficit. UDISE+ reports 246,932,680 enrolments from foundational through secondary education in 2024-25, with 52,312,506 in the foundational stage, 66,115,921 preparatory, 63,695,100 middle and 64,809,153 secondary; these administrative counts are not language-specific unmet need (Department of School Education & Literacy, 2025). In rural ASER 2024, 23.4% of government-school Standard III pupils could read a Standard II text, 33.7% of all-school Standard III pupils could perform the subtraction task, 30.7% of Standard V pupils could perform the division task and 45.8% of Standard VIII pupils could perform the basic arithmetic task. These named-task results make foundational explanation and feedback consequential, but they do not identify language as the cause or translation as the measured remedy (ASER Centre, 2025).

Higher and practical education remain separate. AISHE 2023-24 reports nearly 45 million higher-education enrolments, a gross enrolment ratio of 30.0 for ages 18-23, 10,188,988 STEM enrolments across undergraduate through doctoral levels, and 343,559 PhD enrolments (Department of Higher Education, 2026). NPTEL explicitly describes a transition from regional-language schooling into mainly English technical education and provides translated material in eleven named languages. The frozen catalogue pages contain 292 unique Hindi title labels, 248 Tamil, 212 Gujarati, 209 Telugu, 199 each in Malayalam and Marathi, 188 Kannada, 181 Bengali, 35 Punjabi, 16 Odia and one Assamese. These are catalogue-presence counts, not complete-course, licence, accessibility, uptake, quality or satisfied-need measures. Actual inspected Hindi and Telugu mathematics books already contain explanations, worked examples, results and glossaries, so a generic duplicate prerequisite book is not yet justified; a smaller navigation/terminology/prerequisite bridge remains testable at the exact course endpoint (NPTEL, n.d.).

Delivery evidence makes a one-download autonomous-study route plausible but not universal. The 2025 telecom survey reports 86.3% of households with internet at home and, by adding mutually exclusive smartphone categories, 85.5% with some smartphone. Internet use in the preceding three months was 93.6% among ages 15-29 and 70.0% among ages 15 and older. These measures do not establish personal ownership, adequate storage or screens, accessible input, reliable power, affordable continuous data or successful study. ASER reports that rural smartphone ownership among 14-16-year-olds was 36.2% for boys and 26.9% for girls; among those reporting smartphone ability, 57% used one for education in the preceding week. Small downloads, phone-native layout, resumable offline state, print export and shared-device continuity must therefore be evaluated separately (National Statistical Office, 2025; ASER Centre, 2025).

The resulting package portfolio is stage-specific. At P/F, compare coherent exact child/home-language literacy and numeracy pathways with bounded diagnostics and worked feedback in phone, print, audio and appropriate signed forms. At S, bind a state or board, exact language, subject and transition; fractions-to-algebra, scientific reading, graphs/data and laboratory reasoning are concrete hypotheses. At V, attach finance, records, measurement, health, agriculture, management, computing or safety units to an actual qualification or job task. At U, compare complete open courses with bilingual prerequisite, terminology, formula-reading and offline-lab bridges. At R, compare complete graduate source chains with proof/search/terminology routes around open resources such as Open Logic and the Stacks Project. Indian Sign Language's 10,000-term video dictionary includes academic and vocational categories and records regional variation, but a dictionary is not a curriculum and one standardized sign route does not cover every signed variety (Indian Sign Language Research and Training Centre, n.d.).

No India-wide or Hindi-wide rank follows. All exact audiences, incremental effects, physical compute, currency costs and dominance relations remain null. Every scheduled group, HINDI-group component, signed-language node and other exact variety with missing matched evidence remains a possible outranker. The gate passes 28 source assertions and 11 constructor checks; an independent reconstruction passes 227 checks; four preserved PDFs contribute 2,029 source pages, ten page-level claim checks and nine visually inspected renders; five generated outputs are byte-identical across two complete builds.

4.5.3 Nigeria is 589 language obligations, not one English-policy row

The Nigeria cross-stage checkpoint (source file: `NGA_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) retains all 589 Glottolog language-level nodes whose registry association includes `NG`: 571 non-bookkeeping nodes, eighteen bookkeeping nodes and seven signed-language nodes. The World Bank reports a 2025 country population of 237,527,782, but that is not the audience for English, Hausa, Yoruba, Igbo, Nigerian Pidgin or any educational package. A 2025 federal announcement says that English is the medium of instruction from primary through tertiary education. Policy language is not a measurement of comfortable educational English, and it does not cover hundreds of other exact languages by declaration (Federal Ministry of Information and National Orientation, Nigeria, 2025; World Bank, 2026a).

The strongest evidence identifies serious educational tasks without assigning them to languages. Nigeria's 2021 MICS reports foundational reading skills for 26.8% and foundational numeracy skills for 25.3% of children ages 7-14. Reading tasks were available in English, Hausa, Igbo and Yoruba; 8% of interviewed children could not be evaluated in their home or school language. NALABE 2022 reports minimum proficiency of 50% in Primary 3 mathematics, 34% in Primary 5 mathematics and 14% in Junior Secondary 2 mathematics. At JS2, mathematics reasoning averaged 35% correct and science-and-technology application 36%. These national samples locate high-consequence foundational and secondary tasks. They do not show that language caused the results, that translation alone would solve them or how the burden divides among languages (National Bureau of Statistics of Nigeria & UNICEF, 2022; Universal Basic Education Commission & UNICEF, 2025).

Nor can the factsheet's ethnicity breakdown repair that missing language denominator. Its Hausa, Igbo, Yoruba, Fulani, Kanuri, Ijaw, Tiv, Ibibio and Edo categories refer to ethnicity of household head, not home language, reading ability or an addressable edition audience. The analysis therefore assigns none of those rows to a language. Instead it preserves exact target and script questions. Hausa Boko and Hausa Ajami are separate routes; Ajami naskh and Kano/Maghribi rendering and Nigeria/Niger conventions require their own evidence. Five Nigeria-linked Fulfulde nodes remain distinct, with Latin, Ajami and Adlam only investigation routes. Central and Manga-Dagera Kanuri remain separate. NERDC's unqualified labels `Fulfude`, `Kanuri` and `Edo` do not automatically resolve those identities, and a reported common-core Naijá orthography does not prove that every Nigerian Pidgin community reads it (Nigerian Educational Research and Development Council, n.d.-a; IFRA-Nigeria, n.d.; Evans & Warren-Rothlin, 2018).

The package comparisons differ by stage. At P/F, compare coherent child/caregiver-language oral, reading and numeracy pathways with smaller diagnostic and worked-feedback units in phone, print, audio and appropriate signed forms. At S, test fractions-to-algebra, mathematical reasoning, graphs/data, scientific reading and science application against a named curriculum and exam. At V, bind finance, bookkeeping, agriculture, health, management, digital skills, measurement or safety to an actual qualification or job task. At U, compare complete open courses with prerequisite, terminology, scientific-writing, formula-reading and runnable-computing bridges. At R, compare complete graduate source chains with proof, search, terminology and reproducible-methods layers around open sources. NERDC reports a nine-year basic curriculum, 42 mainstream senior-secondary subjects, 34 trade/entrepreneurship subjects, out-of-school-youth material, visually impaired curricula and instructional sign language; NBTE and TERAS expose vocational, online, repository and subscription routes. Those are supply signals, not proof of complete, openly reusable, accessible or effective provision (Nigerian Educational Research and Development Council, n.d.-b; National Board for Technical Education, n.d.; Tertiary Education Trust Fund, n.d.).

Delivery is a separate part of the intervention. The latest World Bank series reports 41.21139908% internet use and 62.5% electricity access in 2024. UNICEF reports that students ages 5-24 lived in households with 91% phone, 39% internet and 10% computer access, while 41% lacked television, internet or a computer. Household access is not personal ownership, permission, a smartphone, reliable electricity/data, storage, accessibility or successful independent study. Small downloads, phone-native layout, resumable offline state, print, audio, semantic HTML/MathML, screen-reader navigation and exact signed-language video must be costed separately. All seven signed-language nodes remain distinct; Nigerian Sign Language, Hausa Sign Language, Yoruba Sign Language, Bura, Ibokun, Magajingari and the multinational American Sign Language catalogue linkage do not cover one another by name (UNICEF Nigeria, 2023; World Bank, 2026a).

No Nigeria-wide or language-wide rank follows. All exact audiences, incremental effects, physical compute, currency costs and dominance relations remain null. Every one of the 589 linked nodes remains a possible outranker when matched evidence is missing. The gate passes 18 source assertions and 21 additional constructor checks; an independent reconstruction passes 160 checks; two preserved PDFs contribute 117 pages, six page-level claim groups and six visually inspected renders; all five tracked outputs are byte-identical across two complete constructor/checker cycles. Existing local project work has zero effect on Nigeria's educational need or global priority.

4.5.4 Pakistan is 86 language obligations, not one Urdu or English route

The Pakistan cross-stage checkpoint (source file: `PAK_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) retains all 86 Glottolog language-level nodes whose registry association includes PK: 83 non-bookkeeping nodes, three bookkeeping nodes and two signed-language nodes. Twenty-three high-consequence profiles receive explicit source questions; the other 63 remain visible source-research obligations. Country association does not establish current residence, language use, vitality, literacy or package need. The World Bank's 2025 country-population series reports 255,219,554 people, but that is not the audience for Urdu, English, Punjabi, Sindhi, Pashto or any other edition (Hammarström et al., 2026; World Bank, 2026b).

The strongest matched count is foundational rather than language-specific. In the 2023 detailed census frame, 25,373,350 of 71,270,068 children aged 5-16 were out of school: 20,033,448 had never attended and 5,339,902 were recorded as dropouts. Five published mother-tongue margins give logical intersection bounds, not readership or benefit estimates. Urdu is bounded at 0-7,778,163; Punjabi at 0-12,140,704; Sindhi at 3,098,795-8,757,513; Pushto at 0-13,298,209; and Saraiki at 0-12,167,106. The sharp union interval under the ten retained geographic margins is 9,929,282-25,373,350. Adding the five separate upper endpoints would produce an impossible 54,141,695, because each maximum assumes a different allocation of the same children (Pakistan Bureau of Statistics, 2023a, 2023b, 2023c).

Current supply narrows the intervention rather than settling it. The non-formal education census reports 1,073,704 enrolments, including 98,370 in accelerated primary, 38,686 in accelerated middle or middle-tech, 902,485 in non-formal basic primary, 1,231 in non-formal middle and 32,932 in adult literacy. The inclusive scheme specifies thirty-month A/B/C and eighteen-month D/E routes, with mathematics displayed in Urdu. Sindh's portal displays Sindhi, Urdu and English books; Punjab reports 6.2 million Urdu/English textbooks dispatched; Balochistan lists materials from Primer through Grade 12; and federal/KP sites expose curriculum and teacher-material routes. These are programme, portal and dispatch observations. They do not establish complete open supply, exercises and answers, offline access, accessibility or independent-study usability (National Curriculum Council, 2025; Pakistan Institute of Education, 2024a).

The National Achievement Test identifies concrete secondary tasks. Across its public-school sample, Grade 8 mathematics averaged 42% correct and science 51%; Grade 8 statistics and probability averaged 32.9%. Balochistan's Grade 8 mathematics sample averaged 30.6%, with 36.3% of pupils at or below the 25% threshold. The instruments were not one-language national measures: mathematics and science were offered in Urdu and Sindhi as well as the English booklet route described in the report. These results motivate mathematics, statistics and scientific-reading comparisons, but do not identify language as the cause, Balochi as the tested language or translation alone as the remedy (Pakistan Institute of Education, 2024b).

The six-stage portfolio therefore compares complete and bounded packages. At P/F, it tests exact child/caregiver-language early learning, literacy, numeracy and school re-entry against smaller diagnostic and worked-feedback units, with age-respectful prerequisite branches. At S, it compares complete mathematics/science/reading sequences with fractions-to-algebra, statistics/probability, graphs and scientific-reading bridges. At V, it binds applied numeracy, finance/management, digital skills and trade safety to an actual qualification. At U, it compares full degree foundations with gateway-course prerequisites, bilingual terminology and worked practice. At R, it compares complete open graduate source chains with terminology, proof-dependency, search and navigation layers. All nine measured OpenStax sources and the Open Logic/Stacks baselines remain alternatives, not automatic recommendations.

Exact identity remains part of every package. Western Panjabi/Shahmukhi and Eastern Panjabi/Gurmukhi are not assigned the whole census Punjabi interval. Central, Northern and Southern Pashto remain distinct within the broad Pushto margin. Eastern, Southern and Western Balochi remain distinct from one another and from Brahui; Shina and Kohistani Shina, Northern and Southern Hindko, Balti, Kashmiri, Khowar, Burushaski and Gujari remain separate. Pakistan Sign Language and the multinational Indian Sign Language registry linkage are separate nodes. A reported standardized Pakistan Sign Language book, sign interpreters and Urdu Braille activity do not establish a complete signed or tactile curriculum or equivalence between signing communities (Ministry of Human Rights, Pakistan, 2024).

Delivery must be compared independently. The latest World Bank series reports 57.253% internet use, 95.7% electricity access and 76.854 mobile subscriptions per 100 people in 2024. A subscription is not a unique smartphone owner, and a country-level connection or electricity indicator does not establish personal access, permission, reliable service, storage, accessibility or learning success. Phone-native, one-download offline, print, audio, signed video, semantic HTML/MathML, screen-reader and tactile/Braille routes therefore remain separately costed modes (World Bank, 2026b).

University and research needs also remain live. HEC reports 1,964,147 higher-education enrolments in 2022-23, more than 80,000 digital-skills learners, 31,353 PhD-directory registrants, 19,250 public dissertations and more than 30,000 journals. Its own FAQ says that digital-library access is institutional, IP-based, subscription-dependent and subject to cost sharing, not an open individual licence. This supports exact gateway-course and advanced-source comparisons without claiming that Pakistan lacks universities or journals (Higher Education Commission, Pakistan, 2025; Higher Education Commission, Pakistan, n.d.).

No Pakistan-wide or language-wide rank follows. Every exact audience, incremental effect, physical compute, currency cost and dominance relation remains null. The source audit passes 179 checks across eight preserved PDFs containing 1,055 pages, 23 page-bound claim groups, nine local official web surfaces, five infrastructure indicators and fourteen visually inspected renders. The constructor passes 21 source assertions and 22 additional checks; an independent implementation passes 601 checks; two complete constructor replays are byte-identical. Every missing exact-language measurement remains a possible outranker, not a low-need disposition.

4.5.5 Ukraine and Russia are a paired omission check, not one Slavic route

The Ukraine-Russia checkpoint (source file: `UKR_RUS_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) asks the same six stage questions of two separate country contexts. It is a sentinel for a global failure mode, not a comparison of national worth. The country-linked register retains 183 distinct Glottolog language-level nodes: 32 linked to Ukraine, 168 linked to the Russian Federation and seventeen appearing in both registry associations. The union contains one bookkeeping node and two signed-language nodes. A registry association can be historical, regional or cross-border; it does not establish current residence, language use, literacy, educational need or comprehension of a dominant language.

The latest retained World Bank population series reports 38,980,376 people for Ukraine and 143,513,328 for the Russian Federation in 2025. Neither number is the audience for a Ukrainian- or Russian-language edition. Ukraine's 2001 census reported 67.5% Ukrainian, 29.6% Russian and 2.5% other native languages; the dated categories are not multiplied by a current country or displacement total. A 2026 KIIS sample in government-controlled Ukraine reported 62% speaking mainly or always Ukrainian at home, 23% using both and 14% mainly or always Russian, but excluded occupied-territory residents and people who left abroad after February 2022. Those home-use categories do not measure preferred language for school, technical reading or research (State Statistics Committee of Ukraine, 2001; Kyiv International Institute of Sociology, 2026; World Bank, 2026c, 2026d).

Displacement creates a host-curriculum problem rather than one portable language count. Eurostat reported 4.41 million people who had fled Ukraine under temporary protection in the EU on 30 June 2026; 29.6% were minors, giving about 1,305,360 under the report's rounded inputs. That is an administrative and demographic context, not a school-enrolment, speaker or unmet-need count. A useful package hypothesis pairs Ukrainian explanation with the learner's actual host instructional language and offers Russian terminology where the learner chooses it. The separate Estonia evidence instead supports a Russian-Estonian subject-language transition route: only 63% of Russian-medium basic-school graduates reached Estonian B1 during the audited 2005-2020 period, against a 90% target. Neither corridor is transported to another country by analogy (Eurostat, 2026; National Audit Office of Estonia, n.d.).

Stage evidence also resists a single ordering. In the eighteen of twenty-seven Ukrainian regions represented by PISA 2022, 42% of students were below Level 2 in mathematics, 41% in reading and 34% in science; 30% attended schools whose principals reported teaching-staff shortages. The Russian 2025 data book reports 6.3853 million preschool, 17.9881 million general-education, 3.8506 million combined skilled-worker and mid-level vocational, 4.4317 million bachelor's/specialist's/master's and 125,900 postgraduate enrolments. Its separate 2023 national PISA-based assessment reported much lower below-basic shares, but that instrument and frame are not silently compared with the OECD Ukrainian result (OECD, 2023c; HSE University Institute for Statistical Studies and Economics of Knowledge, 2025).

The package implications therefore differ by stage and exact community. P/F compares coherent child- or caregiver-language early learning, literacy and numeracy with bounded oral, picture, diagnostic and worked-feedback units. S compares complete mathematics, science and reading sequences with prerequisite, subject-language and curriculum-continuity bridges. V requires a named trade, qualification and language rather than a national vocational total. U compares full open courses with gateway-prerequisite, terminology, formula-reading and runnable-computing companions. R compares coherent graduate source chains with proof, search, terminology and reproducible-methods layers. UNESCO's report that 12% of Ukrainian scientists and university teachers had relocated and about 30% of scientists were working remotely supports portable and restartable research access; it does not show a language effect (UNESCO & United Nations Ukraine, 2024).

Existing Russian provision materially narrows some proposals. Yandex displays school mathematics and computing resources; Stepik's inspected introductory statistics page displayed 326,869 learners; Open Education displayed 1,459 courses and 419 programmes; MCCME provides full mathematical books; and Math-Net maintains journal archives. These are substantive alternatives, not proof of complete open licences, autonomous-study coherence, accessibility or zero additional need. They require a Russian package to identify the missing topic, prerequisite, answer support, format, rights or delivery advantage. They say nothing by themselves about the exact needs of Chechen, Tatar, Bashkir, Chuvash, Sakha, Tuvinian, Mari, Udmurt, Komi, Moksha, Ossetian, Nenets, Evenki, Chukchi, Nivkh or the other Russia-linked nodes.

Ukrainian, Russian, Belarusian, Polish, Crimean Tatar, several Romani nodes, Eastern Yiddish and every other intersecting node remain separate. Ukrainian Sign Language and Russian-Tajik Sign Language are separate signed routes whose video, notation, regional-variation and low-bandwidth costs are not represented by written token traffic. An Interslavic mathematics edition remains a valid experiment because notation and prepared-reader subject knowledge may reduce some prose burden. It receives no automatic audience or coverage: worked-example use, learning a new definition, following a proof and navigating a research reference remain separate tests. If `q` is the bridge's retained additional benefit relative to matched separate editions and `r` is its total-cost ratio, the bridge is more efficient only when `q > r`; the selected literature supplies no value for `q`.

No Ukraine-versus-Russia, Ukrainian-versus-Russian or interlanguage-versus-native funding order follows. All 183 nodes retain P/F/S/V/U/R obligations, null audience/effect/compute/currency/dominance/rank fields, no automatic coverage and possible-outranker status. Fifty-eight nodes have a first explicit evidence question; the other 125 remain visible rather than being classified as low need. The source audit passes 257 checks across three PDFs containing 359 pages, eleven page-bound claim groups, thirteen local web-source groups, ten World Bank indicators and eleven manually inspected renders. An independent readback passes 1,715 checks, and two fresh constructor runs reproduce all three outputs byte-for-byte.

4.5.6 Japan is not a negative control

The Japan cross-stage checkpoint (source file: `JPN_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) tests a proposition that the earlier comparison left unresolved: strong aggregate educational performance does not make a country or language an appropriate zero-need control. Japan's 2025 population of 123,366,734, its large school and research systems, and its strong national PISA results establish scale and substantial existing provision. None is an exact count of people who would gain access from a particular Japanese, other-language, signed-language or accessibility package (MEXT, 2024a, 2024c; World Bank, 2026e).

One current functional-access population is directly measured. MEXT defines learners needing Japanese-language instruction to include both learners without sufficient everyday Japanese and learners who can converse in Japanese but lack the grade-level academic language needed to participate in learning. The 2023 survey counts 69,123 enrolled learners: 57,718 of foreign nationality and 11,405 of Japanese nationality. Of these, 46,132 were in elementary school, 15,967 in lower secondary and 5,573 in upper secondary, with the balance in compulsory-education, secondary and special-support schools. Among 4,464 lower-secondary graduates in the survey frame, 90.3% progressed to further education, compared with 99.0% for all students in the source comparison; 5.0% were in neither progression nor employment, compared with 0.8%. These are measured transition signals, not a translation effect or a complete census of migrant, minority or multilingual learners (MEXT, 2024b).

The same survey prevents a one-language interpretation. Its two disjoint nationality partitions reconcile to broad daily-life-language counts of 13,754 Chinese, 12,579 Portuguese, 11,121 Filipino, 4,918 English, 4,046 Spanish, 3,978 Vietnamese, 669 Korean, 6,149 Japanese and 11,909 other. These are survey labels, not exact varieties, scripts or production tags. The report also records mother-tongue support routes in 409 local authorities, with further named examples including Nepali, Indonesian, Thai, Russian, Urdu, Mongolian, French, Hindi, Arabic, Bangla, Sinhala, Persian and Ukrainian. Authority counts do not measure staff capacity, learner coverage, material quality or effect.

The six educational stages therefore produce six different comparisons. At preschool and foundational stages, compare a coherent child/caregiver-language early-learning or literacy/numeracy pathway with a bounded diagnostic and academic-Japanese bridge in the exact home, signed or accessible route. At secondary level, compare complete mathematics, science, computing and reading pathways with a transition package for a named bottleneck such as algebra, data literacy, scientific reading or academic Japanese. At vocational level, bind a specific credential or work task before comparing full reskilling sequences with smaller statistics, finance, management, health or computing toolkits. At university level, compare a complete open course with a prerequisite, terminology, formula-reading or offline-laboratory companion. At postgraduate/research level, compare coherent advanced source chains with a source-faithful edition or bilingual proof, terminology and search scaffold for a named missing work.

Existing provision changes these comparisons rather than ending them. MEXT's tables report 607,951 specialized-training students, 2,945,599 university students, 168,706 master's students, 75,841 doctoral students and 910,393 researchers in the retained 2023 frames. J-STAGE reports more than 4,000 publications from more than 2,400 publishers and says that more than 90% of its articles are free to read; only items displaying an open-access licence establish the corresponding reuse permission. A historical 2010-2012 NISTEP analysis also shows a mixed Japanese/English publication ecology: domestic non-open-access journals were 53.2% English and 46.5% Japanese in its Scopus-derived frame, whereas domestic open-access journals were 89.6% English and 10.2% Japanese. Publication language does not measure comprehension, present-day shares or unmet educational need (Japan Science and Technology Agency, n.d.; NISTEP, 2016).

Japanese is not every Japan-linked language. The universal gate retains all 25 `JP`-linked Glottolog language-level nodes. The Agency for Cultural Affairs separately names Ainu, Yaeyama, Yonaguni, Hachijo, Amami, Kunigami, Okinawan and Miyako as endangered language or variety categories. Broad labels such as Ainu and Amami are not forced into one exact node. Amami O Shima Sign Language, Japanese Sign Language and Miyakubo Sign Language remain three separate signed-language nodes; the 2025 statutory recognition of sign language does not establish that one signed route covers the others (Agency for Cultural Affairs, n.d.; Cabinet Office, 2025).

The immediate experimental order is not a funding rank: first test the measured academic-Japanese transition route with an exact home-language, signed-language or accessibility arm; separately test one matched offline/accessibility package, one named advanced Japanese edition or bilingual research scaffold, and one bounded endangered-language or signed-language prestige-domain prototype. All nine measured OpenStax collections, Open Logic, the Stacks Project and other advanced open corpora remain comparison baselines rather than automatic recommendations. Every exact audience, incremental effect, physical compute, currency cost, dominance relation and funding rank remains unset. The source audit passes 232 checks across eighteen retained sources, five PDFs containing 236 pages, five workbooks containing 106 sheets and sixteen visually inspected PDF pages. An independently implemented readback passes 298 checks, and two complete five-process replays reproduce all eight canonical outputs byte-for-byte.

4.5.7 Brazil is not one Portuguese-language audience

The Brazil cross-stage checkpoint (source file: `BRA_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) separates a large Portuguese-medium education system from exact Indigenous-language, signed-language, adult-re-entry, distance-learning, accessibility and research-access questions. The zero-disappearance register retains all 337 Glottolog language-level nodes associated with Brazil, including ten bookkeeping nodes and seven signed-language nodes. The 2022 Census table independently retains 304 census language labels. Those two taxonomies are not forced into a name-based crosswalk, and the sum of the census label counts-475,893-is not treated as a count of unique people because the source hierarchy and reporting categories can overlap (Brazilian Institute of Geography and Statistics [IBGE], 2025a).

The directly observed language-use frame is consequential but still not an intervention audience. Among 1,521,956 Indigenous people aged five or older in the relevant 2022 Census table, 433,980 were recorded as using an Indigenous language at home, 1,327,592 as using Portuguese at home and 181,641 as not using Portuguese at home. Portuguese and Indigenous-language use are overlapping categories, so neither the two yes-counts nor their percentages can be added. A separate simple-note literacy measure records 178,707 of 1,187,246 Indigenous people aged fifteen or older as not literate, or 15.05%, compared with 7% nationally.

It does not identify their exact language, desired instructional route or likely response to a translated package (IBGE, 2025b, 2025c).

The stage denominators show why one national score would be misleading. The 2024 school census reports 47,088,922 basic-education enrolments, including 9,491,894 in early childhood, 26,002,356 in foundational education and 7,790,396 in secondary education. Its 2,576,293 vocational enrolments overlap secondary and adult-education categories and cannot be added as another population. Higher education contains 10,226,873 undergraduate enrolments, of which 5,189,391 are in distance education. CAPES records 419,915 distinct postgraduate people in its 2024 frame. PISA represents about 2.263 million fifteen-year-olds and reports 73% below Level 2 in mathematics, 50% in reading and 55% in science; those subject groups overlap, exclude an estimated 24% of the age cohort and do not establish a language cause (Inep, 2025a, 2025b; OECD, 2023d; CAPES, 2025).

Indigenous and signed-language provision is a distinct comparison rather than a residual after Portuguese. The retained federal report records 3,608 schools offering Indigenous education and 294,249 enrolments in 2024, plus teacher, higher-education and language-documentation programmes. Its reported count of fifty co-officialized oral and signed languages is a policy count, not an identity crosswalk or proof of complete curricular supply. Federal law establishes Libras and written Portuguese as a bilingual deaf-education route, but it supplies no equivalence between Libras and Macushi, Marajo, Maxakali, Papiu Yanomama, Terena or Urubú-Kaapor Sign Language. Each of the seven registry nodes therefore remains its own video, notation, regional-variation, transcript, caption and low-bandwidth research obligation (Ministry of Planning and Budget, Brazil, 2026; Federal Senate of Brazil, 2021).

The six educational stages consequently support different matched experiments. P compares coherent child/caregiver-language early learning with one bounded phone, print, audio or signed activity pack. F compares complete literacy, numeracy and science pathways with an exact-language or academic-Portuguese diagnostic and worked-feedback bridge. S compares full mathematics, statistics, computing and science sequences with one prerequisite, subject-language or data-literacy transition package. V requires an actual occupation or qualification before testing finance, management, statistics, computing, health or safety material. U compares complete degree-core sources with a prerequisite, terminology, accessibility or offline companion, particularly in the large distance route. R compares coherent advanced source chains with a source-faithful Portuguese edition or bilingual proof, terminology and search scaffold for a named missing work. Research education remains education; strong journal and university systems do not supply every advanced open work.

Eleven target profiles and all nine measured OpenStax source packages are now bound to these questions, alongside Open Logic, forallx and the Stacks Project as advanced baselines. The near-term experiment sequence starts with one exact Indigenous-language F route, one exact signed-language mathematics or science route, one Portuguese or bilingual S prerequisite route, one EJA/V practical route, one low-bandwidth U route and one named R comparison. This sequence is workflow, not funding rank. Exact audience, incremental effect, physical compute, currency cost, dominance and global rank remain unset. The source audit passes 858 checks across 26 retained source slots and nineteen visually inspected PDF pages; an independent reconstruction passes 5,310 checks; two complete six-process replays reproduce all nine canonical outputs byte-for-byte.

4.5.8 Mexico is not one Spanish-language audience

The Mexico cross-stage checkpoint (source file: `MEX_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) preserves three identity systems without pretending that they are interchangeable. Glottolog contributes 339 language-level nodes associated with Mexico, including seventeen bookkeeping nodes and five signed-language nodes. INALI independently specifies eleven families, 68 groupings, 364 catalogue variants and 476 self-denominations. Its own methodology says that a grouping can contain speech forms that are not mutually intelligible. INEGI's selected census chart instead reports ten broad labels plus an “other languages” category. A label such as Náhuatl, Maya, Mixteco or Zapoteco is therefore neither one automatic production audience nor a source-audited match to one Glottolog node (Instituto Nacional de Lenguas Indígenas [INALI], 2008; Instituto Nacional de Estadística y Geografía [INEGI], 2022).

The scale is consequential but does not settle the language boundary. The 2020 Census records 7,364,645 people aged three or older who spoke an Indigenous language, 6.1% of the same-age population. The 2023-2024 Indigenous-education report records 48,290 initial, 417,934 preschool and 779,514 primary enrollments: 1,245,738 in total, with 60,597 teachers or group-facing personnel and 22,315 schools. It also reports 569,547 Indigenous-language primary material copies distributed across twenty broad language columns. Copies are not unique titles, learners reached, exact variants served, open licenses, usability or learning effects. The material table therefore changes the next supply question; it does not erase any of the 364 exact-variant obligations (Dirección General de Educación Indígena, Intercultural y Bilingüe [DGEIIB], 2024; INEGI, 2022).

The six stages require different denominators. Current SEP figures report 3,996,767 preschool, 12,838,201 primary, 6,293,530 secondary and 5,492,491 upper-secondary enrollments in 2024-2025. The 43,597 professional-technical enrollments are nested within upper secondary. Higher education contains 5,519,791 enrollments, including 4,913,471 university and technological undergraduates and 478,382 postgraduate enrollments. PISA represents about 1,393,700 fifteen-year-olds-an estimated 64% of that age population-and reports 66% below Level 2 in mathematics, 47% in reading and 51% in science. Those three burdens overlap in the same represented cohort and cannot be added as separate people or attributed to language without a matched test (Organisation for Economic Co-operation and Development [OECD], 2023e; Secretaría de Educación Pública [SEP], 2025).

Observed supply is substantial but unevenly informative. CONALITEG exposes current preschool, primary, secondary and telesecundaria textbook routes. Cursos @prende.mx displays 2,708,965 cumulative registrations, 1,359 courses and 95 partner institutions; its help page describes Spanish-language courses that are free and available online around the clock, phone/tablet viewable, but better on a desktop or laptop. That is not evidence of downloadable offline supply, exact-language access or item-level open licensing. SciELO Mexico lists 288 journals and 183 current serials, while ANUIES exposes a current higher-education yearbook route. Current federal law separately names Indigenous, intercultural and plurilingual education, open and distance options, Mexican Sign Language, Braille, audio, large print, captions and audiodescription. Legal recognition and catalogue presence are intervention context, not proof of complete implementation or coverage of the four non-MSL signed-language nodes (Asociación Nacional de Universidades e Instituciones de Educación Superior [ANUIES], 2026; Cámara de Diputados, 2024, 2026; CONALITEG, 2026; SciELO México, 2026; SEP, 2026).

The matched package questions follow from those distinctions. P compares an exact child/caregiver-language early-learning sequence with one bounded phone, print, audio or signed activity pack. F compares coherent literacy, numeracy and science pathways with a variant-specific diagnostic and bilingual transition. S compares complete mathematics, statistics, computing and science sequences with one prerequisite or academic-language bridge while preserving PISA's coverage limit. V requires an actual task or qualification before testing finance, management, statistics, computing, health or safety. U compares degree-core sources with one gateway-course, terminology, accessibility or offline companion. R compares coherent advanced source chains with full translation, source-facing parallel text, bilingual navigation or one bounded research sidecar for a named field. Thirteen target profiles, all nine measured OpenStax sources, Open Logic, the Stacks Project and advanced research corpora are bound to these questions. No exact audience, incremental effect, physical compute, currency cost, dominance or global rank is assigned. The source audit passes 144 checks; an independent reconstruction passes 11,793; and two six-process runs reproduce nine canonical outputs byte-for-byte. This is a workflow checkpoint, not a country or funding order.

4.5.9 Ethiopia cannot be reduced to an Amharic- or English-language audience

The Ethiopia cross-stage checkpoint (source file: `ETH_CROSS_STAGE_SUPPLY_AND_PACKAGE_v1.md`) preserves the country's multilingual educational problem before asking which package should receive compute. The zero-disappearance register contains all 105 Glottolog language-level nodes associated with Ethiopia, including two bookkeeping nodes and Ethiopian Sign Language. Fourteen exact first-question profiles cover Amharic, Afar, three separate Oromo nodes, Somali, Tigrinya, Sidamo, Wolaytta, Hadiyya, Nuer, Berta, Harari and Ethiopian Sign Language. The remaining 91 nodes are still visible possible outrankers. Amharic, English, a broad source label or an interlanguage is not credited with covering another node without evidence of usable comprehension.

The available population evidence is consequential but cannot be converted into current language audiences. The 2007 census defines mother tongue through childhood acquisition or the language most frequently used in the household and reports a national total of 73,750,932. Its selected historical labels include Oromigna 24,930,424, Amarigna 21,634,396, Somaligna 4,609,274, Tigrigna 4,324,933, Sidamigna 2,981,471, Welaitigna 1,627,955, Affarigna 1,281,284 and Hadiyigna 1,253,894. Those are 2007 census categories, not current language-edition audiences. They are not projected by multiplying their old shares into the World Bank's 2025 national population estimate of 135,472,051, and broad labels such as Oromigna are not silently assigned to one of the three retained Oromo nodes (Ethiopian Statistical Service, 2008; World Bank, 2026).

The foundational evidence makes exact-language experiments urgent enough to test, but it does not supply a ready-made rank. A 2021 national early-grade reading assessment covered nine named language groups, 484 primary schools, 968 teachers and 19,360 students. It reported zero-reader classifications for 68% of Grade 2 learners and 51% of Grade 3 learners nationally, with large differences across language groups. A different 2024 midterm evaluation in Afar and Oromia reported weighted means of 37.8 letters, 10.7 familiar words and 10.3 oral-reading words per minute, plus 10.8% comprehension. The report itself cautions that observed associations are not necessarily causal. The two studies have different samples and cannot be merged into one national language effect (World Food Programme, 2024).

Delivery conditions also change what counts as a useful edition. The 2022/23 statistics abstract reports that only 11% of respondent secondary schools had Internet access, while 69.7% had electricity and 62.4% had computers. Functional biology, chemistry and physics laboratories were reported at 34.6%, 35.9% and 31.9%, respectively. The World Bank's national series place Internet use at 21.94% and electricity access at 56.6% in 2024. These measurements use different frames, but together they rule out an online-only default. The same statistics abstract warns that new-curriculum textbooks were not completely printed and distributed and contains an internal reporting-year inconsistency that remains explicit (Ethiopia Ministry of Education, 2023a; World Bank, 2026).

Current law and strategy define additional routes without proving implementation. General Education Proclamation No. 1368/2025 provides for mother-tongue instruction from pre-primary to an endpoint chosen by regional education bodies, makes English compulsory as a medium from Grade 9, and names sign language, Braille and adaptive technology. The inclusive-education strategy identifies sign-language, Braille, audio-visual and assistive-material needs, while the digital strategy calls for online and offline delivery and open educational resources. The vocational strategy emphasizes community-led labour-market evidence, digital skills, entrepreneurship, recognition of prior learning, refugees and host communities. Each is a source-bound design requirement, not a measured beneficiary count (Federal Democratic Republic of Ethiopia, 2025; Ethiopia Ministry of Education, 2023b, n.d.-a; Ethiopia Ministry of Labor and Skills, 2025).

The resulting package comparison spans all six stages. P tests an exact child/caregiver-language, audio or signed early-learning sequence. F tests coherent literacy, numeracy and introductory science with worked feedback in the exact language and script. S tests offline mathematics, statistics, computing and science plus the transition to English-medium study. V starts from a named occupation or qualification and compares practical finance, management, agriculture, health, safety, computing and entrepreneurship resources. U compares a complete degree-core route with a course-specific prerequisite, terminology, accessibility or low-bandwidth companion. R compares a named advanced source chain with a source-faithful edition, bilingual proof scaffold, terminology and search support. Research education remains eligible; it is not traded away merely because foundational need is severe.

The comparison maps nine measured OpenStax books and ten other open baselines, including TESSA, OpenRN Nursing Fundamentals, soil science, climate-smart agriculture, the Carpentries, the Turing Way, open research methods, MIT OpenCourseWare, Open Logic and the Stacks Project. Their combined 4,463,123-token OpenStax text scale is a workload reference, not a compute bill or benefit score. Teacher availability is not assumed: phone-downloadable, printable, low-bandwidth, worked-answer, semantic-HTML, audio, Braille and signed-video variants are separate delivery designs. The 630-row language-by-stage register keeps every audience, incremental effect, physical compute, currency cost, dominance relation and funding rank unset. The source audit passes 233 checks across twelve new and eight reused source files and 39 visually inspected pages; an independent reconstruction passes 620 checks; two complete replays reproduce all six canonical outputs byte-for-byte.

4.5.10 The United States is not one English-language audience

The United States checkpoint shows why national wealth and abundant English material cannot be used as exclusion rules. The discovery register retains 332 Glottolog language-level nodes linked to the United States or one of its territories, including four bookkeeping nodes and six name-identified signed-language nodes. A registry link is only a research locator: it does not prove current residence, present-day use, or the size of an educational audience. The separate resident-language evidence uses 28 selected Census labels, and those labels are not silently forced into exact varieties or scripts.

The multilingual scale is large even before exact educational need is known. The Census Bureau's 2017-2021 national table reports 67.2 million people age five or older speaking a language other than English at home; 25.54 million reported speaking English less than “very well.” The second figure is 38.006% of the first, but neither home language nor self-rated conversational English is a measure of academic comprehension or the effect of a translated course. A separate 2019 report gives 67.8 million people speaking a non-English language at home. The periods and publications remain separate rather than being averaged into false precision (U.S. Census Bureau, 2022a, 2022b).

School evidence identifies more specific starting points. NCES reports about 5.0 million English learners in public schools in fall 2020, or 10.3% of enrollment. Its leading administrative home-language labels include Spanish/Castilian, Arabic, Chinese, Vietnamese, Portuguese, Russian, Haitian/Haitian Creole, Hmong and Urdu. These categories are non-exhaustive and sometimes broad. They identify routes for exact-language investigation; they do not establish that one edition serves everyone under a label. The report's Portuguese count is 43,426-the page image shows a comma where text extraction produced a period (National Center for Education Statistics, 2023a).

National totals also hide locally important communities. In Alaska's state table, the source labels Central Yupik, Inupiaq and Central Siberian Yupik have estimated home-language populations of 15,680 ± 746, 5,722 ± 504 and 1,032 ± 116. Hawaiʻi's selected labels include Iloko at 59,050 ± 3,269, Hawaiian at 26,000 ± 1,916, Hawaiʻi Creole English at 11,600 ± 1,337, Chuukese at 7,861 ± 1,403 and Marshallese at 6,243 ± 1,163. These are Census labels, estimates and margins of error-not automatic Glottolog matches or course audiences. American Samoa, Guam, the Northern Mariana Islands, Puerto Rico, the U.S. Virgin Islands and the U.S. Minor Outlying Islands remain separate profiles. Retained Island Areas table shells prove that jurisdiction-specific language variables exist, but they do not contain the populated language counts; that missing evidence remains a priority rather than a zero (U.S. Census Bureau, 2022c, 2023a, 2023b).

Tribal education is a full-pathway question, not an “Indigenous-language add-on.” Bureau of Indian Education and National Advisory Council evidence identifies early childhood, Native-language and cultural maintenance, assessment, school improvement, technology, workforce preparation, entrepreneurship and tribal-college pathways. The 2023-2024 BIE report card records 35,138 school-year enrollment, 5,888 English learners under a different report-card denominator, 16.58% mathematics proficient or advanced and 74.7% four-year graduation. Those figures use different frames and cannot be added or interpreted as translation effects. They make matched, community-specific comparisons testable without granting any one Native language, English, Spanish or interlanguage automatic coverage (Bureau of Indian Education, 2024a, 2024b; National Advisory Council on Indian Education, 2023).

Signed language and accessibility are separate, sometimes intersecting routes. The register retains American Sign Language, Hawaiʻi Sign Language, Keresan Pueblo Indian Sign Language, Martha's Vineyard Sign Language, Plains Indian Sign Language and Puerto Rican Sign Language. Black ASL remains a documented variety question without inventing a seventh Glottolog node. IDEA service counts, deaf-student evidence and a 180-student postsecondary access study show why ASL, other signed languages, captions, transcripts, speech-to-text, MathML, braille, tactile graphics, audio, plain language, device access and offline delivery cannot be collapsed into one “accessible” file. Disability counts are not signed-language counts, and a small study is not national prevalence (Gallaudet University Press, n.d.; National Deaf Center on Postsecondary Outcomes, 2019, 2024; U.S. Department of Education, 2025).

The comparison therefore spans the whole educational pathway. P covers preschool and early childhood; F foundational and primary education; S secondary; V vocational, technical and practical education; U taught university and other tertiary education; and R postgraduate study and research. National context includes 61.0% preschool enrollment among ages three to five in 2023, 49.62 million public-school students in 2022, 16.39 million undergraduates and 3.27 million postbaccalaureate students in 2024, and 57,862 research doctorates in 2023. None is a language-specific beneficiary denominator. The useful unit is an exact language or variety × stage × subject × format × delivery route × common educational endpoint (National Center for Education Statistics, 2023b, 2024a, 2026; National Center for Science and Engineering Statistics, 2025).

At each stage the checkpoint compares three alternatives: a coherent open pathway, a smaller diagnostic/transition/terminology/accessibility package, and verified existing supply that reaches the same learners and endpoint. The subject menu includes the nine measured OpenStax baselines-prealgebra, statistics, biology, chemistry, physics, management, finance, anatomy/physiology and Python-plus Open Logic, forall x: Calgary and the Stacks Project for advanced work. Existing English supply is relevant to additionality, but it is not evidence that every exact learner can use it. Existing local work has zero effect on global need.

The result is a 1,992-row language-by-stage ledger in which every one of the 332 nodes remains a possible outranker. All exact audience, incremental effect, physical compute, currency cost, dominance and funding-rank fields remain unset. The source audit passes 340 checks across 28 retained files and 14 visually inspected pages; the constructor passes 57 checks; independent readback passes 54 checks with 21,912 CSV/JSON field comparisons and no mismatches; and two complete constructor runs reproduce all four outputs byte-for-byte. This closes a bounded evidence gate. It does not assign the United States or any U.S.-linked language a global rank.

4.6 Practical skills and continuity beyond a uniform mathematics package

Seven further profiles generate eleven concrete package hypotheses. These are different educational tasks, not a new language league table. The complete source cards retain national versus language populations, observation dates, existing alternatives and uncertainty: mainland Southeast Asia (source file: `agent_reports/MAINLAND_SOUTHEAST_ASIA_PACKAGE_NEEDS_20260904.md`) and Brazilian Portuguese and Latin-American Spanish (source file: `agent_reports/PORTUGUESE_SPANISH_PACKAGE_NEEDS_20260904.md`).

Thailand provides particularly direct motivation for a practical-skills route. Its assessment of people aged 15-64 reports 64.7% below the foundational-reading threshold and 74.1% below the foundational-digital threshold. Basic digital tasks include using a pointer or keyboard and identifying a displayed price. These are different skill measures, not percentages to add or multiply. They motivate a Thai reading-and-digital-operations package, with device-based practice distinguished from paper exercises; they do not identify the number of Thai-language beneficiaries (World Bank, n.d.).

Myanmar illustrates a different mechanism. UNICEF's 2026-framed programme page describes around 3.5 million children outside schooling or learning, without providing an observation year or language breakdown. That is contextual evidence for interrupted-learning routes, not a Burmese-language count. A print-first sequence of independent science-reading units could serve Burmese-using adolescents with interrupted study. Existing ministry textbooks and curriculum videos are alternatives; other Myanmar languages remain separate candidates (UNICEF Myanmar, n.d.).

Language profile	Concrete hypothesis to compare	Evidence that narrows, but does not settle, the additional benefit
Vietnamese	A six-unit science-reading and diagram companion for entry to Grade 6	The publisher's existing science book already supports reading, practice and self-assessment. The specific technical-reading bottleneck is not measured.
Thai	Twelve short reading/digital-operation tasks, with optional offline practice	Thai MOOC already supplies digital-literacy courses and tests. Entry difficulty and device requirements need a task-level comparison.
Burmese	Twelve stand-alone science-reading restart units	Existing books/videos do not remove disruption; neither their presence nor disruption alone measures this package's effect.
Central Khmer	Eight technical-reading tasks connecting Grade 9 study and vocational entry	BEEP already combines equivalency, science, workplace and review material. A new bridge cannot claim those features are absent.
Lao	Twelve age-respectful comprehension stories with explained answers	Khang Panya Lao already offers readers and advertises offline delivery. Exact feedback overlap remains uninspected.
Brazilian Portuguese	Adult science-reading, enterprise-records and reproducible-data alternatives	Encceja and IFRS provide substantial existing study and self-paced routes; each proposed delivery or prerequisite advantage is provisional.
Latin-American Spanish, territory-specific	Country-aligned science-reading, business-case and data-laboratory alternatives	INEA, SENA and UNAM already supply relevant content. Mexican, Chilean and Colombian evidence is not a representative regional denominator.

The package sizes are bounded design scenarios, not established minimum doses or optima. The eleven designs yield fifteen stage-specific routes because some span more than one stage; those variants are not disjoint populations. UNAM's public catalogue, for example, already includes R-based data science, finance and scientific thinking. Catalogue inspection establishes alternatives, not their full contents, offline usability or universal access (UNAM, n.d.).

A further methodological prediction follows: a search concentrated on existing textbook catalogues may favor supplement proposals simply because supplements are easier to distinguish from a listed title. The comparison must therefore retain full open editions, integrated learning packages and smaller additions under the same audience, quality and cost assumptions. The current evidence does not establish that small companions are globally optimal. Existing provision can change a particular intervention's expected additional benefit; it cannot erase the population's measured educational need.

4.7 Philippine home languages require separate transition and self-study comparisons

The Philippine Statistics Authority reports language generally spoken at home by household, not by person. Its 2020 table gives 10,522,507 Tagalog households, 4,214,122 Bisaya/Binisaya households, 1,933,512 Hiligaynon/Ilonggo households, 1,863,409 Ilocano households and a separate 1,716,080 Cebuano households. The separate Bisaya/Binisaya and Cebuano rows are not merged. Nor is Tagalog household use automatically equated to comfortable use of the national Filipino educational standard. Hiligaynon/Ilonggo, Ilocano, Bikol/Bicol, Waray, Kapampangan, Maguindanao and Pangasinan/Panggalato are retained as possible-outranker profiles; the residual category and missing responses remain visible without becoming fake edition targets (Philippine Statistics Authority, 2023a).

The 2024 FLEMMS reports functional literacy for 60.17 million of an estimated 85.00 million people aged 10-64. Their displayed difference is 24.83 million, but both figures are rounded and the result has no language attribution. The national PISA 2022 represented cohort similarly cannot be divided among home languages by household shares. These national burdens justify investigation at foundational, secondary and practical stages; they are not Cebuano, Filipino or any other language's beneficiary count (Philippine Statistics Authority, 2025a).

DepEd Order No. 020 makes Filipino and English the primary media for Kindergarten to Grade 3 in school year 2025-2026, includes Filipino Sign Language for deaf and hard-of-hearing learners, and gives regional languages an auxiliary translanguaging role, with specified exceptions. This supports three comparisons: a complete 24-36-unit exact-home-language foundational transition sequence versus an 8-12-unit sidecar; a localized multi-strand independent-study sequence versus a language/navigation/feedback layer over existing ALS material; and a complete signed FSL route versus a smaller signed terminology and feedback layer. Existing ALS resources already span communication, science and critical thinking, mathematics, life and career skills and digital citizenship, so a generic claim that no curriculum exists would be false (Department of Education, n.d.-a, 2025).

Current catalog evidence makes the supply question more specific. The Learning Resources Management and Development System displayed 2,053 Grade 7 records, including 717 mathematics and 383 science records. Every one of the first fifteen rows in each of two bounded Grade 7 subject-list samples was labelled English; this is a sample, not a language census of the full catalog. Across the complete current Basic Literacy, Elementary and Secondary ALS catalog endpoints, 485 stage rows represented 482 unique resource identifiers: 283 rows were labelled English, 192 Tagalog and ten one of Hiligaynon, Kalanguya, Mamanwa, Manobo or Meranaw. Those labels do not prove file contents or exact-variety coverage; “Manobo,” in particular, is not an adequate production identity (Department of Education, n.d.-b).

The catalog advertises a download link on 480 of the 485 ALS rows, but one exact anonymous test returned an authorization barrier. Two sampled Grade 7 learner records likewise showed disabled anonymous View and Download controls. One failed test cannot be generalized to every record, yet it shows why a catalog entry is not the same as a learner obtaining an offline file. The resulting secondary comparison is an exact-language autonomous mathematics, reading/scientific-language and science companion versus a complete autonomous sequence at the same endpoint. Separate comparisons remain necessary for exact first-language-to-Filipino/English transitions, ALS pathways, FSL signed study and non-additive accessibility delivery. All retain unknown matched audiences, effects, physical compute, currency costs and ranks; the detailed source inventory and six alternatives are reported in the Philippines target-specific checkpoint (source file: `PHL_SECONDARY_SUPPLY_AND_PACKAGE_v1.md`).

Delivery also changes the comparison. The 2024 National ICT Household Survey reports home internet for 13.56 million households (48.8%) and internet use for 61.46 million people aged ten or above (67.3%). Among internet users, 98.8% used a cellphone and 76.9% used the internet daily; those conditional percentages are not population phone-ownership rates. Home-internet access was 21.2% in Zamboanga Peninsula and 27.7% in BARMM, while internet use among people aged ten or above was 40.0% in BARMM. These marginals support phone-readable, print, local-share and download-once offline alternatives, but cannot be multiplied by household-language or PISA rows (Philippine Statistics Authority, 2025b).

4.8 Central Asia requires national-language and bridge-language comparisons

Three different kinds of evidence must not be collapsed. Kazakhstan's 2021 census table reports 17,194,712 people in its frame, including 13,750,439 who freely read Kazakh, 14,356,787 who freely read Russian and 5,670,216 who freely read English. Those columns overlap. The table cannot be subtracted into monolingual populations, and its nationality rows do not measure preferred educational language. It shows that Russian is a consequential bridge; it does not show that Russian serves everyone (Bureau of National Statistics of Kazakhstan, 2021).

Uzbekistan provides a different construct: 2021/22 school enrollment by language of instruction. The table reports 5,355,701 students in Uzbek, 642,636 in Russian, 125,850 in Karakalpak, 65,358 in Tajik, 53,558 in Kazakh, 10,902 in Turkmen, 8,183 in Kyrgyz and 1,538 in an other-language category. Kyrgyzstan's 2024 table reports 695,209 students in single-language organizations and 841,424 in multilingual organizations; the largest displayed multilingual configuration is Kyrgyz-Russian at 660,688. These are schooling contexts, not speaker counts, shortages or individual multilingual proficiency (National Statistical Committee of the Kyrgyz Republic, 2024; State Committee of Uzbekistan on Statistics, 2022).

The learning evidence supports real stage investigation without assigning a national burden to one language. PISA 2022 reports mathematics below Level 2 for 49.6% of Kazakhstan's represented cohort, 80.7% of Uzbekistan's and 51.1% of Mongolia's. World Bank/UIS learning-poverty briefs report national values of 10%, 32%, 64% and 39% for Kazakhstan, Uzbekistan, Kyrgyz Republic and Mongolia; the inspected Turkmenistan comparison reports no country value. No value is a language-specific translation effect. Missing Turkmen evidence is a research priority, not a low-need result.

The resulting comparison adds full-course and smaller-bridge alternatives for Uzbek, Kazakh, Uyghur, Kyrgyz, Turkmen, Mongolian/Khalkha and Karakalpak from foundational through later stages where current evidence supports a route. Russian receives cross-border secondary, practical, university and research comparisons. Tajik is not duplicated because an earlier source packet already retains all six stages. Xinjiang's census figure of 11,624,257 people of Uyghur ethnicity is retained only as a high-consequence scale flag: ethnicity is not Uyghur use, literacy or educational need (Xinjiang Statistics Bureau, 2021).

Script is part of the intervention. Uzbek Latin/Cyrillic, Kazakh Cyrillic and future transition cohorts, Uyghur Perso-Arabic and diaspora scripts, Kyrgyz Cyrillic, Turkmen Latin/legacy Cyrillic, Mongolian Cyrillic/traditional script, and Karakalpak Latin/Cyrillic require separate comprehension and production assumptions. Southern Uzbek, Tuvan varieties in Mongolia, Pamiri languages and Central Asian signed languages remain explicit unresolved profiles rather than being treated as covered by a national standard. The Central Asian evidence comparison (source file: `CENTRAL_ASIA_PACKAGE_NEEDS_20260904.md`) retains all counts, residuals and uncertainty.

4.9 West Africa requires exact-language and stage-specific comparisons

Nigeria's 2021 MICS reports foundational reading skills for 26.8% and foundational numeracy for 25.3% of assessed children aged 7-14. The north-east and north-west results are lower still, but they are regional rather than Hausa, Kanuri or Fulfulde rates. Reading tasks were available in English, Yoruba, Hausa and Igbo; 8% of interviewed children could not be evaluated in their home or school language. This directly motivates foundational recovery and language-access investigation, but it does not divide a national burden among those four languages or the hundreds not offered in the assessment (National Bureau of Statistics of Nigeria & UNICEF, 2022).

Ghana shows a different measurement problem. Among 26,210,872 people aged six or older, 7,924,409 were reported not literate in any language and 18,286,463 literate. Of the latter, 9,662,938 were reported literate in a Ghanaian language, 17,610,870 in English and 183,443 in Hausa. These overlapping columns cannot be subtracted into exclusive readerships. The questionnaire separately names Akuapem Twi, Asante Twi, Fante, Nzema, Ga, Dangme, Ewe, Dagbani, Gonja, Dagaare, Kasem, Gruni and Hausa, but the inspected national table publishes most of them only inside the aggregate “Ghanaian language.” A separate ethnic table reports large Akan, Mole-Dagbani, Ewe and Ga-Dangme populations. Ethnic identity is not an exact language, literacy status, comfortable educational language or beneficiary count (Ghana Statistical Service, 2021a, 2021b, 2021c).

Mali reports gross/net enrolment of 62.8/44.9% in fundamental 1, 43.8/22.4% in fundamental 2 and 36.4/15.2% at secondary level. Literacy among people aged 15+ is 34.6%; 61.2% of people aged 6+ are recorded with no instruction level and 2.9% at higher level. These are consequential stage indicators. Yet the inspected tables collapse national languages into one unspecified category. Even that category differs between the report's synoptic table (15.5%) and Table 4.7 (19.9%); both observations are retained without selecting a convenient value. Neither can be assigned to Bambara, N’Ko, Fulfulde, Soninke, Songhay, Dogon, Senufo or another exact language (INSTAT Mali, 2025).

Senegal supplies a more detailed language table. Among 13,108,118 residents aged 10+, it reports 8,151,034 literate in at least one language. The age-10+ literacy rates are 37.5% in French, 14.9% in Wolof, 11.9% in Arabic, 6.1% in Pulaar, 2.0% in Sereer, 0.8% in Mandingue, 0.6% in Diola and 0.3% in Soninke. They may overlap and cannot be added. The same report records 20.9% of compulsory-age children as never having attended school, 40.7% as out of school, and 91.0% of residents aged 6+ as having had no vocational training. Those are distinct needs, not three measurements of one population (ANSD, 2025).

The package comparison therefore spans six stages. It tests oral/picture/audio early learning; independent foundational reading, arithmetic and science with explained answers; complete secondary STEM versus English- or French-facing subject bridges; practical bookkeeping, finance, agriculture, public health, data and digital skills; university prerequisites and scientific writing; and complete advanced editions versus research-navigation layers. Every route compares a complete independently usable course with a smaller addition to existing provision. It does not assume a teacher is continuously available, or that official use of English, French or Arabic establishes comfortable independent study.

Manding, Fula and script-sharing are empirical overlap hypotheses, not regional shortcuts. Bambara, Mandinka and Dyula remain separate while an N’Ko layer is tested for actual readers. Nigerian Fulfulde and Senegal Pulaar remain separate until orthographic and comprehension transfer are measured. Hausa Boko/Ajami, Wolof Latin/Wolofal/Garay, and Akuapem Twi/Asante Twi/Fante likewise require explicit reader and script evidence. Mathematical structure could make a smaller shared register more useful, but supplies no automatic comprehension multiplier. The West African evidence comparison (source file: `WEST_AFRICA_PACKAGE_NEEDS_20260904.md`) retains the exact labels, source discrepancies and route-less high-consequence profiles.

4.10 Indigenous Americas, Pacific and circumpolar needs are not one residual category

The next pass applies a selection rule that does not depend on population size or data completeness. It includes all sixty-one Haiti, Indigenous-Americas and named Canadian Cree/Ojibwe/Inuktut rows in the validated Americas evidence table; all seventeen Pacific rows in the validated Oceania table; and sixteen named Alaska, Greenland, Sámi and Russian-Arctic profiles already present in the regional mapping. Ninety-four profiles receive P/F/S/V/U/R package hypotheses. Seventeen unresolved omission profiles remain route-less but retain every stage. This is an inclusion checkpoint, not a claim that all 564 new routes are equally urgent or supported by equally strong evidence.

Some rows give unusually direct stage clues. In Peru's Grade 4 intercultural-bilingual assessment, 3.1% of participating Quechua Chanka pupils reached the satisfactory first-language reading category, compared with 18.9% in Quechua Collao; 59.4% and 48.1%, respectively, were “in process,” and 37.5% and 33.0% were “at start.” These figures are exact variety/assessment labels, not all-speaker rates. They support early reading, explained mathematics/science and Spanish-bridge comparisons while leaving exact variety population and later-stage need unknown. Guatemala's 2018 childhood-language count for K’iche’ is 1,054,818, but the observed fall in intercultural-bilingual enrolment from primary to later levels is published across languages and cannot be assigned to K’iche’. This motivates continuation and secondary-transition research without fabricating a K’iche’ attrition rate (Guatemala Instituto Nacional de Estadística, 2019; Guatemala Ministry of Education, 2022; Peru Ministry of Education, 2018).

Paraguay demonstrates why bilinguality is not a zero-need state. In a 2022 home-language frame, 33.4% reported Guaraní only, 34.4% Spanish plus Guaraní and 29.6% Spanish. PISA-D evidence cited in the national plan associates home Guaraní with a higher below-adequate-reading share than home Spanish (84% versus 56%). That is an association, not a causal effect or a count of standard-Guaraní readers; standard written Guaraní and mixed Jopara remain unresolved. It nevertheless justifies comparing a complete Guaraní route with a smaller bilingual subject-and-academic-literacy bridge (Paraguay Instituto Nacional de Estadística, 2022; Paraguay Ministry of Education, 2022).

Haiti combines a large scale flag with a delivery problem. The official 2024 territorial estimate of 11,867,032 is only a ceiling, not a Kreyòl speaker count. The Constitution identifies Kreyòl as the common language, while a 2014 observation study in 97 schools found substantially more French than Kreyòl subject time despite Kreyòl initial-literacy policy. Recent school closures and armed-group occupation make print, download-once and local-server delivery material. The useful comparison therefore includes a complete Kreyòl foundational-to-secondary sequence and a smaller diagnostic/feedback and French-transition layer; neither the territorial ceiling nor common-language status supplies the benefit denominator (Institut Haïtien de Statistique et d’Informatique, 2024; World Bank, 2015; UNICEF, 2025).

Canada and the Arctic expose a different continuity problem. Statistics Canada reports 12,005 people able to converse in Plains Cree and 41,675 in Inuktitut, while retaining separate mother-tongue, active-use and silent-speaker constructs. Alberta provides a Plains Cree Y-dialect Grade 7-9 guide; it is not all Cree. Nunavut reports staged/draft K-6 Inuktut curricula and territory-wide attendance/graduation indicators, but those outcomes are not variety-specific. Greenland's 56,740 territorial population cannot become a Kalaallisut speaker count. Kalaallisut, Tunumiisut, Inuktun, Iñupiaq, St. Lawrence Island Yupik, Unangam Tunuu, each Sámi language and the two Russian-Arctic Yupik profiles therefore remain separate. Their packages emphasize exact script/variety, teacher and settlement/offline support, majority-language transitions, climate/land/marine/health STEM, tertiary terminology and research continuity (Government of Alberta, n.d.; Government of Nunavut, 2024, 2025; Statistics Canada, 2025a, 2025b; Statistics Greenland, 2026).

Pacific evidence likewise changes the content question. The Solomon Islands census reports 101,588 people aged five or older for whom Pijin was the first language learned; a sampled early-grade study observed Pijin in 86% of lessons while English remained the formal school language. Māori has 213,849 people reporting everyday conversational ability and a much smaller Māori-medium enrolment pathway; Hawaiian has an estimated 25,490 home-language users and an established K-12 immersion path, with teacher and postsecondary continuity constraints. Marshallese has a rounding-compatible 35,318-35,354 age-five-plus reported-speaking interval; Chuukese has 27,677 age-five-plus home-language respondents. These constructs are different and nonadditive. They motivate bilingual secondary transitions, teacher resources, university prerequisites and advanced terminology in addition to foundational content (Federated States of Micronesia Statistics, 2024; Marshall Islands Economic Policy, Planning and Statistics Office, 2022; Solomon Islands National Statistics Office, 2023; State of Hawaiʻi DBEDT, 2019; Stats NZ, 2024; World Bank, 2019).

The package content is stage-specific. P/F alternatives combine oral language, early literacy/numeracy, audio, diagnostics, exercises and explained answers usable without a continuously available teacher. S compares complete STEM with a bilingual examination-language bridge. V includes health, household and enterprise finance, climate, land/marine knowledge and digital skills. U includes prerequisites, teacher education and scientific writing. R compares complete advanced editions with terminology, theorem/navigation and reproducible-methods layers. Every stage also requires low-data semantic HTML/EPUB or tagged PDF, print/offline packages, captions/transcripts and screen-reader/keyboard support.

The identity rules are strict. “Central Nahuatl” receives no generic route; fifteen exact INALI varieties receive separate profiles and retain dated/shared-locality cautions. Quechua, Maya, Guaraní, Cree, Ojibwe, Inuktut and Sámi are not single educational languages. Tok Pisin, Solomon Islands Pijin and Bislama are distinct languages and do not cover Papua New Guinea tok ples, Solomon local languages or Vanuatu's other languages. Australian Indigenous languages, Pacific signed languages, exact diaspora cohorts and unswept Amazonian, Caribbean and circumpolar languages remain explicit possible-outranker/evidence obligations. The regional packet (source file: `AMERICAS_PACIFIC_PACKAGE_NEEDS_20260904.md`) and its machine-readable evidence preserve every number and boundary.

4.11 More specific needs in Africa, Arabic/Iranic communities and signed languages

Nineteen additional profiles now have source-linked package comparisons. Their contribution is not nineteen new positions on a league table. They show why a useful publication can be a full course for one learner and a smaller addition for another. Complete courses and smaller alternatives remain distinct in the calculation, rather than allowing catalogue inspection to decide the size of every intervention.

Learner situation	Concrete educational material worth comparing	Important existing alternative or boundary
Amharic- or Oromo-using early learners	A coherent low-equipment literacy/numeracy sequence with diagnostic questions and explanatory feedback	Regional guides and diagnostic tools already exist; a smaller reteaching layer is an alternative, not an automatic ceiling.
Kinyarwanda-using learners not adequately served by English text	Scientific reading, measurement, fractions, graphs and practical data, with separate school and adult entry points	English-medium textbooks and distribution programmes exist, but do not establish comfortable English study for every learner.
Somali- or Tigrinya-using learners returning after interruption	A complete prerequisite-to-school-continuity course	Existing accelerated or resumed instruction may make a practice companion more useful for some learners.
Kirundi-using school and practical learners	Applied science, observation, measurement, tables and records	Local STEM projects supply a concrete use case; their anecdotes are not a measured population effect.
isiZulu- or Sepedi-using Grade-4 learners	Independently usable bilingual mathematical instruction	DBE supplies substantial language-specific plans, guides and worksheets; do not claim an empty textbook market.
MSA, Dari and Kurdish learners with interrupted or changed-medium schooling	Sequential science/reading courses, with exact national curriculum and written-variety routes	Madrasa, Begum, Darakht-e Danesh and eWane supply substantial independent provision. Offline continuity and prerequisite fit are specific comparisons.
Persian- or Tajik-using learners	Early-learning, scientific-reading and practical computing/data courses matched to entry level	Existing school books and free course providers require a subject-level comparison, not a nationality-based scarcity assumption.
ASL, Libras, ISL, Kenyan Sign Language and SASL users	Distinct algebra, computing, practical numeracy, scientific-data and qualification-continuity courses	Signed dictionaries, course videos, workbooks and institutional training are different types of provision; none automatically establishes a full autonomous route.

Rwanda supplies a particularly concrete language-access anchor. Table C.25 of its 2022 census education report covers 8,289,582 residents aged 15+ and reports 54.0% literate in Kinyarwanda alone: approximately 4.48 million under that reported category. Because the percentage is rounded, this is not an exact published person count; the nearest-tenth rounding interpretation gives 4,472,230-4,480,519 people. Nor is it the number who need a science course. Alongside the education board's account of English-medium instruction from lower primary since 2021, it supports investigating Kinyarwanda scientific and practical learning rather than assuming English provision serves everyone. Exact tables, language overlap and narrative discrepancies are retained in the African evidence comparison (source file: `AFRICAN_PACKAGE_NEEDS_20260904.md`) (NISR, 2023; Rwanda Basic Education Board, n.d.).

Displacement and exclusion also require full educational pathways, not only basic arithmetic. The Arabic/Iranic comparison separates national out-of-school observations from actual language readership, preserves MSA's spoken-to-written learning distinction, and distinguishes Iranian Persian, Dari, Tajik, Sorani and Kurmanji writing routes. It compares full science and practical-data courses with smaller additions to actual independent providers. Existing courses are evidence about the intervention's alternatives; their mere existence does not reduce a population's educational need. See the six-profile report (source file: `agent_reports/ARABIC_IRANIC_PACKAGE_NEEDS_20260904.md`) and its source-specific dates and population frames.

Signed-language access is not written-language translation plus a generic avatar. The five-profile comparison points to ASL algebra/computing, Libras fraction/ratio and research-data learning, ISL secondary/programming transitions, Kenyan Sign Language literacy/practical numeracy, and SASL matric mathematical literacy/digital-work preparation. Exact provision, replayable explanation, notation, exercises and explained answers matter. Deafness totals are not counts of sign-language users. The inspected sources do not provide matched beneficiaries or measured signed-course production costs; those remain unknown rather than zero. See the signed-language comparison (source file: `agent_reports/SIGNED_LANGUAGE_PACKAGE_NEEDS_20260904.md`).

These designs preserve independent-study and classroom routes. Preschool material normally needs a caregiver or educator, while older learners can use a complete self-contained course without assuming a teacher is available. Text, audio, signing, MathML, printable versions and offline delivery are distinct resources and costs. Keeping all six stages visible prevents the strongest evidence at one stage from silently excluding the others.

4.12 Country totals are omission alarms, not language denominators

A separate zero-disappearance audit now retains all 247 country, additional-locator and unlocated rows in the validated universal matrix. Thirty-four have no current country-population value; eighty have no observed registry node anywhere in the present source set; and 142 have at least one typed signed-language node but no locally linked signed-language observation. Each remains an explicit research obligation. None receives a zero need, a low rank or an implied “served” state because evidence is missing.

The high-consequence gate covers all 31 country contexts with a 2025 World Bank population estimate of at least fifty million, including both billion-person contexts and all sixteen at or above one hundred million: India, China, the United States, Indonesia, Pakistan, Nigeria, Brazil, Bangladesh, Russia, Ethiopia, Mexico, Japan, Egypt, the Philippines, the Democratic Republic of the Congo and Viet Nam. It records exact population estimates and current route visibility only to detect a dangerous omission. A country population is not a language population, a count of comfortable readers, unmet educational need or expected beneficiaries. A route for one national language, stage or subject does not discharge a country's regional, minority, signed, accessibility or displacement obligations.

Among the 31 contexts, twenty-one currently have at least one explicit country/profile route, five have only context-linked routes, and five have none. These are workflow states, not educational coverage rates. The audit explicitly keeps Indonesian stage and subject comparisons beyond the current secondary route, Chinese written standards separate from spoken Sinitic varieties, Japanese needs open by stage and subject, Nigeria's named languages separate from national or zonal measurements, and Russia's many exact language profiles separate from Russian. The same rule applies everywhere, including countries below the threshold and communities that country tables do not enumerate. The complete sentinel table and nonexhaustive next tests appear in the country and territory omission audit (source file: `GLOBAL_COUNTRY_TERRITORY_OMISSION_AUDIT_v1.md`).

4.13 Signed-language identity and displacement require their own ledgers

The universal register contains 227 typed signed-language nodes and seven possibly signed but taxonomically unresolved nodes. All 234 survive the signed-language audit as separate identities. Only six currently crosswalk to any route in the bounded package calculator, and 228 have none. That count describes the incompleteness of this research, not the educational importance of the six. Two of the 227 exact nodes have a numeric source-label observation in the current universal matrix, thirty-four have at least one scope-unverified direct-target relation, and the separate named-source table has eighteen evidence records, fifteen of which currently link through an exact edition-target relation.

Identity mismatches are exposed instead of smoothed over. The present "Kenyan Sign Language" package route possibly mismatches the universal Kenya-Somali Sign Language node and therefore receives no exact-language coverage credit. The Filipino/Philippine Sign Language label pair remains a scope-to-verify link. Hawaiʻi Sign Language, Langue des signes québécoise and the Spanish sign-language-use observation remain three explicit evidence-graph repair obligations. Deafness or hearing-disability totals never substitute for sign-language user counts. The global signed-language audit (source file: `GLOBAL_SIGNED_LANGUAGE_OMISSION_AUDIT_v1.md`) retains all exact identities, source links, current routes and unresolved relations.

Displacement is modeled separately because a refugee, migrant, internally displaced, enrolled-displaced-learner or stateless cohort is not a language population. Ten source-bound portfolios now compare a complete multilingual continuity sequence with a smaller host/origin-language bridge at every P/F/S/V/U/R stage: DRC, Afghan, Palestinian, Syrian, Rohingya, Sudanese, Somali, South Sudanese, Ukrainian school-continuity and Venezuelan cases. This adds 120 unranked routes. Six portfolios retain a source-stated status or enrolled-learner count only as context; four retain no population value. Language, script, host-country and stage distributions remain unresolved, so no route inherits a status-cohort total and the totals are not summed. The displacement package comparison (source file: `DISPLACEMENT_PACKAGE_NEEDS_20260904.md`) preserves the language/script arms, delivery requirements and exact missingness.

Those ten examples are now checked against an all-row end-2025 UNHCR snapshot rather than treated as the global displacement universe. The global displacement omission audit (source file: `GLOBAL_DISPLACEMENT_OMISSION_AUDIT_v1.md`) retains 6,290 origin-host population rows, 10,084 population-type-specific demographic rows and 110 official source footnotes, aggregating them without loss into 209 origin contexts and 178 host contexts. Its omission alarms flag 46 origin and 75 host contexts; 34 named high-consequence origin contexts have no prior portfolio, while two further alarms are special source codes for unknown origin and statelessness rather than country-language identities. The thresholds catch missing research and do not rank funding.

The exact API/annex totals are 28,461,306 refugees under UNHCR mandate, 8,998,097 asylum-seekers, 7,177,473 other people in need of international protection and 64,239,352 IDPs of concern to UNHCR. The separate 5,964,782 Palestine refugees under UNRWA mandate remain external. `Refugees + asylum-seekers + other people in need of international protection` is exposed as a transparent cross-border display sum only; it is not added to IDPs, statelessness, returnees, origin and host views, or UNRWA populations. The demographic endpoint contains 28,856,667 observed records aged 0-17 across selected status types, but incomplete age classification prevents treating that as a complete displaced-child denominator. Every consequential context therefore remains an exact language, script, host, stage and existing-provision research obligation rather than inheriting a national or status label.

A combined corridor language-and-education pass (source file: `DISPLACEMENT_CORRIDOR_LANGUAGE_EDUCATION_EVIDENCE_v3.md`) preserves the first fourteen origin-host rows cell for cell and adds fourteen further high-volume pathways. The 28 exact end-2025 UNHCR rows are now bound to 37 education sources and contain 26,361,921 selected current cross-border status observations. That selected-row sum is still only an inventory: it is not a unique global population, language denominator, beneficiary forecast or rank. The evidence distinguishes direct corridor observations, host-system context and the special `UNK` origin code. Venezuela-to-Colombia, Peru, Chile and Ecuador show that shared Spanish can coexist with enrolment, cost, disability, curriculum, prerequisite, placement and credential barriers. Lebanon adds a direct Arabic-to-English/French mathematics-and-science transition; Jordan adds a direct case in which shared Arabic did not prevent severe foundational-reading gaps. South Sudan-to-Sudan and Sudan-to-South Sudan require different Arabic/English curriculum bridges and cannot be merged.

Four former host-system-only gaps now have bounded origin-host evidence. A Hamburg administrative panel follows 2011-2015 public-school starting cohorts and separately identifies Syrian- and Afghan-born pupils; its comparison suggests that a year-long, German-heavy separate preparatory class can underperform immediate standard-curriculum integration with added language support, especially in fifth-grade mathematics and German. It is historical Hamburg evidence, not a current Germany-wide language distribution or a general argument against language support. A World Bank case study in Aw-Barre and Sheder identifies a Somali-medium grades 1-4 to English-medium grades 5-12 transition alongside textbook scarcity, curriculum discontinuity and English-comprehension barriers; it does not measure current Somalia-Ethiopia corridor prevalence or exact Somali varieties. Houston district data identify 306 Venezuela-origin immigrant pupils and concrete F/S transition mechanisms, but its 88% English Learner and 76% Spanish figures apply to all 11,834 immigrant pupils and cannot be assigned to the Venezuelan subset. `UNK-USA` is consequently the only remaining source-gap row and remains prohibited from receiving any language assignment. Forty-seven evidence checks and consecutive byte-identical builds preserve all population cells, keep language distribution unmeasured, and leave funding ranks and corridor-sized benefit claims empty.

4.14 Somali-region supply shows why “exists” is not “usable”

The first corridor supply audit separates four questions that a catalogue count cannot answer: whether material has existed, whether it is current and practically reachable, whether it can lawfully be reused, and whether a new package improves learning. A saved Ethiopia Learning Library index contains 196 Somali-tagged PDF links. Eighty-four are mathematics links: eight complete student books, eight teacher guides and sixty-eight chapter files spanning grades 1-8. Readable grade-5 and grade-8 samples prove historical content existence, but their notices reserve copying, publication and electronic-distribution rights. The mirror itself describes the books as eight years old or more and out of date; its file host also failed current HTTPS name validation. Those facts neither erase the books nor establish a current, open, reliable learner supply.

The official Ethiopian 2021/22 statistical abstract reports Somali-region textbook-to-pupil ratios of 0.30 in primary and 0.42 in middle school, and fewer than one secondary textbook per pupil. The abstract warns that missing reports and classification errors can bias those values. A selected 2019 study in Aw-Barre and Sheder supplies a mechanism rather than a prevalence estimate: grades 1-4 used Somali, later schooling used English, and 83% of twelve observed classrooms had textbook problems. Somalia’s 2022-2026 sector plan independently documents severe origin-system scarcity-some schools with no textbooks and subject ratios reaching 43:1 in primary and 73:1 in secondary-but it is context, not evidence about every Ethiopia-hosted learner.

For a like-for-like source-traffic comparison, the audit measures the lawful CC BY-NC-SA OpenStax Prealgebra 2e source. The complete eleven-chapter collection contains 75 modules and 364,270 content-text tokens. A declared six-chapter arithmetic-to-algebra bridge-algebra language, fractions, decimals, percents, linear equations and graphs-contains 38 modules and 186,978 tokens, or 51.33% of the measured source scale. With equal per-token variable cost, no common setup cost and equal endpoints, the smaller package is more efficient only if it retains more than 51.33% of the full package’s positive yield; equivalently, the full sequence must yield more than 1.948 times as much. These are algebraic traffic thresholds, not observed effects. The FECU scenario counts input, output and reasoning token events; it does not price physical compute, money, rendering, accessibility work, deployment or maintenance.

Both alternatives therefore remain live. If a bounded first build had to be selected before matched outcomes existed, the six-chapter Somali-English bridge would be the provisional experiment because it addresses the directly observed language and curriculum transition at roughly half the measured source traffic. It would pair Somali explanation with English subject language, prerequisites, worked examples, exercises, answers and feedback, and phone-readable offline delivery. A complete open sequence can still be the better intervention if present usable supply is absent or its added functional yield clears the threshold. Exact variety, script, literacy, stage audience, current open supply, effect, end-to-end compute and global rank remain unmeasured. The full source inventory, arithmetic and 96 deterministic checks are in the Somali-region foundational/secondary checkpoint (source file: `SOM_ETH_FOUNDATIONAL_SECONDARY_SUPPLY_AND_TRAFFIC_v1.md`).

4.15 Germany shows a bridge problem inside substantial existing supply

The Germany F/S comparison is not a blank-supply case. Serlo reports 930 articles, 20 courses, 105 videos and 5,000 exercises with model solutions across school and higher mathematics, available without charge under CC BY-SA 4.0. Its separate SchlaU-Werkstatt collection directly targets newly arrived adolescents and young adults with slow, language-sensitive mathematics; the saved collection has four top-level topics and thirteen immediate second-level branches. Hamburg also publishes a 133-page IVK mathematics handbook with four grade 5-10 teaching sequences, worksheets, some solutions, bilingual glossary instructions and explicit German academic-language work. Binogi advertises more than 1,000 grade 5-10 videos with quizzes across eleven subjects and up to fifteen languages, but the free tier is limited and full individual access is listed at €199 per year. These provider inventories prove real supply, not complete exact-language, curriculum, offline or accessibility coverage.

The directly observed mechanism is the connection between a learner's strongest exact language and German mathematical participation. Hamburg's handbook treats symbols and nonverbal representations as useful scaffolds but not self-explanatory substitutes for definitions, structure words, word problems and mathematical discussion. Its 2024 framework ordinarily moves learners from an IVK into a regular class within twelve months, adds another year of language support and assigns academic-language development across subjects. A historical Hamburg administrative study found worse short-term fifth-grade outcomes, especially mathematics and German, after separate German-heavy preparatory classes than after direct integration with additional language support. This local historical result does not show that language support is harmful or determine current Germany-wide policy; it makes an integrated subject-language and curriculum-continuity package a serious comparator.

On the same OpenStax Prealgebra 2e source basis used for the Somali-region test, the full sequence remains 75 modules and 364,270 content-text tokens. A five-chapter bridge using algebra language, fractions, percents, linear equations and graphs contains 30 modules and 143,859 tokens, or 39.49% of the full measured scale. With equal per-token variable cost, no common setup cost and the same endpoint, the bridge must retain more than 39.49% of the full package's positive yield to be more efficient; the full sequence must yield more than 2.532 times as much to win. Arabic, Dari, Pashto, Kurdish and German FLORES profiles provide declared traffic scenarios only. Origin is not language, and none of those profiles is assigned to a learner population.

The provisional first experiment is therefore an open exact-language/script bridge that pairs German classroom and mathematical language with origin-language explanation, prerequisite diagnosis, curriculum mapping, word problems, exercises, answers, feedback, offline phone delivery and screen-reader or signed-language companions. A complete parallel sequence remains live where a measured language lacks current usable provision, autonomous learners need a coherent offline path, or measured outcomes clear the threshold. Hamburg's saved 2025/26 chart total of 2,412 state-school base/IVK pupils and the Ukraine-, Syria- and Afghanistan-to-Germany corridor counts remain context-not language, learner-benefit or rank denominators. Exact audience, variety/script/literacy distribution, current supply by language, effect, physical compute, currency and global rank remain null. The source inventory, scenario arithmetic and 204 deterministic checks are in the Germany F/S checkpoint (source file: `DEU_F_S_SUPPLY_AND_TRAFFIC_v1.md`).

4.16 Houston exposes a grade-6-to-grade-7 supply breakpoint, not a binary language answer

Houston is neither a blank-supply case nor a case in which the presence of Spanish materials settles access. The current Texas Bluebonnet portal lists free digital Spanish mathematics for kindergarten through grade 6. On the same official surface, grade 7, grade 8, Algebra I, Geometry and Algebra II are listed only in English. The state describes Bluebonnet as open educational resources, with complete subject materials, teacher editions, student workbooks, assessments and downloadable PDF components. Those claims establish a substantial current state-aligned supply; they do not prove campus adoption, learner use, offline usability, component-by-component derivative rights or effect. A separate Texas standards page likewise states that its Spanish TEKS coverage runs from kindergarten through grade 6 (Texas Education Agency, n.d.-a, n.d.-b).

Houston ISD's current public description adds three program mechanisms: dual-language education, grade-level content taught through ESL, and transitional bilingual education at elementary level. Its intake starts with a home-language survey and English-proficiency testing. The public process does not show academic-mathematics literacy, prior course credit, interrupted schooling or device/offline access. Nor is the district accurately summarized by one elementary-only label: Wharton K-8 reports Spanish dual-language instruction through grade 8 and Algebra I for high-school credit. That one campus is a counterexample to a blanket cutoff, not proof of districtwide coverage (Houston Independent School District, n.d.-a, n.d.-b; Wharton K-8 Dual Language Academy, n.d.).

The latest locally frozen district evaluation in this checkpoint is historical but unusually informative. It reports 12,655 cumulatively enrolled immigrant pupils in 2021-2022: 11,241 (89%) classified as emergent bilingual and 10,756 (85%) economically disadvantaged. It separately reports 405 Venezuela-origin pupils and 9,615 pupils with Spanish as the home-language label. These are two marginal distributions, not a country-by-language table, so the Venezuela cohort cannot be assigned Spanish. The same table also reports Pashto, English, Arabic, Farsi, Swahili, Dard, Vietnamese, French, Telugu, Japanese, Hindi, Mandarin and an `Other` category. Several are broad or ambiguous labels rather than exact varieties and scripts. None can be converted into a current language-by-grade audience without further evidence (Houston Independent School District, 2022).

The historical outcomes locate the stage problem without identifying its cause. Immigrant pupils met the 2022 English mathematics STAAR grade-level threshold at 34%, compared with 59% for all emergent-bilingual pupils and 64% districtwide. Spanish mathematics values were 51% and 59% for immigrants and all emergent-bilingual pupils. Algebra I end-of-course values were 38%, 49% and 61%; AP course-enrolment values for grades 8-12 were 7.1%, 14.6% and 24.0%. The report itself calls out persistent secondary and postsecondary-readiness gaps. These group contrasts are not randomized effects of translation, curriculum or language of instruction.

The bounded first experiment therefore has two coupled arms. The measurement arm records exact language or variety, script, literacy, prior mathematics, interrupted schooling, completed courses and credits, and phone/offline access. The production arm tests an exact-language-to-English secondary transition package. For Spanish-literate learners, the visible grade-6-to-grade-7 breakpoint makes a Spanish-English package directly testable: academic mathematics register, prerequisite diagnosis, Algebra I, geometry and graphs, worked examples and answers, state-test orientation, course placement and credit navigation. Every non-Spanish learner retains a separate exact-language arm. The design never assigns a language from country of origin.

On the same OpenStax Prealgebra 2e source basis, the complete sequence is 75 modules and 364,270 content-text tokens. A four-chapter secondary-transition comparator-language of algebra, linear equations, mathematical models and geometry, and graphs-is 24 modules and 119,602 tokens, or 32.83% of the complete measured scale. With equal variable cost per source token, zero common setup and the same endpoint, the bridge must retain more than 32.83% of the full package's positive yield to be more efficient; the full sequence must yield more than 3.046 times as much to win. Neither yield is observed. A complete exact-language sequence remains a live alternative for autonomous or offline learners and for any stratum whose added outcome justifies the larger source scale. Exact audience, effect, full-workflow compute, currency and global rank remain null. The frozen sources, traffic scenarios, visual source check and 256 deterministic checks are in the Houston F/S checkpoint (source file: `HOU_F_S_SUPPLY_AND_TRAFFIC_v1.md`).

4.17 Source traffic differs by subject; one mathematics book is not a universal cost proxy

The first complete-versus-smaller comparisons used one measured prealgebra source. A cross-subject checkpoint now applies the same extraction and tokenizer protocol to nine pinned OpenStax collections: prealgebra, introductory statistics, biology, chemistry, college physics, management, finance, anatomy and physiology, and introductory Python. Together they contain 1,444 CNXML modules and 4,463,123 `cl100k_base` content-text tokens. Complete-source size ranges from 70,994 tokens for the Python collection to 950,860 for College Physics-a 13.39-fold difference. This is a workload-scale result, not evidence that physics is thirteen times as important, difficult or expensive to translate (OpenStax, 2026a-2026i).

The bounded alternatives differ just as much. The selected source share ranges from 5.71% for Biology's first publisher-named unit to 69.06% for the finance chapters explicitly marked Core by the publisher. The other measured shares are 11.69% for the first anatomy-and-physiology unit, 17.79% for four chemistry-foundation chapters, 18.56% for sampling and descriptive statistics, 22.59% for eight mechanics chapters, 43.34% for Python through functions, 47.34% for an analyst-selected management toolkit, and 47.76% for the earlier six-chapter prealgebra scenario. These packages do not have a common endpoint. Their ratios therefore cannot be placed in a benefit-per-cost league table without target-specific audience, supply and effect evidence.

The source representations also omit different kinds of work. Python's comparatively small text count excludes its integrated code runner, executable checking and videos. Biology, chemistry, physics and anatomy exclude figures and other media, while formula, terminology, laboratory, safety and curricular checks remain outside the count. The management selection is explicitly an analyst scenario; the finance selection is publisher-defined but concerns undergraduate corporate finance rather than personal financial literacy. Anatomy and physiology is an allied-health foundation, not public-health guidance or clinical training. These distinctions prevent a convenient source title from silently becoming an answer to “what does this population need?”

All nine records keep audience, educational effect, physical compute, currency cost and global priority null. Their stage labels are applicability tags rather than scores, and the original source-course stage is recorded separately. Twenty-one deterministic checks pass, and two consecutive builds reproduce the structured JSON, readable Markdown and QA bytes exactly. The full identities, subsets, source hashes and break-even scenarios are in the cross-subject source-scale checkpoint (source file: `OPENSTAX_SUBJECT_SOURCE_SCALE_v1.md`).

4.18 Accessibility is a separate marginal-access axis

Language count alone cannot represent educational access. A source-linked accessibility register now distinguishes seventeen delivery interventions, forty-six source records and eighteen non-additive population or supply-context observations. Six are architecture or content-type baselines: semantic responsive HTML, reflowable EPUB 3.3, native MathML, text descriptions for meaningful images, offline download packaging and lower-bitrate media variants. Eleven are conditional parallel modes: synchronized text/audio, prerecorded captions, live captions, downloadable transcripts, one exact named sign language, localized plain language, Easy Read, declared-code braille, tactile graphics, audio description and large print. “Conditional” does not mean optional for a learner who needs that mode.

Every mode remains visible at P/F/S/V/U/R in a comparison separate from the person-language route calculator. That produces 102 mode-stage cells and 204 unranked alternatives: complete integration throughout a coherent package and a bounded high-consequence addition with an exact coverage manifest. A bounded baseline is a migration checkpoint, not an accessible-edition equivalence claim. A bounded parallel mode leaves uncovered units as an explicit gap. Where the existing synthesis does not directly name a stage, the cell is marked research-required rather than either useful or useless by assumption.

The population anchors cannot be added or assigned to modes. The report includes, among other contexts, Internet non-use, hearing-loss, vision-loss, adult-literacy and supply-side accessibility estimates. They overlap and do not measure screen-reader use, caption preference, sign-language identity, braille literacy, need for plain language or additional educational access. All mode-specific need, effect and cost values therefore remain unknown. Written input-token traffic is not the full cost of narration, captioning, sign performance, braille, tactile graphics, descriptions, compatibility work or device testing. Carrier standards likewise do not remove the need to localize text, metadata, pronunciation, descriptions, captions, narration and interface labels. The complete accessibility package comparison (source file: `ACCESSIBILITY_PACKAGE_NEEDS_20260904.md`) retains these boundaries.

5. Shared registers and mathematics

An interlanguage can be useful without reproducing native-language comprehension perfectly. Suppose separate native-language editions yield positive combined net benefit B at positive cost C_s, and a shared-register edition yields r times that benefit at positive cost C_b. The shared edition has greater benefit per cost when $r > C_b/C_s$. A zero or negative comparator benefit requires a direct net-benefit comparison instead of this ratio. This compares complete alternatives; a fixed-budget portfolio choosing only some native editions is a separate optimization.

Formula-rich mathematics offers a specific hypothesis: prepared readers may recover more of an argument through notation, diagrams, familiar definitions and proof structure than through prose alone. But negation, quantifier scope, definitions and inferential dependencies remain consequential. The analysis should compare exact readers, scripts and tasks rather than awarding full language-family coverage or zero benefit by default.

Native-language and shared-register editions can complement each other. A shared introductory reference, an advanced mathematical register and a native-language foundational course may address different tasks. Their overlapping audiences must be accounted for in a portfolio.

5.1 What the intelligibility literature does and does not establish

Related-language comprehension varies by direction, task, preparation and script. Golubović and Gooskens (2015) studied six West/South Slavic language communities, using several spoken and written tasks. They found substantial asymmetries, and written Bulgarian tasks depended on reported Cyrillic literacy. Those results motivate exact community/script comparisons. They do not supply an Interslavic mathematics comprehension rate or an East Slavic population estimate.

Mathematical notation is also not automatically an advantage. In Österholm's (2006) group-theory reading experiment, a prose version produced higher median comprehension than the symbolic version. The unfamiliar subject matter and closed-text assessment differ from a prepared researcher's use of familiar mathematics. The finding therefore weakens a blanket claim that more notation always makes reading easier; it does not settle the proposed advantage of a carefully designed interlanguage for knowledgeable readers.

The useful hypothesis distinguishes advanced reference use, learning a new definition, following a proof and completing a worked example. These tasks may have different benefit retention. The expanded comparison retains ten design alternatives and forty task rows, including seven forecast bridges and three targeted

Germanic alternatives. A direct Interslavic survey adds evidence about cloze comprehension, not mathematical-edition effectiveness (Merunka et al., 2019).

The production comparison charges common source preparation once in both alternatives and retains community checking, extra script outputs and bridge-design effort. Under four declared scenarios, the ten-edition Interslavic comparison requires 24.6%-46.0% benefit retention; the two-edition Malay-Indonesian comparison requires 72.2%-137.0%. These are cost-derived thresholds, not predicted comprehension or measured compute. A threshold above 100% means that cost saving alone does not justify that design. The full assumptions and exact communities appear in `SHARED_REGISTER_PORTFOLIO_COMPARISONS_v1.md`.

Every native-subset size remains a comparator: a bridge that beats a full ten-edition bundle might still lose to a well-targeted one- or two-edition portfolio. Hybrid portfolios use the bridge's additional benefit after the native editions, rather than counting its standalone audience again. This supports a meaningful interlanguage hypothesis without making family size or an assumed common effect decide the result.

If a bridge reaches people who receive no benefit from the separate-edition comparator, its total benefit ratio can exceed one. Such people are counted directly, not removed because a within-stratum retention ratio is undefined. Benefit weights also need not equal population weights. The comparison uses additional usable access for the specified task and horizon.

6. Initial intervention comparisons

The integrated calculation now contains 1,022 stage-and-mode routes: the earlier 338, 564 Indigenous Americas/Pacific/circumpolar routes and 120 displacement/status routes. These are alternative designs and stage applications, not 1,022 disjoint populations or equally strong findings. All 100 forecast profiles and 169 additional challengers retain all six stages, giving 1,614 explicit profile-stage cells. There are current package designs in 224 of the 600 forecast stage cells; empty cells are unfinished investigations, never lower ranks. One hundred seven constructor checks preserve all earlier 902 route results and all 36 pairwise secondary-mathematics comparisons. A known source cohort and a measured tokenization profile permit a numerical threshold; otherwise the missing quantity remains symbolic. The Bangla route retains all three age-matched scenarios. Five Pakistan routes retain intervals and joint constraints rather than invented midpoints. New regional, country and status-cohort context counts are not converted into language-specific need. Normalized text-traffic values remain fixed-size scenarios, not the actual relative costs of a full course, short companion or signed-video edition.

The accessibility calculator is deliberately not added to those language-route totals. It separately retains 17 modes across 102 P/F/S/V/U/R cells and 204 complete-versus-bounded alternatives; twenty-one source, identity, hash, stage, pairing, nonadditivity and no-rank checks pass. Adding those routes to the language count would falsely suggest disjoint populations and comparable production units. An eventual portfolio must instead attach applicable modes to each language/content package, count shared architecture once and estimate additional mode-specific access and cost without duplicating overlapping learners.

The displacement-corridor evidence now has its own matched-endpoint decision surface. Every stage actually named across the twenty-eight originhost rows becomes one of 92 corridor-stage comparison cells. Ninety ordinary cells each compare a complete continuity sequence with a smaller bridge or companion; the two `UNK-USA` stage cells permit evidence resolution only and propose no language production. This yields 180 conditional production alternatives and two identity-resolution actions. Each ordinary pair holds the learner stratum, stage, task, exact language/script profile, delivery conditions and time horizon fixed. The new direct evidence narrows four comparisons: integrated German academic-language and curriculum continuity for Syrian- and Afghan-origin learners; a Somali-to-English prerequisite, catch-up and offline bridge; and measurement-first English transition and credit/navigation support for Venezuelan-origin learners. Audience, baseline usable supply, incremental effect, standardized end-to-end compute, dominance and global rank remain unknown rather than inheriting the corridor's status count. The exact alternatives and evidence needed to discriminate them are published in the corridor intervention comparison register (source file: `DISPLACEMENT_CORRIDOR_INTERVENTION_COMPARISONS_v2.md`); thirty checks pass, including exact preservation of all comparison cells outside the four revised corridors.

Intervention	Initial evidence-motivated routes	Intended educational contribution
Secondary quantitative catch-up	Indonesian, Japanese, Ukrainian, Brazilian Portuguese, Filipino, Thai, Vietnamese, Central Khmer, Malaysian Malay, Central/West Asian and African profiles, and exact Indigenous-American/Pacific/circumpolar routes	Diagnostic, answer-supported work on numerical and algebraic prerequisites; language fit remains a route-specific condition.
Foundational learning and school re-entry	Bangla, Urdu and Pakistan's regional languages; exact African, Haitian Creole, Quechua, Nahuatl, Maya, Guaraní, Cree/Ojibwe/Inuktut and Pacific profiles	Age-appropriate reading/numeracy support, oral/audio explanation and short practice, rather than repurposing a college text as early learning.
Subject-language transitions	Kiswahili/English, Hausa/French or English, Russian/Estonian, Ukrainian/host-language, Central Asian national-language/Russian/script, exact West African English/French, Indigenous-language/Spanish-English-French-Danish and Pacific bridge-language routes	Paired explanations plus practice in the teaching or assessment language.
Tertiary-entry companions	Hindi, Tamil, Telugu, Marathi, Central/West Asian, African, Indigenous-American, Pacific and circumpolar pathways	Prerequisites, teacher education, terminology and scientific-writing support that complement existing courses.
Statistics and practical skills	Prepared Chinese readers; Bangla, Hausa and Kiswahili routes; Ukrainian/Russian, Central/West Asian, African, Indigenous-American and Pacific finance/health/climate/land-marine/data alternatives	Reproducible analyses or specific financial, recordkeeping, health, environmental and planning tasks, with existing provision compared explicitly.
Technical preparation and research access	Ukrainian recovery training; exact Ukrainian, Russian, Chinese, Japanese, Central/West Asian, African, Indigenous-American, Pacific, circumpolar and interlanguage research profiles	Preparatory calculations, scientific writing, terminology or source-specific arguments, at stated prerequisites.
Displacement and stateless continuity	DRC, Afghan, Palestinian, Syrian, Rohingya, Sudanese, Somali, South Sudanese, Ukrainian school-continuity and Venezuelan portfolios	Complete multilingual continuity sequences versus smaller host/origin-language bridges, with offline resumption, credential navigation and exact language/script arms; 92 source-bound corridor-stage cells now define matched endpoints, while status-cohort totals remain context rather than language denominators.
Accessibility and delivery architecture	Every language/content package, with exact user, locale, device and modality requirements	Six baseline and eleven conditional modes spanning semantic HTML, EPUB, MathML, descriptions, offline/low-bitrate delivery, audio, captions/transcripts, named sign languages, plain language/Easy Read, braille, tactile graphics and large print; kept in a separate non-additive comparison lane.

These are intervention recommendations for further comparative development, not a declaration that every listed package outranks every unlisted one. Source selection is concrete: relevant OpenStax topics, Open Logic/forall x and exact advanced open sources are starting points. The source package's learner assumptions, examples, complete-solution coverage and distribution terms are part of the comparison.

The secondary calculation illustrates the scale of the decision. Under its fixed-size text-traffic scenario, an Indonesian edition has higher additional access per traffic than a Japanese edition if its unknown incremental yield per learner exceeds approximately 3.45% of the Japanese route's yield. The conditional full-age-15 coverage envelopes widen this threshold to approximately 2.84-5.95%. These values concern one stage and endpoint, not total language priority.

Coverage can be decisive in a closer comparison. For Japanese and Ukrainian routes, the represented-cohort point threshold is approximately 0.559. Using Ukraine's national coverage envelope and Japan's full-age-15 envelope yields a range of approximately 0.325-2.414. Equal-yield scenarios can therefore imply different directions within those bounds. The national envelope does not copy the participating regions' observed rate to unassessed people.

The comparison fixes source population estimates and uses explicit subset assumptions. It does not identify actual package yields, complete physical production cost or future uptake. Its immediate contribution is a transparent threshold calculation with no silent population removal. The full route definitions, source identities, coverage conditions and sensitivity checks accompany the manuscript.

The first actual-source-scale comparison changes the earlier bookkeeping materially. A complete OpenStax prealgebra source is about 3.64 times the normalized 100,000-token canon used in the earlier table. For Indonesian, the scenario therefore rises from 1.34 million to 4.89 million token events before extra work. This is a correction to the package-size input, not a change in Indonesian need or eligibility. A selected six-chapter route is less than half the measured text, but it is preferable only if it retains enough benefit on the same outcome. A complete course remains a serious candidate when it substantially improves independent progression, supplies otherwise missing outcomes, or amortizes shared infrastructure across later units.

For the Bangladesh four-domain comparison, two deliberately broad design budgets make the unknowns visible. A twelve-item entry-and-feedback companion is represented by 2,720-7,840 English/source-reference tokens; a 24-unit explanatory and practice sequence by 12,800-30,400. Assuming the same per-token workflow and no unequal extra work, the small package's positive-yield threshold spans approximately 8.9%-61.3% of the fuller sequence's yield across the crossed budget extremes. These are planning bounds, not confidence intervals, observed Bangla lengths or instructional specifications. They cannot establish that the companion wins. Authoring, answer verification, formula checking, accessibility and audio/video work remain separate terms. The underlying 15.3-15.7 million national numeracy scenario is unchanged because prior production and package size do not alter population need.

7. Remaining empirical synthesis

7.1 What existing evaluations suggest about package design

One actionable hypothesis is that matching prerequisites and supplying useful feedback will often matter more than adding another undifferentiated textbook. In a Delhi lottery evaluation, an adaptive after-school programme increased independently assessed mathematics scores by 0.37 standard deviations and Hindi scores by 0.23 over 4.5 months. The programme combined computer activities with small-group instruction; it did not isolate a translation or a teacher-free resource (Muralidharan et al., 2019). The design implication to test is a diagnostic entry point, appropriately sequenced practice and explanations-not assigning every proposed publication those effects.

Two other evaluations help distinguish access to an object from access to learning. Conventional textbook provision in Kenyan primary schools did not establish an average test-score improvement, though initially stronger pupils benefited more; language and prerequisite mismatch were possible explanations, not separately randomized treatments (Glewwe et al., 2009). Peru's One Laptop per Child evaluation found increased computer use but no established mathematics or language test-score effect after 15 months (Cristia et al., 2017). Neither finding establishes that open material, offline delivery or translation is unhelpful. They make the package's usability a substantive research question.

These findings generate three explicit predictions for the proposed packages. First, catch-up resources should be more useful when their entry level matches the learner's prerequisites. Second, a subject-language bridge should supply practice and explanations, not just substitute terminology. Third, the benefit of an offline edition should depend on whether a learner can navigate, study and check work with the available device. The comparisons will preserve evidence that supports or weakens each prediction. Classroom findings can inform these mechanisms without making teachers a requirement for the autonomous-learning pathway.

The study keeps functional access and subsequent learning outcomes separate. A test-score standard deviation is not a percentage of people newly able to study. None of these evaluations supplies a universal translation-effect multiplier or a measure of AI production cost. The detailed evidence register also distinguishes the 2012 working-paper and 2017 published versions of the Peru evaluation. Their sample sizes and coefficients are not mixed.

7.2 Work still needed for the global portfolio

The globally decisive portfolio is not yet complete. The current quantitative surface now has a 100-row worldwide potential-reach table and a distinct 100-row fixed-package workload proxy, alongside the bounded ten-program hedge and the nonordinal 100-profile research inventory. The two ranked tables answer real but deliberately conditional questions; the profile inventory does not. Required remaining work includes stage-specific addressable audiences; item-level comparisons of independently usable subject provision; measured incremental effects; standardized physical-compute scenarios; delivery and accessibility costs; beneficiary overlap; global omission and sensitivity tests; and comparisons against possible outrankers. Signed-language and accessibility interventions, smaller communities, and vocational and research opportunities remain substantive lanes rather than residual categories.

The present findings already distinguish useful package hypotheses from raw speaker-count priorities. The final synthesis must carry that distinction through every recommendation and remain understandable without knowledge of any particular translation project.

7.3 A global synthesis universe replaces the population-dominant shortcut

The current synthesis does not start from the languages for which convenient population figures happened to be available. It starts from 10,625 nonadditive analytical entities: all 8,618 Glottolog language-level identities; all 1,077 prior edition targets; the frozen 100-candidate hypothesis roster; 169 evidence challengers; 17 accessibility modes; 10 interlanguage or shared-register designs; 247 country and territory omission contexts; 209 displacement-origin contexts; and 178 displacement-host contexts. These rows overlap and are not people totals. Their purpose is to prevent a language, variety, modality or displaced route from vanishing while better evidence is sought.

Ten completed country or jurisdiction identity gates contain 2,966 assignments covering 2,854 distinct Glottocodes. That is progress in finding identities, not proof that those 2,854 languages have measured audiences or that the other 5,764 matter less. Country association does not establish present residence or use. A source label does not establish an exact variety. A prior edition target does not establish unmet need. An existing intervention route does not establish adequate supply or benefit.

The previous 186-row `GLOBAL_FUNDING_PORTFOLIO_v1.csv` is therefore historical evidence of the population-dominant shortcut, not an input to the replacement analysis. Its common-prior estimates and ordinal positions are not current recommendations. The replacement register gives every entity an evidence state and next research action while leaving final package-specific funding rank empty. That does not delete prioritization research: the conditional reach and fixed-package workload Top 100s remain active screening outputs, and the finite-budget scenario portfolios below are explicit compute-allocation calculations. Every entity remains a possible outranker; missing population, need, supply, effect, compute or overlap evidence never becomes a zero or an ordinal penalty.

The robust global partial order currently has zero asserted dominance edges. That is not a finding that every intervention has equal value. A dominance claim requires the same educational endpoint, an exact addressable audience, current usable supply, an incremental-effect interval, standardized physical-compute and delivery costs, and a rule that counts overlapping beneficiaries once. Those quantities are not yet jointly bound for a global pair. Substituting common midpoints or putting unknown cases last would manufacture an order from evidence availability.

The requested Top 10 and Top 100 now exist in two explicitly conditional, ordinal senses. The reach order compares population plausibly helped under one common missing-data scenario. The fixed-package proxy divides that numerator by the modeled token-event workload for one identical package. Neither is the final package-specific funding order. Separately, the ten-program tranche is one feasible hedge across large audiences, regional depth, displacement, multiple scripts and practical subject needs, while the 100-profile set is a nonordinal communication and investigation inventory. Its member numbers are stable identifiers, not positions. The 169 current challengers and all entities outside those displays remain possible outrankers. Accessibility formats and interlanguages remain parallel, nonadditive design layers; all 209 origin and 178 host displacement contexts remain visible rather than being reduced to ten examples.

This universe passes 34 constructor checks, an independent 265,625-field CSV/JSON comparison with no mismatches, and two complete byte-identical rebuilds of all four outputs. The result establishes a safe global comparison boundary. It does not establish educational benefit, compute efficiency or a final allocation order.

7.4 What is-and is not-comparable now

The replacement synthesis now has both a visible global screening layer and a route-level evidence funnel. The screening layer preserves 775 non-English language-and-script profiles, publishes the first 100 by potential reach, and compares the same fixed text package across all 775 using measured or common-envelope token-event workloads. The route layer preserves 1,022 proposed language-stage-package routes. Fifteen have a numeric need interval or point in a source-defined context; thirty-two have a declared base text-traffic scenario; and nine routes participate in thirty-six same-endpoint threshold comparisons. The global screening tables are useful conditional orders, and the route cells are useful conditional comparisons, but neither alone identifies global winners. A national cohort, census category or status population is not automatically the exact audience that can use one edition. A token-traffic scenario is not standardized physical compute, energy, money or achieved quality.

No current route jointly measures the six decisive elements: exact edition audience, independently usable baseline supply, incremental effect on a common educational endpoint, standardized physical compute, delivery/accessibility cost and portfolio overlap. In particular, all 1,022 incremental-yield fields remain unmeasured logical ranges, not effect estimates. The robust global dominance graph therefore still has zero edges. This does not mean that all routes are equal, useless or low priority. It means that current evidence cannot honestly decide a global winner on compatible units.

The machine-readable readiness register assigns every route one of four states. Nine are conditional same-endpoint threshold cells with no winner. Six more contain a numeric context bound but lack comparable effect and compute. Nine hundred ninety-five contain a source-referenced package hypothesis but are not globally rankable. Twelve retain a route hypothesis without an embedded package design. Every row remains a possible outranker, has no funding rank and cannot be excluded because of missing evidence.

The source-and-package layer currently binds seventy design entries across thirteen country or jurisdiction checkpoints. These establish concrete questions about current supply, stages, subjects and full-versus-bounded packages. They do not award those places higher priority for having more documentation. The 100-profile forecast remains a communication portfolio: sixty-three profiles have at least one route and thirty-seven remain explicit research gaps, while 169 evidence challengers sit outside the forecast. All are eligible to displace a provisional selection when compatible evidence supports it.

The next allocation step is therefore specific, not vague: use the global reach and fixed-package tables as an omission-resistant screen; complete comparable cells for exact language/script/community × stage × subject × package × delivery interventions; estimate standalone and overlapping benefits; normalize the complete production and delivery workflow to physical compute; and then publish interval-supported dominance relations and budget frontiers. Every Top 10 or Top 100 must name the quantity being ordered, show its assumptions, identify candidates that can still outrank it and state the evidence most likely to change the result.

For clarity, P means preschool/early childhood, F foundational/primary, S secondary, V vocational/technical/practical, U university/taught tertiary and R postgraduate/research. These letters label educational endpoints, not scores or priority.

7.5 The high-reach evidence frontier

The first high-reach synthesis contains 83 deliberately separate evidence cells: 45 typed language-population bases, seven parallel or common-core package hypotheses, and 31 country omission sentinels. Fifty-two language or package cells remain possible outrankers. None has a funding rank or an exclusion disposition. The country cells are not translation candidates; they are alarms against silently overlooking large national contexts while the exact languages, varieties, scripts, displaced populations, signed languages and delivery formats inside them are resolved.

The largest rows demonstrate both the stakes and the danger of casual comparison. The register retains a mixed global English lower-bound estimate of 1.5 billion users; a Mainland Standard Written Chinese exposure ceiling of 1.412 billion; reported Putonghua communicative ability around 1.140 billion; a Spanish potential-user lower bound of 630 million and a separate 520 million native-domain estimate; 503 million people in an MSA educational-exposure aggregate; a 450 million Arabic umbrella context; 396 million institutionally estimated Francophones; 322.230 million raw Hindi mother-tongue returns; 280.404 million Mainland formal-education enrolments; 265 million Portuguese speakers; a 264.589-264.606 million Bangladesh-and-India Bangla/Bengali parallel-package context; more than 250 million Russian users; 248.502 million Indonesian users by the cited BPS conversational-ability definition; a 240.458 million Urdu national-exposure ceiling in Pakistan; and more than 200 million cross-border Kiswahili users. These are different measures, often overlapping. Their order of magnitude makes them impossible to dismiss; it does not make them a benefit ranking.

The distinction is substantive. Reported ability does not prove comfortable academic reading. A mother-tongue count does not measure current educational supply. Formal enrolment is an opportunity population, not unmet need. A territorial ceiling is not an exact edition audience. A world speaker estimate cannot be copied into one country, script or standard. Country learning, literacy, completion and connectivity indicators are ecological context and are never multiplied into a language total as if they were conditional language-specific rates.

The seven shared-work cells make possible efficiencies visible without awarding blanket interlanguage coverage. They include Bangladesh/India Bangla, Gurmukhi/Shahmukhi Punjabi, Hindi/Urdu, Iranian Persian/Dari/Tajik, six West African Pidgin profiles, multiscript Hausa and multistandard Kiswahili. Each is a hypothesis about parallel production, common terminology, reusable formal structure or a bounded shared register. A child language never inherits the package population. One shared text serves the whole context only if its exact users can actually understand and use it at the stated mathematical and linguistic prerequisites.

Every intervention cell is therefore expressed as $P \times q$, where P retains the source's typed population measure and q is the still-unmeasured expected per-person change in functional educational access. The attached 100,000-source-token text envelope-approximately 0.771 million, 1.575 million and 4.044 million fresh-equivalent token events under low, base and high scenarios-is a common traffic sensitivity test. It is not physical compute, energy, money, achieved quality or a target-specific production measurement.

This frontier changes the next research question. Large cohorts are now visible without being promoted for having convenient data. The next comparisons must bind, for one exact community and stage-subject-format package, an actual usable audience, existing open supply, delivery constraints, a common access endpoint, incremental effect, standardized physical compute and beneficiary overlap. Only then can intervals support a dominance relation or a conditional portfolio. Until then, Indonesia, Simplified Standard Written Chinese, Bangla, Hindi, Ukrainian and every other measured or unmeasured candidate remain eligible to displace a provisional choice.

7.6 What the effect evidence changes-and what it cannot decide

The access commitment is not an empirical hurdle placed in front of learners. This study treats inherited familiarity with academic language as an institutionally distributed advantage and treats a source-faithful translation into an audience-specified adult register as an eligible intervention whenever any credible nonzero educational benefit is plausible. The literature can help estimate the size and distribution of that benefit. It can show that a particular source, audience, register or delivery design was ineffective or counterproductive. It cannot convert family education, schooling history, income, racialization, disability, measured attainment or register familiarity into an innate-capacity judgment or a reason to withhold access.

The closest direct evidence is encouraging but narrow. Bälter et al. (2024) randomized 2,263 learners between Swedish and English versions of the same eight-module programming course. Among course starters, performance was 16.9 versus 14.3 correct answers, with a small Cliff's delta of .085; dropout was 57% in Swedish and 71% in English. In a four-arm trial with 111 Scandinavian family physicians, Gulbrandsen et al. (2002) found median recall of four of thirteen facts after reading in the mother tongue and three after reading in English; paper versus screen made no detected difference. These are unusually close to the proposed language-edition contrast, but they concern highly educated samples, short or bounded material and immediate or course-level outcomes. They are anchors, not global multipliers.

Adult intralingual register experiments show both meaningful gains and strong heterogeneity. Leroy et al. (2013) preserved source information while adding definitions and familiar wording. In a 99-person experiment, correct comprehension was 63% for the translated passages and 52% for the originals; learning gains were 18% and 9%, while free recall did not differ. Martínez, Mollica, and Gibson (2022) found a 5.8-percentage-point comprehension advantage for equivalent-meaning ordinary English over legalese in short contracts. Sayfi et al. (2024) found a 19.8-point advantage for a plain-language WHO recommendation but only 3.9 points for a CDC text, whose confidence interval included no effect. The large difference between two related sources is as important as either headline number: source structure, audience and the exact translation choices matter.

Null and adverse comparisons are therefore design information. They are not a vote on whether a language or social group is capable of using educational material. One bilingual physics comparison produced smaller gains than its German-only counterpart, while a paper-reading interface added perceived ease without a detected objective-comprehension gain. Such results require analysis of terminology, cognitive load, prerequisites, layout, teaching context and audience fit. They justify a different design or a smaller first deployment, not permanent ineligibility.

Open access, offline delivery and instructional support also answer different questions. Clinton and Khan's (2019) meta-analysis found essentially equal average learning outcomes for open and commercial textbooks ($g=.01$, 95% CI [-.16, .19]) and lower course-withdrawal odds for open-textbook conditions ($OR=.71$, 95% CI [.56, .90]). That comparison supports substituting free books for paid books without assuming an achievement penalty. It does not estimate the effect of a usable book versus no book, or the effect of translation. Likewise, making a file downloadable changes acquisition; making it reflowable, screen-reader-native, printable and offline changes practical usability; worked examples, retrieval practice, explanatory feedback, prerequisites and self-regulation support change learning conditional on use.

For planning, the analysis consequently separates four quantities:

1. the population for whom the edition is relevant;
2. the probability that the edition is reached, acquired and usable;
3. the probability of sustained active use; and
4. the effect among active users on a named endpoint.

For a passive release, those stages can be multiplied only when their conditions and denominators are compatible. They must not be multiplied into a study's intent-to-treat effect, which already includes uptake and nonuse. Nor may a comprehension percentage be multiplied directly by a worldwide language population without an exposure and use model.

The current evidence register therefore publishes scenarios rather than a single promised effect. For a well-audited full-content adult-register translation, the immediate-comprehension scenarios range from an adverse -3-to-0 percentage points, through near-null 0-to-3 and central 3-to-10, to a 10-to-20-point high-mismatch case. For an actively used static package with worked examples, retrieval, answers and explanatory feedback, the planning range is 0.05-0.25 standard deviations over roughly a term or year. A merely published file receives a 0-0.02 scenario. Preschool remains unmeasured rather than being assigned zero. These are stress tests assembled from distinct evidence classes, not confidence intervals, pooled global estimates or eligibility thresholds.

No controlled study located so far evaluates a complete, proposition-preserving mathematics curriculum translated into another language or adult register. That is a reason to measure the first well-audited editions carefully. It is not a reason to presume no benefit. The full 53-study evidence register, 18 planning scenarios, source-to-scenario transport rules and independent reproducibility checks are published in `INTERVENTION_EFFECT_EVIDENCE_v1.md` and its machine-readable companions.

7.7 An actionable first tranche without pretending it is a world league table

A decision-maker may still need to act before every comparable cell is measured. The current response is a ten-program measurement-and-production tranche. It deliberately hedges across very large audiences, multilingual systems, multiple scripts, displacement, offline access, foundational learning, practical subjects and advanced study. Every program remains eligible independently of the evidence gathered so far. The evidence selects its first package and its scale; it does not decide whether its people count.

Program slot	Why it is in this hedge	First bounded package to compare
Bahasa Indonesia	Very large first- and additional-language reach, major measured secondary learning burdens and an unusually informative open-production case study	Independent secondary mathematics/science prerequisite repair with exercises, solutions and offline/mobile use, followed by statistics, computing, finance/business and tertiary bridges
Bangladesh and India Bangla/Bengali	Large country-disjoint parallel context, severe foundational-learning evidence and potential reuse across two explicit national interfaces	Foundational numeracy plus reading-supported mathematics, then secondary science, statistics, health and practical finance
Hindi and Urdu	Hundreds of millions across related but distinct educational standards and scripts, with potential shared formal work that must not erase either native output	Separate Devanagari and Nastaliq foundational/secondary mathematics, science and finance, plus an optional measured shared mathematical layer
Simplified Standard Written Chinese	Enormous written-standard and formal-enrolment opportunity alongside substantial existing provision, making targeted additionality more useful than a generic school-book assumption	Audit and fill open, downloadable and accessible advanced mathematics, science and computing gaps; include semantic HTML/MathML, reflow and offline formats
Philippine multilingual profiles	High country-level learning burdens and a sharply multilingual home-language landscape that one Filipino or English label cannot discharge	Early numeracy and reading-supported mathematics, secondary science, health/disaster learning and financial capability across separately named interfaces
Ethiopian multilingual profiles	Severe foundational, transition, connectivity and practical-learning questions across several large language communities	Foundational literacy/numeracy, secondary science, agriculture, health, technical and small-enterprise modules with offline delivery
Democratic Republic of the Congo multilingual and displacement routes	Very large population, severe learning/connectivity context and school-continuity needs that cross origin and host settings	Resumable mathematics/science, public health and livelihood modules with credential and continuity navigation

Program slot	Why it is in this hedge	First bounded package to compare
Hausa in Boko and Ajami	A very large cross-border community, meaningful script choice and practical subject reach	Foundational numeracy/literacy, secondary science, health, agriculture, finance and enterprise in low-bandwidth text/audio and both literacy interfaces
Multistandard Kiswahili	Very large cross-border L1/L2 reach and potential reuse across several education systems without assuming one uniform standard	Secondary mathematics/science, statistics/computing, health, agriculture, finance and management with national terminology overlays
Modern Standard Arabic plus vernacular and displacement interfaces	A large learned written register across many systems, with diglossic, local-language and displacement needs that an Arabic umbrella cannot collapse	Reading-supported bridges into MSA, mathematics/science, health, finance and management, plus advanced open sources and explicit vernacular interfaces

The shared measurement protocol uses a 100,000-source-token benchmark to learn target-specific production, quality-assurance, accessibility and delivery costs. That benchmark is not the first educational package and does not force every language into a 100,000-token book. The first packages above differ because the best available evidence points to different stages, subjects and delivery problems. Complete courses and smaller additions are compared at the same educational endpoint.

The wider 100-profile inventory is a communication set, not ranks 1-100. Its member numbers are record identifiers. This differs from the newly published 100-row potential-reach order and 100-row fixed-package workload proxy, whose columns are explicitly labelled as ordinal positions and whose quantities are fully stated. All analytical entities outside the profile inventory remain possible outrankers. The current target-specific partial order still contains zero robust global dominance edges because no pair jointly binds compatible audience, same-endpoint effect, measured physical compute and nonoverlapping beneficiaries. The tranche is therefore an expert-constructed feasible hedge, not the output of a falsely precise optimizer.

This is also why the list contains both very large educational languages and regional, signed, displacement and shared-register lanes. A population-only portfolio would suppress smaller high-severity and prestige-domain needs. A scarcity-only portfolio would suppress hundreds of millions of people in large languages. A documentation-readiness portfolio would reward communities that are already easiest to measure. The conditional portfolio keeps those failure modes visible and publishes the evidence that could change each slot. Its full Top 10 rationales, nonordinal Top 100, accessibility overlays, displacement portfolio, possible-outranker fields and deterministic checks are in `CONDITIONAL_GLOBAL_PORTFOLIOS_v1.md` and its CSV/JSON companions.

7.8 From a population screen to a target-specific allocation frontier

The first target-specific frontier corrects a category mistake that the profile catalogue can otherwise invite. A language profile is a search key. It is not an intervention, a priority or a unit to which compute can be allocated. Compute is allocated to a production-and-delivery plan that names the target audience, educational endpoint, source package, language/register/script, formats, delivery route, quality threshold and horizon.

Fifteen current routes contain a numeric need/context interval. Nine of them share the declared secondary quantitative-catch-up endpoint and can therefore enter a conditional comparison. Their point-scenario coefficients under equal target-specific effect are Indonesian 2,305,230; Brazilian Portuguese 1,247,193; Filipino 951,470; Thai 246,160; Vietnamese 181,776; Standard Malay 146,227; Japanese 79,644; Central Khmer 47,980; and Ukrainian 44,546 context members per million fresh-equivalent token events. These values are `N/C`, not people known to have learned, and the denominator is workload traffic rather than physical compute. The six other numeric routes remain visible but are not cross-ranked against a different endpoint.

For every directional pair, the frontier publishes the exact effect-ratio threshold at which the comparison reverses. With positive effects, route A exceeds B when `q_A/q_B > (N_B/N_A)(C_A/C_B)`. This produces 72 directional thresholds for the nine-route subset. A direct estimate can therefore replace the equal-effect placeholder without rebuilding the comparison or silently removing candidates. The independent review recomputes each threshold from the source registers and verifies that no route receives a funding rank or exclusion disposition.

Existing supply changes the proposed intervention, not the underlying need. Indonesia has official Grade VII student books and separate teacher guides, so the matched comparison is an autonomous diagnostic, worked-feedback, remediation and offline companion versus a complete independent sequence-not a claim that no Indonesian material exists. Brazil has a large represented secondary-proficiency context, but the current evidence does not yet identify whether language, prerequisites, pedagogy or delivery is the main constraint. The Philippines has extensive K-12 and alternative-learning-system records, fragmented access paths, a directly observed anonymous-download barrier and unresolved allocation of burden among Filipino, English and exact local languages. Japanese and Ukrainian have stage-matched package hypotheses, while their most specific local frames do not equal the shared catch-up context. Thai, Central Khmer and Standard Malay currently lack a source-bound package aligned to this exact endpoint; that is a high-value research gap, not evidence of zero need. Vietnamese has a related secondary science-reading proposal rather than the exact quantitative-catch-up package.

Delivery evidence also changes the intervention. Indonesia's cited 2024 national marginals report 72.78% internet use, 68.65% mobile-phone ownership and 18.52% household-computer ownership. Philippine sources report 67.3% internet use among people aged ten and older and 48.8% household home internet, with large regional differences. These are not learner × language × device cross-tabs and are not multiplied into effect. They support offering phone-readable, printable, download-once and offline forms together rather than assuming continuous web access.

Package size must be judged against the same endpoint. At the pinned OpenStax Prealgebra 2e source commit, the measured learning-content text contains 364,270 `cl100k_base` tokens across 75 modules. A declared six-chapter quantitative selection contains 173,958 tokens, or 47.755% of the complete source; a fractions-only unit contains 41,336 tokens, or 11.348%. With no common setup cost and a genuinely common endpoint, the six-chapter package is more efficient only if it delivers more than 47.755% of the full package's positive yield. The threshold rises to 52.505% when common setup is one tenth of full variable production cost and to 73.878% when it equals that cost. A fractions-only unit cannot be credited with the complete catch-up endpoint; its lower arithmetic threshold applies only to an explicitly narrow fractions outcome.

The immediate empirical comparison is therefore not “which language wins?” It is: for Indonesian, Brazilian Portuguese and Filipino, specify one identical quantitative endpoint and horizon; compare a bounded prerequisite-and-feedback companion with a complete autonomous sequence; model passive publication, active static use, offline use and supported use separately; and measure the full accepted-package production work on a declared reference system. That experiment is the shortest route from the current population screen to a defensible compute-allocation frontier. In parallel, unresolved high-reach and high-severity routes remain explicit possible outrankers so evidence scarcity cannot turn into exclusion.

7.9 What the fixed-budget calculation actually allocates

The allocation question is not abandoned when profile identifiers are shown. A finite-budget calculation has now been run across all 775 retained edition rows. It uses a declared FECU logical-workload budget, one common 100,000-source-token package per selected edition, the current potential-reach coefficient, and an equal unknown net-yield factor `q_common`. Therefore each objective is a standalone reach coefficient multiplied by `q_common`, not observed people helped. Overlap, stage, endpoint and delivery must still be supplied before the coefficient can be interpreted as unique additional understanding.

Workload budget	Editions in the exact point-scenario portfolio	Standalone coefficient
5m FECU	Simplified Written Chinese, Hindi, Spanish	2.126b × q_common
10m FECU	Simplified Written Chinese, Hindi, Spanish, Arabic, French, Portuguese, Indonesian	3.210b × q_common
25m FECU	17 editions, including Chinese, Hindi, Spanish, Arabic, French, Urdu, Portuguese, Indonesian, Bangla, Russian, Kiswahili, Punjabi, German, Japanese, Javanese, Vietnamese and Telugu	4.746b × q_common
50m FECU	33 editions in the exact point optimum	5.812b × q_common

These portfolios are conditional choices, not a claim that the listed languages are more deserving. They make the research question operational: given a named workload budget and a common scenario, which indivisible editions fit and what standalone opportunity coefficient do they cover? At 10m FECU, the complete source plan selects seven editions in the base scenario; a bounded six-chapter plan selects Chinese, Hindi, Spanish and French plus other high-reach editions under its own source-scale costs. The shorter plan is not credited with the complete-course outcome: it is advantageous only when its endpoint-specific net yield clears the published package break-even threshold.

Twenty-five inherited rows have no population coefficient because their evidence base is unresolved. They are represented as unknown audiences, not as zero benefit. Each receives an entry threshold: the unknown candidate would enter a budget portfolio if its coefficient exceeded the opportunity cost of replacing the current selected item. This keeps Ancient Greek, Balinese, Caddo, Chickasaw, Church Slavic, Fula, Geez, Ido, Jutish, Kawi, Literary Chinese, Lojban, Mycenaean Greek, Obolo, Osage, Palatine German, Pali profiles, Romagnol, Samogitian, Samtao, Vai and Votic visible as possible outrankers.

The accompanying `GLOBAL_BUDGET_SCENARIOS_v1.md`, JSON and QA receipt preserve all 775 candidates, six exact scenario runs, budget bounds and unknown-entry thresholds. This is the current compute-allocation result under a transparent logical-workload scenario; it is not the final package-specific global order.

References used in this draft

Bälter, O., Kann, V., Mutimukwe, C., & Malmström, H. (2024). English-medium instruction and impact on academic performance: A randomized control study. Applied Linguistics Review, 15(6), 2373-2396. https://doi.org/10.1515/applirev-2022-0093

Clinton, V., & Khan, S. (2019). Efficacy of open textbook adoption on learning performance and course withdrawal rates: A meta-analysis. AERA Open, 5(3). https://doi.org/10.1177/2332858419872212

Gulbrandsen, P., Schroeder, T. V., Milerad, J., & Nylenna, M. (2002). Paper or screen, mother tongue or English: Which is better? JAMA, 287(21), 2851. https://doi.org/10.1001/jama.287.21.2851

Hammarström, H., Forkel, R., Haspelmath, M., & Bank, S. (2026). Glottolog 5.3. Max Planck Institute for Evolutionary Anthropology. https://doi.org/10.5281/zenodo.15525265

Leroy, G., Endicott, J. E., Kauchak, D., Mouradi, O., & Just, M. (2013). User evaluation of the effects of a text simplification algorithm using term familiarity on perception, understanding, learning, and information retention. Journal of Medical Internet Research, 15(7), e144. https://doi.org/10.2196/jmir.2569

Martínez, E., Mollica, F., & Gibson, E. (2022). Poor writing, not specialized concepts, drives processing difficulty in legal language. Cognition, 224, 105070. https://doi.org/10.1016/j.cognition.2022.105070

Martínez, E., Mollica, F., & Gibson, E. (2023). Even lawyers do not like legalese. Proceedings of the National Academy of Sciences, 120(23), e2302672120. https://doi.org/10.1073/pnas.2302672120

Sayfi, S., Charide, R., Elliott, S. A., Hartling, L., Munan, M., Stallwood, L., et al. (2024). A multimethods randomized trial found that plain language versions improved adults' understanding of health recommendations. Journal of Clinical Epidemiology, 165, 111219. https://doi.org/10.1016/j.jclinepi.2023.11.009

China Disabled Persons' Federation. (2021). 14th Five-Year Plan for the Protection and Development of Disabled People. https://www.cdpf.org.cn/zwgk/ghjh/af85c8a4edd744159b1a60f659a36d9b.htm

Ministry of Education of the People's Republic of China. (2021). Overview of China's language and script situation. https://hudong.moe.gov.cn/jyb_sjzl/wenzi/202108/t20210827_554992.html

Ministry of Education of the People's Republic of China. (2025). National Smart Education Public Service Platform inventory and users. https://www.moe.gov.cn/fbh/live/2025/77791/mtbd/202512/t20251231_1425330.html

Ministry of Education of the People's Republic of China. (2026). 2025 National Education Development Statistical Communique. https://www.moe.gov.cn/jyb_sjzl/sjzl_fztjgb/202607/t20260706_1442870.html

National Bureau of Statistics of China. (2026a). Communique of the 2025 national 1 percent population sample survey. https://www.stats.gov.cn/english/PressRelease/202605/t20260522_1963789.html

National Bureau of Statistics of China. (2026b). 2025 National Migrant Worker Monitoring Survey Report. https://www.stats.gov.cn/sj/zxfb/202604/t20260430_1963472.html

National Bureau of Statistics of China, UNICEF, & UNFPA. (2023). What the 2020 Census Can Tell Us About Children in China. https://www.stats.gov.cn/zs/tjwh/tjkw/tjzl/202304/P020230419425666818737.pdf

National Press and Publication Administration. (2026). Accessible reading resources for people with visual and hearing disabilities. https://www.nppa.gov.cn/xxfb/ywdt/202607/t20260709_998348.html

Wang, et al. (2025). English academic writing needs of Chinese science and engineering academics. PLOS One. https://doi.org/10.1371/journal.pone.0324760

Agence Nationale de la Statistique et de la Démographie du Sénégal. (2025). RGPH-5 2023, Thème II: Alphabétisation, scolarisation, niveau d'instruction et formation professionnelle. https://anads.ansd.sn/index.php/catalog/311/download/3095

Federal Ministry of Information and National Orientation, Nigeria. (2025, November 13). FG reaffirms English as medium of instruction, launches inclusive language policy. https://fmino.gov.ng/fg-reaffirms-english-as-medium-of-instruction-launches-inclusive-language-policy-to-promote-national-unity-2/

Ghana Statistical Service. (2021a). 2021 Population and Housing Census General Report, Volume 3C: Background characteristics. https://census2021.statsghana.gov.gh/

Ghana Statistical Service. (2021b). 2021 Population and Housing Census General Report, Volume 3D: Literacy and education. https://census2021.statsghana.gov.gh/

Ghana Statistical Service. (2021c). 2021 PHC-1B questionnaire. https://census2021.statsghana.gov.gh/

Institut National de la Statistique du Mali. (2025). RGPH5 2022: Scolarisation, instruction et alphabétisation. https://www.instat-mali.org/laravel-filemanager/files/shares/rgph/rapport-scolarisation-instruction-alphabetisation-rgph5_rgph.pdf

National Bureau of Statistics of Nigeria, & UNICEF. (2022). 2021 Nigeria Multiple Indicator Cluster Survey and National Immunization Coverage Survey: Statistical snapshots report. https://www.nigerianstat.gov.ng/pdfuploads/2021%20MICS%20Statistical%20Snapshots%20Report.pdf

Evans, L. P., & Warren-Rothlin, A. (2018). Proposal to encode additional Arabic script characters for Hausa to the UCS (Unicode L2/18-094). Unicode Consortium.

https://www.unicode.org/L2/L2018/18094-arabic-hausa.pdf

IFRA-Nigeria. (n.d.). Guide to standard Naijá orthography: An NLA harmonized writing system for common Naijá publications. Retrieved September 5, 2026, from https://ifra-nigeria.org/research/former-projects/89-guide-to-standard-naija-orthography-an-nla-harmonized-writing-system-for-common-naija-publications

National Board for Technical Education. (n.d.). Digest of statistics of TVET institutions in Nigeria. Retrieved September 5, 2026, from https://www.nbte.gov.ng/nbte/digest%20of%20statistics

Nigerian Educational Research and Development Council. (n.d.-a). Language Development Centre: Achievements recorded. Retrieved September 5, 2026, from https://www.nerdc.gov.ng/content_manager/ldc_achievements.html

Nigerian Educational Research and Development Council. (n.d.-b). Curriculum Development Services: Scope of service and achievements. Retrieved September 5, 2026, from https://www.nerdc.gov.ng/content_manager/cd.html

Tertiary Education Trust Fund. (n.d.). TERAS services. Retrieved September 5, 2026, from https://teras.tetfund.gov.ng/services

UNICEF Nigeria. (2023). Nigeria education fact sheets: Analyses for learning and equity using MICS data. https://www.unicef.org/nigeria/media/9211/file/Nigeria%20Education%20Fact%20Sheets.pdf

Universal Basic Education Commission, & UNICEF. (2025). National Assessment of Learning Achievement in Basic Education: NALABE 2022 report. https://www.unicef.org/nigeria/reports/national-assessment-learning-achievement-basic-education

World Bank. (2026a). World Development Indicators: Nigeria [Data set]. Retrieved September 5, 2026, from https://data.worldbank.org/country/nigeria

World Bank. (2026b). World Development Indicators: Pakistan [Data set]. Retrieved September 5, 2026, from https://data.worldbank.org/country/pakistan

World Bank. (2026c). World Development Indicators: Ukraine [Data set]. Retrieved September 5, 2026, from https://data.worldbank.org/country/ukraine

World Bank. (2026d). World Development Indicators: Russian Federation [Data set]. Retrieved September 5, 2026, from https://data.worldbank.org/country/russian-federation

OpenStax. (2026a). Prealgebra 2e source collection (Git commit `38cae454e644abf9f0a623e876994553881597c9`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-prealgebra-bundle/tree/38cae454e644abf9f0a623e876994553881597c9

OpenStax. (2026b). Introductory Statistics 2e source collection (Git commit `1f6a35825395bb4aa2834cf1eca37512655f920c`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-introductory-statistics-bundle/tree/1f6a35825395bb4aa2834cf1eca37512655f920c

OpenStax. (2026c). Biology 2e source collection (Git commit `63f8b6f8d129dd1582989bb755011e9a6d523471`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-biology-bundle/tree/63f8b6f8d129dd1582989bb755011e9a6d523471

OpenStax. (2026d). Chemistry 2e source collection (Git commit `3be4b60ff501f29a445f0cacf003e5f5cc16244d`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-chemistry-bundle/tree/3be4b60ff501f29a445f0cacf003e5f5cc16244d

OpenStax. (2026e). College Physics 2e source collection (Git commit `fd1b25dfd5d8c6580c6e2b2b34a19e29cc69ada9`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-college-physics-bundle/tree/fd1b25dfd5d8c6580c6e2b2b34a19e29cc69ada9

OpenStax. (2026f). Principles of Management source collection (Git commit `f2b749648a176dbe114dbc4e35ec20c030be4713`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-principles-of-management-bundle/tree/f2b749648a176dbe114dbc4e35ec20c030be4713

OpenStax. (2026g). Principles of Finance 2e source collection (Git commit `e3e5135ae36e21282d7afb4cccd5ce09f421d830`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-principles-finance/tree/e3e5135ae36e21282d7afb4cccd5ce09f421d830

OpenStax. (2026h). Anatomy and Physiology 2e source collection (Git commit `716383a4c6c16037b14d75a156c65145e75e895e`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-anatomy-physiology/tree/716383a4c6c16037b14d75a156c65145e75e895e

OpenStax. (2026i). Introduction to Python Programming source collection (Git commit `d215dd3b99c33db855c49315152ebd5fbadc8534`) [CNXML source]. GitHub. https://github.com/openstax/osbooks-introduction-python-programming/tree/d215dd3b99c33db855c49315152ebd5fbadc8534

Houston Independent School District. (2022). Immigrant student program evaluation report 2021-2022. https://www.houstonisd.org/fs/resource-manager/view/a76fe050-7086-4b96-9bdc-540738502595

Houston Independent School District. (n.d.-a). Multilingual program models. Retrieved September 5, 2026, from https://www.houstonisd.org/schools-academics/academics/multilingual/multilingual-program-models

Houston Independent School District. (n.d.-b). Testing & identification. Retrieved September 5, 2026, from https://www.houstonisd.org/schools-academics/academics/multilingual/testing-identification

Texas Education Agency. (n.d.-a). Bluebonnet Learning: K-5 math and secondary mathematics. Retrieved September 5, 2026, from https://tea.texas.gov/curriculum-and-instruction/instructional-materials/bluebonnet-learning/bluebonnet-learning-k-5-math-and-secondary-mathematics

Texas Education Agency. (n.d.-b). Texas Essential Knowledge and Skills in Spanish. Retrieved September 5, 2026, from https://tea.texas.gov/laws-and-rules/state-board-education/teks/texas-essential-knowledge-and-skills-spanish

Wharton K-8 Dual Language Academy. (n.d.). Academic offerings. Retrieved September 5, 2026, from https://wharton.houstonisd.org/academics-programs/academic-offerings

World Bank. (n.d.). Fostering foundational skills in Thailand: From a skills crisis to a learning society. https://www.worldbank.org/en/country/thailand/publication/fostering-foundational-skills-in-thailand

UNICEF Myanmar. (n.d.). Learning. Retrieved September 4, 2026, from https://www.unicef.org/myanmar/learning

Universidad Nacional Autónoma de México. (n.d.). MOOC UNAM: Catálogo de cursos. Retrieved September 4, 2026, from https://mooc.unam.mx/

Badan Pusat Statistik. (2022a). Kemampuan berbahasa Indonesia: Long Form SP2020 tables. https://sensus.bps.go.id/topik/tabular/sp2022/196/1/0

Badan Pusat Statistik. (2022b). Bahasa pertama: Long Form SP2020 tables. https://sensus.bps.go.id/topik/tabular/sp2022/197/0/0

Badan Pemeriksa Keuangan. (2003). Undang-Undang Republik Indonesia Nomor 20 Tahun 2003 tentang Sistem Pendidikan Nasional. https://peraturan.bpk.go.id/Download/32160/UU%20Nomor%2020%20Tahun%202003.pdf

Badan Pusat Statistik. (2025). Telecommunication statistics in Indonesia 2024. https://www.bps.go.id/id/publication/2025/08/29/beaa2be400eda6ce6c636ef8/telecommunication-statistics-in-indonesia-2024.html

Kementerian Pendidikan Dasar dan Menengah. (n.d.). Sistem Informasi Perbukuan Indonesia: Katalog buku. Retrieved September 5, 2026, from https://buku.kemendikdasmen.go.id/katalog

Pusat Standar dan Kebijakan Pendidikan. (2023). Optimising the use of local languages in early grades (Policy Brief No. 1). https://pskp.kemendikdasmen.go.id/assets_front/images/produk/1-gtk/kebijakan/Policy_Brief-Optimising_the_Use_of_Local_Languages_in_Early_Grades_FIN_%281%29.pdf

Bangladesh Open University. (n.d.). BBA course materials, third-level course contents. https://www.ebookbou.edu.bd/wp/OS/osbba3_1.php

Bangladesh Bureau of Educational Information and Statistics. (2026). Bangladesh education statistics 2024. https://banbeis.gov.bd/pages/static-pages/6a338d7fcb6cb358ff185b6c

Bangladesh Bureau of Statistics. (2026). ICT access and use by households and individuals survey 2024-25. https://bbs.gov.bd/pages/files/6a609418b859b8767ff841e0

Islam, M. T., Hasan, M. K., & Akhand, M. M. (2025). Challenges in English-medium instruction classes among engineering major students at Bangladeshi private universities. Studies in English Language and Education, 12(1), 285-300. https://doi.org/10.24815/siele.v12i1.38407

Karim, A., Kabilan, M. K., Ahmed, Z., Reshmin, L., & Rahman, M. M. (2023). The medium of instruction in Bangladeshi higher education institutions: Bangla, English, or both? Journal of Language, Identity & Education, 22(3), 232-246. https://doi.org/10.1080/15348458.2020.1871353

Sultana, S., & Fang, F. (2024). English as the medium of instruction and mother-tongue-based translanguaging: Challenges and prospects for tertiary education in Bangladesh and China. International Journal of Educational Development, 104, 102951. https://doi.org/10.1016/j.ijedudev.2023.102951

OECD. (2023a). PISA 2022 results (Volume I): The state of learning and equity in education. https://doi.org/10.1787/53f23881-en

OECD. (2023b). PISA 2022 statistical tables: Annex A2 and Chapter 3. https://stat.link/hpg9nd ; https://stat.link/znxau8

OECD. (2023c). PISA 2022 results: Ukrainian regions (18 of 27) country note. https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/11/pisa-2022-results-volume-i-and-ii-country-notes_2fca04b9/ukrainian-regions-18-of-27_de3ad3b1/78043794-en.pdf

Eurostat. (2026, August 11). 4.41 million under temporary protection in June 2026. https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20260811-1

HSE University Institute for Statistical Studies and Economics of Knowledge. (2025). Education in figures: 2025. https://www.hse.ru/en/primarydata/oc2025

Kyiv International Institute of Sociology. (2026, June 15). Dynamics of attitude to the status of the Russian language in Ukraine during 1997-2026. https://www.kiis.com.ua/?cat=reports&id=1619&lang=eng&page=1

National Audit Office of Estonia. (2023, June 1). Estonian language training in a Russian-language basic school. https://www.riigikontroll.ee/en/audits/estonian-language-training-russian-language-basic-school

State Statistics Committee of Ukraine. (2001). All-Ukrainian population census 2001: Results. https://2001.ukrcensus.gov.ua/d/mono_eng.pdf

UNESCO, & United Nations Ukraine. (2024, March 11). Analysis of war damage to the Ukrainian science sector and its consequences. https://ukraine.un.org/en/262987-analysis-war-damage-ukrainian-science-sector-and-its-consequences

Pakistan Bureau of Statistics. (2026). Household Integrated Economic Survey 2024-25: Social report. https://www.pbs.gov.pk/wp-content/uploads/2020/07/PSLM_Report_2024-25-janurary-26-Social-Reportupdated.pdf

UNICEF Bangladesh. (2026). Data on the situation of children in Bangladesh: MICS 2025 indicators. https://www.unicef.org/bangladesh/en/data-situation-children-bangladesh

Virtual University of Pakistan. (n.d.). Open courseware: Mathematics. https://ocw.vu.edu.pk/Courses.aspx?cat=Mathematics

University of Tokyo. (n.d.). Introduction to Python programming [Introduction to Python programming]. https://utokyo-ipp.github.io/

Xu, Z. (2023). Probability and statistics with Python solutions [Probability and statistics with Python solutions]. Tsinghua University Press. https://www.tup.tsinghua.edu.cn/booksCenter/book_09529801.html

Golubović, J., & Gooskens, C. (2015). Mutual intelligibility between West and South Slavic languages. Russian Linguistics. https://doi.org/10.1007/s11185-015-9150-9

Österholm, M. (2006). Characterizing reading comprehension of mathematical texts. Educational Studies in Mathematics, 63, 325-346. https://www.diva-portal.org/smash/get/diva2:22664/FULLTEXT01.pdf

Cristia, J., Ibarrarán, P., Cueto, S., Santiago, A., & Severín, E. (2017). Technology and child development: Evidence from the One Laptop per Child program. American Economic Journal: Applied Economics, 9(3), 295-320. https://doi.org/10.1257/app.20150385

Glewwe, P., Kremer, M., & Moulin, S. (2009). Many children left behind? Textbooks and test scores in Kenya. American Economic Journal: Applied Economics, 1(1), 112-135. https://doi.org/10.1257/app.1.1.112

Muralidharan, K., Singh, A., & Ganimian, A. J. (2019). Disrupting education? Experimental evidence on technology-aided instruction in India. American Economic Review, 109(4), 1426-1460. https://doi.org/10.1257/aer.20171112

Bangladesh Bureau of Statistics, & UNICEF Bangladesh. (2026). Bangladesh Multiple Indicator Cluster Survey 2025: Final report. https://www.unicef.org/bangladesh/en/reports/bangladesh-multiple-indicator-cluster-survey-2025-0

UNICEF Data. (n.d.). Child-related SDG progress assessment for Bangladesh: Population overview. Retrieved September 4, 2026, from https://data.unicef.org/sdgs/country/bgd/

Ministry of Statistics and Programme Implementation. (2020). Household social consumption on education in India: NSS 75th Round, July 2017-June 2018 (Report No. 585). https://mospi.gov.in/sites/default/files/publication_reports/Report_585_75th_round_Education_final_1507_0.pdf

ASER Centre. (2025). ASER 2024 national findings. https://asercentre.org/wp-content/uploads/2022/12/ASER-2024-National-findings.pdf

Department of Higher Education, Ministry of Education, Government of India. (2026). All India Survey on Higher Education 2023-24: Final report. https://cdnbbsr.s3waas.gov.in/s392049debbe566ca5782a3045cf300a3c/uploads/2026/07/202607131602421770.pdf

Department of School Education & Literacy, Ministry of Education, Government of India. (2025). UDISE+ report 2024-25: NEP structure. https://www.dsel-education.gov.in/statistics

DIKSHA. (n.d.). Features and supported languages. Retrieved September 5, 2026, from https://diksha.gov.in/help/faqs/user/features.html

Indian Sign Language Research and Training Centre. (n.d.). Frequently asked questions: ISL Dictionary. Retrieved September 5, 2026, from https://islrtc.nic.in/faq/

National Council of Educational Research and Training. (n.d.). Indian-language primers. Retrieved September 5, 2026, from https://ncert.nic.in/primers.php

National Statistical Office. (2025). Comprehensive Modular Survey: Telecom, 2025 (NSS Report No. 593). https://mospi.gov.in/sites/default/files/publication_reports/CMST_report_m.pdf

NPTEL. (n.d.). Translation programme and translated books. https://nptel.ac.in/translation

Agency for Cultural Affairs, Japan. (n.d.). Languages and dialects in critical situations. Retrieved September 5, 2026, from https://www.bunka.go.jp/seisaku/kokugo_nihongo/kokugo_shisaku/kikigengo/

Cabinet Office, Japan. (2025). Promotion of measures concerning sign language. https://www8.cao.go.jp/shougai/suishin/jsl.html

Japan Science and Technology Agency. (n.d.). J-STAGE overview. Retrieved September 5, 2026, from https://www.jstage.jst.go.jp/static/pages/JstageOverview/-char/en

Ministry of Education, Culture, Sports, Science and Technology, Japan. (2024a). Statistical abstract 2024: School education and science-and-technology tables [Data set]. https://www.mext.go.jp/b_menu/toukei/002/002b/1417059_00009.htm

Ministry of Education, Culture, Sports, Science and Technology, Japan. (2024b). Results of the 2023 survey on the acceptance of students requiring Japanese-language instruction. https://www.mext.go.jp/content/20240808-mxt_kyokoku-000037366_4.pdf

Ministry of Education, Culture, Sports, Science and Technology, Japan. (2024c). Japan's student performance for PISA 2022. https://www.mext.go.jp/content/20240524-mxt_chousa02-100002205_2.pdf

Ministry of Education, Culture, Sports, Science and Technology, Japan. (2024d). The fourth basic plan for the promotion of education [English translation]. https://www.mext.go.jp/content/20240228-soseisk02-100000597_09.pdf

National Institute of Science and Technology Policy. (2016). Characteristics of research-article publication in major countries, focusing on journals (Research Material No. 254, press summary). https://www.nistep.go.jp/wp/wp-content/uploads/NISTEP-RM254-PressJ.pdf

World Bank. (2026e). World Development Indicators: Japan [Data set]. Retrieved September 5, 2026, from https://data.worldbank.org/country/japan

Office of the Registrar General & Census Commissioner, India. (2011a). C-16 population by mother tongue. https://censusindia.gov.in/nada/index.php/catalog/10191

Office of the Registrar General & Census Commissioner, India. (2011b). C-17 population by bilingualism and trilingualism. https://censusindia.gov.in/nada/index.php/catalog/10262

Department of Education, Republic of the Philippines. (n.d.-a). ALS learning materials. https://www.deped.gov.ph/als-learning-materials/

Department of Education, Republic of the Philippines. (n.d.-b). Learning Resources Management and Development System. Retrieved September 5, 2026, from https://lrmds.deped.gov.ph/k_to_12

Department of Education, Republic of the Philippines. (2025). Policy on the medium of instruction for Kindergarten to Grade 3 effective school year 2025-2026 (DepEd Order No. 020, s. 2025). https://www.deped.gov.ph/wp-content/uploads/DO_s2025_020r_8.pdf

Philippine Statistics Authority. (2023a). Tagalog is the most widely spoken language at home (2020 Census of Population and Housing) (Reference No. 2023-42). https://psa.gov.ph/content/tagalog-most-widely-spoken-language-home-2020-census-population-and-housing

Philippine Statistics Authority. (2023b). More than 50 million have access to the Internet (2020 Census of Population and Housing) (Reference No. 2023-73). https://psa.gov.ph/content/more-50-million-have-access-internet-2020-census-population-and-housing

Philippine Statistics Authority. (2025a). For every 10 Filipinos, 9 have basic literacy, while 7 have functional literacy (Reference No. 2025-118). https://psa.gov.ph/content/every-10-filipinos-9-have-basic-literacy-while-7-have-functional-literacy

Philippine Statistics Authority. (2025b). Percentage of households with internet connection increased to 48.8 percent in 2024; two in every three individuals aged 10 years and over used the internet. https://psa.gov.ph/content/percentage-households-internet-connection-increased-488-percent-2024-two-every-three

Bureau of National Statistics of Kazakhstan. (2021). National composition, religion and language proficiency in the Republic of Kazakhstan. https://stat.gov.kz/en/national/2021/

Government of Mongolia. (2020). Mongolian State Report on the implementation of the 1960 Convention and Recommendation against Discrimination in Education. https://media.unesco.org/sites/default/files/webform/r2e002/mongolian-state-report-on-the-implementation-of-the-1960-convention-and-recommendation-against-discrimination-in-education-15-nov-2020.pdf

National Statistical Committee of the Kyrgyz Republic. (2024). Students in daytime general educational organizations by language of instruction. https://stat.gov.kg/kg/opendata/category/4263/

State Committee of the Republic of Uzbekistan on Statistics. (2022). Main statistical indicators of the activity of general secondary educational institutions. https://stat.uz/images/umumtalim-eng-30_03_2022.pdf

World Bank & UNESCO Institute for Statistics. (2024). Learning Poverty Briefs: Kazakhstan, Kyrgyz Republic, Mongolia, and Uzbekistan. https://www.worldbank.org/en/topic/education/brief/learning-poverty

Xinjiang Uyghur Autonomous Region Statistics Bureau. (2021). Seventh National Population Census Bulletin No. 1. https://tjj.xinjiang.gov.cn/tjj/tjgn/202106/e3605637963f4f7280654a6f07af0b0e.shtml

Pakistan Bureau of Statistics. (2023a). Population by single year age, sex and rural/urban: Census 2023 (Table 4). https://www.pbs.gov.pk/wp-content/uploads/census_tables/tables/Table_4_national.pdf

Pakistan Bureau of Statistics. (2023b). Population by mother tongue, sex and rural/urban: Census 2023 (Table 11). https://www.pbs.gov.pk/wp-content/uploads/census_tables/tables/table_11_national.pdf

Pakistan Bureau of Statistics. (2023c). Literacy rate, enrolment and out-of-school population by sex and rural/urban: Census 2023 (Table 12). https://www.pbs.gov.pk/wp-content/uploads/census_tables/tables/table_12_national.pdf

Higher Education Commission, Pakistan. (n.d.). Digital library frequently asked questions. Retrieved September 5, 2026, from https://www.hec.gov.pk/english/services/students/DL/Pages/FAQs.aspx

Higher Education Commission, Pakistan. (2025). Annual report 2023-24. https://www.hec.gov.pk/english/news/AnnualReports/Annual-Report-2023-24.pdf

Ministry of Human Rights, Pakistan. (2024). Year book 2022-23. https://mohr.gov.pk/SiteImage/Misc/files/Final%20Draft%20Year%20Book%202022-23%20ver%202nd%20April,%202024.pdf

National Curriculum Council. (2025). Inclusive scheme of studies, version I. https://ncc.gov.pk/SiteImage/Misc/files/Inclusive%20Scheme%20of%20Studies%20Version%20I%281%29.pdf

Pakistan Institute of Education. (2024a). Pakistan non-formal education statistics 2022-23. https://pie.gov.pk/SiteImage/Publication/NFE.pdf

Pakistan Institute of Education. (2024b). National Achievement Test 2023: Findings report. https://pie.gov.pk/SiteImage/Downloads/NAT%202023%20Findings%20Report%2006.03.2024%20-Final-%20v6.pdf

Uwezo Tanzania. (2019). Are our children learning? Tanzania learning assessment report. Twaweza East Africa. https://twaweza.org/wp-content/uploads/2021/01/Tanzania-Report-FINAL3-2019-13-12-2019-Print-Readyzm.pdf

National Curriculum and Textbook Board. (2026). Primary mathematics, grade 2 [Primary mathematics, grade 2] (2026 academic-year reprint). https://drive.egovcloud.gov.bd/index.php/s/VeBQB1ZuzSZPRBJ

Merunka, V., van Steenbergen, J., Yordanova, L., & Kocór, M. (2019). The Interslavic language as a tool for supporting e-democracy in Central and Eastern Europe. International Journal of Electronic Governance, 11(3/4), 260-288. https://interslavic-language.org/doc/ijeg-2019-interslavic.pdf

Khan Academy. (n.d.). Early mathematics: Counting [Bangla learning materials]. https://bn.khanacademy.org/math/early-math/cc-early-math-counting-topic

National Institute of Statistics of Rwanda. (2023). Fifth Rwanda Population and Housing Census: Education characteristics of population. Tables 7.7, C.25 and C.31. https://statistics.gov.rw/sites/default/files/documents/2025-02/Education.pdf

Rwanda Basic Education Board. (n.d.). Translated books are now available for students and teachers. https://www.reb.gov.rw/news-detail/translated-books-are-now-available-for-students-and-teachers

Federated States of Micronesia Statistics. (2024). 2023 FSM municipality workbook. https://www.pacificweb.org/library/preview.php?file=fsm/states/2023FSMMunicipality.xlsx

Government of Alberta. (n.d.). Plains Cree Y-dialect Grade 7-9 guide to implementation [Draft guide]. https://open.alberta.ca/dataset/ac9f2505-2b2d-4e4f-919e-41769c41d6ca/resource/4c805ee1-02bf-4721-a62e-c78ba6875038/download/cree-12y-guide-to-implementation-7-9-draft.pdf

Government of Nunavut. (2024). Department of Education annual report 2022-23. https://www.assembly.nu.ca/sites/default/files/2024-11/EDU%20Annual%20Report%202022-23%20EN.pdf

Government of Nunavut. (2025). Department of Education annual report 2023-24. https://www.assembly.nu.ca/sites/default/files/2025-09/TD-463-6(2)-EN-Education%20Annual%20Report%202023-24.pdf

Guatemala Instituto Nacional de Estadística. (2019). XII Censo Nacional de Población y VII de Vivienda: Principales resultados. https://censo2018.ine.gob.gt/archivos/Principales_resultados_Censo2018.pdf

Guatemala Ministry of Education. (2022). Memoria de labores 2022. https://edu.mineduc.gob.gt/documents/memoriaLabores/Memoria_de_Labores_2022.pdf

Institut Haïtien de Statistique et d'Informatique. (2024). Projections et estimations de la population. https://www.ihsi.gouv.ht/statistiques/statistiques_demographiques_et_sociales/projections_et_estimations_de_la_population

Marshall Islands Economic Policy, Planning and Statistics Office. (2022). 2021 census report. https://prdrse4all.spc.int/sites/default/files/marshall_islands_census_report_2021.pdf

Paraguay Instituto Nacional de Estadística. (2022). Día Internacional de la Lengua Materna: EPHC 2022. https://www.ine.gov.py/ende/contenido.php?c=1484&title=21_de_febrero,_D%C3%ADa_Internacional_de_la_Lengua_Materna

Paraguay Ministry of Education. (2022). Plan Nacional de Transformación Educativa Paraguay 2040. https://siteal.iiep.unesco.org/sites/default/files/sit_accion_files/plan_nacional_de_transformacion_educativa_paraguay_2040.pdf

Peru Ministry of Education. (2018). Resultados 2018: Evaluación Censal y Muestral de Estudiantes. https://repositorio.minedu.gob.pe/bitstream/handle/20.500.12799/6562/Resultados%202018%20Evaluaci%C3%B3n%20Censal%20de%20Estudiantes%20y%20Evaluaci%C3%B3n%20Muestral%20de%20Estudiantes%20evaluaciones%20de%20logros%20de%20aprendizaje.pdf?isAllowed=y&sequence=1

Solomon Islands National Statistics Office. (2023). English is Solomon Islands' most predominant language in communication.

https://statistics.gov.sb/english-is-solomon-islands-most-predominant-language-in-communication/

State of Hawaiʻi Department of Business, Economic Development & Tourism. (2019). State of Hawaii Data Book 2018, Section 1. https://files.hawaii.gov/dbedt/economic/databook/db2018/section01.pdf

Statistics Canada. (2025a). Algonquian languages, 2021 Census. https://www150.statcan.gc.ca/n1/pub/41-20-0002/412000022025002-eng.htm

Statistics Canada. (2025b). Inuktut languages, 2021 Census. https://www150.statcan.gc.ca/n1/pub/41-20-0002/412000022025004-eng.htm

Statistics Greenland. (2026). Befolkningsstatus 1. januar 2026. https://stat.gl/publ/da/BE/202601/oversigt/Befolkningsstatus.htm

Stats NZ. (2024). 2023 Census ethnic-group summary: Māori. https://tools.summaries.stats.govt.nz/ethnic-group/maori

UNICEF. (2025). Haiti's children confront a polycrisis. https://www.unicef.org/child-alert/haitis-children-confront-polycrisis

World Bank. (2015). Why do students learn so little? Seeking answers inside Haiti's classrooms. https://files.eric.ed.gov/fulltext/ED573569.pdf

World Bank. (2019). Solomon Islands early grade reading assessment results report. https://documents1.worldbank.org/curated/en/178621563252790030/pdf/Solomon-Islands-Early-Grade-Reading-Assessment-Results-Report.pdf

Coordination for the Improvement of Higher Education Personnel. (2025). Metadata for stricto sensu postgraduate students in Brazil, 2021-2024. https://dadosabertos.capes.gov.br/dataset/c6bd4dca-a0fb-499a-9f7f-df0740563333/resource/70633e5c-5294-4cd4-abea-79341c63e406/download/metadados_discentes_pos_graduacao_2021_2024.pdf

Federal Senate of Brazil. (2021). Law No. 14,191 of 3 August 2021: Bilingual education for deaf people. https://legis.senado.gov.br/norma/34639654

Brazilian Institute of Geography and Statistics. (2025a). 2022 Demographic Census: Indigenous ethnicities and languages-Supplementary Table 21 [Data set]. https://ftp.ibge.gov.br/Censos/Censo_Demografico_2022/Etnias_e_Linguas_Indigenas_principais_caracteristicas_sociodemograficas_Resultados_do_universo/Tabelas_complementares/xlsx/Tabela_complementar_21.xlsx

Brazilian Institute of Geography and Statistics. (2025b). 2022 Demographic Census: Indigenous ethnicities and languages-Selected Tables 7-9 [Data set]. https://ftp.ibge.gov.br/Censos/Censo_Demografico_2022/Etnias_e_Linguas_Indigenas_principais_caracteristicas_sociodemograficas_Resultados_do_universo/Tabelas_selecionadas/xlsx/

Brazilian Institute of Geography and Statistics. (2025c). 2022 Demographic Census: Indigenous people-Literacy. https://biblioteca.ibge.gov.br/visualizacao/periodicos/3111/cd_2022_indigenas_alfabetizacao.pdf

Instituto Nacional de Estudos e Pesquisas Educacionais Anísio Teixeira. (2025a). Censo Escolar da Educação Básica 2024: Resumo técnico. https://download.inep.gov.br/publicacoes/institucionais/estatisticas_e_indicadores/resumo_tecnico_censo_escolar_2024.pdf

Instituto Nacional de Estudos e Pesquisas Educacionais Anísio Teixeira. (2025b). Censo da Educação Superior 2024: Notas estatísticas. https://download.inep.gov.br/publicacoes/institucionais/estatisticas_e_indicadores/notas_estatisticas_censo_escolar_da_educacao_superior_2024.pdf

Ministry of Planning and Budget, Brazil. (2026). Agenda Transversal Povos Indígenas 2026: Ano-base 2025. https://www.gov.br/planejamento/pt-br/assuntos/orcamento/agendas-transversais-e-multisetoriais/arquivos/2026/2026_05_05_atpovosindigenas.pdf/@@display-file/file

OECD. (2023d). PISA 2022 results: Brazil country note. https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/11/pisa-2022-results-volume-i-and-ii-country-notes_2fca04b9/brazil_764ee0aa/61690648-en.pdf

Asociación Nacional de Universidades e Instituciones de Educación Superior. (2026). Anuario estadístico de educación superior. https://www.anuies.mx/informacion-y-servicios/informacion-estadistica-de-educacion-superior/anuario-estadistico-de-educacion-superior

Cámara de Diputados. (2024). Ley General para la Inclusión de las Personas con Discapacidad (última reforma DOF 14-06-2024). https://www.diputados.gob.mx/LeyesBiblio/pdf/LGIPD.pdf

Cámara de Diputados. (2026). Ley General de Educación (última reforma DOF 15-01-2026). https://www.diputados.gob.mx/LeyesBiblio/pdf/LGE.pdf

Comisión Nacional de Libros de Texto Gratuitos. (2026). Catálogo de libros, ciclo escolar 2026-2027. https://www.conaliteg.sep.gob.mx/

Dirección General de Educación Indígena, Intercultural y Bilingüe. (2024). Estadística indígena, ciclo 2023-2024. Secretaría de Educación Pública.

Instituto Nacional de Estadística y Geografía. (2022). Principales resultados del Censo de Población y Vivienda 2020: Estados Unidos Mexicanos. https://www.inegi.org.mx/programas/ccpv/2020/

Instituto Nacional de Lenguas Indígenas. (2008). Catálogo de las lenguas indígenas nacionales: Variantes lingüísticas de México con sus autodenominaciones y referencias geoestadísticas. https://www.inali.gob.mx/clin-inali/

Organisation for Economic Co-operation and Development. (2023e). PISA 2022 results: Mexico country note. https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/11/pisa-2022-results-volume-i-and-ii-country-notes_2fca04b9/mexico_51978f68/2d35d9e0-en.pdf

SciELO México. (2026). Lista alfabética de revistas. https://www.scielo.org.mx/scielo.php?script=sci_alphabetic&lng=es&nrm=iso

Secretaría de Educación Pública. (2025). Principales cifras del Sistema Educativo Nacional 2024-2025. https://www.planeacion.sep.gob.mx/Doc/estadistica_e_indicadores/principales_cifras/principales_cifras_2024_2025_bolsillo.pdf

Secretaría de Educación Pública. (2026). Cursos @prende.mx. https://dev.mexicox.gob.mx/

Ethiopia Ministry of Education. (2021). Education Sector Development Programme VI, 2020/21-2024/25. https://moe.gov.et/storage/Books/Ethiopian%20Education%20Sector%20Development%20Program%206%20%28ESDP%20VI%29.pdf

Ethiopia Ministry of Education. (2023a). Education statistics annual abstract 2022/23. https://moe.gov.et/storage/Books/ESAA%202022-23%20Final.pdf

Ethiopia Ministry of Education. (2023b). Digital education strategy and implementation plan 2023-2028. https://www.dpgethiopia.org/wp-content/uploads/2024/03/Digital-Education-Strategy-and-Implementation-Plan-for-Ethiopia-.pdf

Ethiopia Ministry of Education. (n.d.-a). Special needs/inclusive education strategy. https://www.moe.gov.et/storage/Books/Special%20Needs%20Inclusive%20Education%20Strategy.pdf

Ethiopia Ministry of Education. (n.d.-b). Strategic policy for national science, technology and mathematics education. https://www.moe.gov.et/storage/Books/Strategic%20Policy%20for%20National%20Science%20Technology%20and%20Mathematics%20Education_Full.pdf

Ethiopia Ministry of Labor and Skills. (2025). Ethiopia technical and vocational training strategy 2025. https://mols.gov.et/wp-content/uploads/2025/12/TVT-Strategy-English-Version-12-Nov-2025_Print.pdf

Ethiopian Statistical Service. (2008). Summary and statistical report of the 2007 Population and Housing Census. https://ess.gov.et/wp-content/uploads/2007/09/Population-and-Housing-Census-2007-National_Statistical.pdf

Federal Democratic Republic of Ethiopia. (2025). General Education Proclamation No. 1368/2025. https://justice.gov.et/wp-content/uploads/2025/04/1368.pdf

World Bank. (2026). World Development Indicators: Ethiopia population, individuals using the Internet, and access to electricity [Data set]. https://api.worldbank.org/v2/country/ETH?format=json

World Food Programme. (2024). McGovern-Dole school feeding programme in Afar and Oromia Regions 2019-2025: Mid-term evaluation, Volume 2-A light-touch early-grade reading assessment report. https://wfp.tind.io/record/127524

Bureau of Indian Education. (2024a). Strategic direction, 2024-2030. https://www.bie.edu/sites/default/files/documents/bie_strategic_direction_2024-2030.pdf

Bureau of Indian Education. (2024b). School year 2023-2024 BIE SEA report card. https://www.bie.edu/sites/default/files/documents/sy_2023-2024_bie_sea_report_card.pdf

Bureau of Indian Education. (n.d.). Performance report. https://www.bie.edu/sites/default/files/documents/bie_performance_report.pdf

Gallaudet University Press. (n.d.). The hidden treasure of Black ASL. https://gupress.gallaudet.edu/Books/T/The-Hidden-Treasure-of-Black-ASL

National Advisory Council on Indian Education. (2023). Annual report to Congress, 2022-2023. https://www.ed.gov/sites/ed/files/2023/08/NACIE-AnnualReport2023-508_completed.pdf

National Center for Education Statistics. (2023a). English learners in public schools. Condition of Education. https://nces.ed.gov/programs/coe/pdf/2023/cgf_508.pdf

National Center for Education Statistics. (2023b). Enrollment in public elementary and secondary schools, by level and grade: Selected years, fall 1980 through fall 2031 [Data set]. https://nces.ed.gov/programs/digest/d23/tables/xls/tabn203.10.xlsx

National Center for Education Statistics. (2024a). Percentage of three-, four-, and five-year-old children enrolled in school, by age and selected child and family characteristics: 2022 and 2023 [Data set]. https://nces.ed.gov/programs/digest/d24/tables/xls/tabn202.20.xlsx

National Center for Education Statistics. (2024b). Career and technical education in the United States. https://nces.ed.gov/programs/coe/pdf/2024/tob_508c.pdf

National Center for Education Statistics. (2026). Total fall enrollment in degree-granting postsecondary institutions, by level of enrollment, control and level of institution, attendance status, and sex of student: Selected years, 1970 through 2034 [Data set]. https://nces.ed.gov/programs/digest/d25/tables/xls/tabn306.10.xlsx

National Center for Science and Engineering Statistics. (2025). Doctorate recipients from U.S. universities: 2023 (NSF 25-300). https://ncses.nsf.gov/pubs/nsf25300/assets/nsf25300.pdf

National Deaf Center on Postsecondary Outcomes. (2019). Undergraduate enrollment of deaf students in the United States. https://nationaldeafcenter.org/wp-content/uploads/2022/11/Undergraduate20Enrollment2020of20Deaf20Students20in20the20United20States.pdf

National Deaf Center on Postsecondary Outcomes. (2024). Access, belonging, and affirmation: Deaf college student data. https://nationaldeafcenter.org/resources/research-data/deaf-college-student-data/access-belonging-and-affirmation/

U.S. Census Bureau. (2022a). Language use in the United States: 2019 (ACS-50). https://www.census.gov/content/dam/Census/library/publications/2022/acs/acs-50.pdf

U.S. Census Bureau. (2022b). 2017-2021 ACS language tables: Nation [Data set]. https://www2.census.gov/programs-surveys/demo/tables/language-use/2017/2017-2021-acs-lang-tables-nation.xlsx

U.S. Census Bureau. (2022c). 2017-2021 ACS language tables: States [Data set]. https://www2.census.gov/programs-surveys/demo/tables/language-use/2017/2017-2021-acs-lang-tables-state.xlsx

U.S. Census Bureau. (2023a). 2020 Island Areas Censuses detailed cross-tabulations: Technical documentation. https://www2.census.gov/programs-surveys/decennial/2020/technical-documentation/island-areas-tech-docs/detailed-cross-tabulations/2020-iac-dct-technical-documentation.pdf

U.S. Census Bureau. (2023b). 2020 Island Areas Censuses detailed cross-tabulations: Guidance. https://www2.census.gov/programs-surveys/decennial/2020/technical-documentation/island-areas-tech-docs/detailed-cross-tabulations/2020-iac-dct-guidance.pdf

U.S. Department of Education. (2025). 47th annual report to Congress on the implementation of the Individuals with Disabilities Education Act. https://sites.ed.gov/idea/files/47th-ARC-for-IDEA.pdf

World Bank. (2026). World Development Indicators: United States population, individuals using the Internet, and access to electricity [Data set]. https://api.worldbank.org/v2/country/USA?format=json

8. Worldwide reach and common-package allocation checkpoint

Status: ongoing research checkpoint. This restores the quantitative allocation chain; it is not the final package-specific allocation result.

What is being answered

The research question remains: how should AI compute be distributed so that educational-material availability produces the greatest additional functional understanding for as many people as possible?

The answer needs more than one list. The first table estimates potential reach. The second asks how that reach changes when every target receives exactly the same one 100,000-source-token, text-only educational package translated and checked to the independent-context-plus-deterministic-qa-v2 target. Later target-specific tables must replace the held-equal assumptions with evidence about the actual stage, subject, baseline supply, learning effect, delivery conditions, accessibility work, reusable setup, and overlapping beneficiaries.

Profile IDs are never ranks. Ordinal positions appear only in columns explicitly named Reach position or Common-package proxy position.

Result 1: potential reach under one common missing-data scenario

The point multiplier is 89.57%; the displayed sensitivity range is 70%-99%. Missing comparable academic-register evidence keeps the common scenario instead of deleting or lowering a target. Chinese in the Simplified-script profile is position 1. Indonesian is position 9 after retaining the direct 2022 BPS communicative-ability population as its point population. Russian is position 10; Ukrainian is position 62 in the pinned CLDR reach layer. Those are potential-reach positions, not judgments about people or final funding positions.

Reach position	Exact edition profile	Population point	Plausibly helped point	Sensitivity range	Population basis
1	Simplified Standard Written Chinese profile	1.286 billion	1.152 billion	900.2 million-1.273 billion	CLDR48_RESOLVED_SCRIPT_PROFILE
2	Hindi, Devanagari-script profile	580.3 million	519.8 million	406.2 million-574.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
3	Spanish, Latin-script profile	507.6 million	454.7 million	355.3 million-502.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
4	Arabic, Arabic-script profile	378.8 million	339.3 million	265.2 million-375.0 million	CLDR48_RESOLVED_SCRIPT_PROFILE
5	French, Latin-script profile	333.0 million	298.2 million	233.1 million-329.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
6	Urdu, Arabic-script profile	313.1 million	280.4 million	219.2 million-310.0 million	CLDR48_RESOLVED_SCRIPT_PROFILE
7	Bangla, Bangla-script profile	279.9 million	250.7 million	195.9 million-277.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
8	Portuguese, Latin-script profile	249.5 million	223.5 million	174.6 million-247.0 million	CLDR48_RESOLVED_SCRIPT_PROFILE
9	Indonesian, Latin-script profile	248.5 million	222.6 million	126.4 million-246.0 million	DIRECT_BPS_2022_INDONESIAN_COMMUNICATIVE_ABILITY_POINT_WITH_CLDR_CONTEXT
10	Russian, Cyrillic-script profile	201.2 million	180.2 million	140.8 million-199.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
11	Swahili, Latin-script profile	194.1 million	173.9 million	135.9 million-192.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
12	Punjabi, Shahmukhi/Perso-Arabic-script profile	176.7 million	158.2 million	123.7 million-174.9 million	CLDR48_RESOLVED_SCRIPT_PROFILE
13	German, Latin-script profile	141.9 million	127.1 million	99.3 million-140.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
14	Japanese, Japanese-script profile	117.6 million	105.3 million	82.3 million-116.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
15	Telugu, Telugu-script profile	101.5 million	90.9 million	71.0 million-100.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
16	Western Panjabi, Arabic-script profile	101.0 million	90.5 million	70.7 million-100.0 million	CLDR48_RESOLVED_SCRIPT_PROFILE
17	Marathi, Devanagari-script profile	98.6 million	88.4 million	69.0 million-97.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE
18	Javanese, Latin-script profile	96.1 million	86.1 million	67.3 million-95.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
19	Vietnamese, Latin-script profile	92.4 million	82.7 million	64.7 million-91.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
20	Tamil, Tamil-script profile	90.6 million	81.2 million	63.4 million-89.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE

Reach position	Exact edition profile	Population point	Plausibly helped point	Sensitivity range	Population basis
21	Persian, Arabic-script profile	89.2 million	79.9 million	62.4 million-88.3 million	CLDR48_RESOLVED_SCRIPT_PROFILE
22	Wu Chinese, Simplified-script profile	85.0 million	76.1 million	59.5 million-84.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
23	Turkish, Latin-script profile	82.4 million	73.8 million	57.7 million-81.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
24	Korean, Korean-script profile	79.3 million	71.0 million	55.5 million-78.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
25	Cantonese, Simplified-script profile	73.6 million	66.0 million	51.5 million-72.9 million	CLDR48_RESOLVED_SCRIPT_PROFILE
26	Filipino, Latin-script profile	73.2 million	65.6 million	51.2 million-72.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
27	Egyptian Arabic, Arabic-script profile	71.2 million	63.8 million	49.8 million-70.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
28	Italian, Latin-script profile	70.5 million	63.1 million	49.3 million-69.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
29	Gujarati, Gujarati-script profile	65.6 million	58.8 million	45.9 million-65.0 million	CLDR48_RESOLVED_SCRIPT_PROFILE
30	Pashto, Arabic-script profile	58.1 million	52.0 million	40.6 million-57.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
31	Thai, Thai-script profile	55.9 million	50.1 million	39.2 million-55.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
32	Kannada, Kannada-script profile	52.1 million	46.7 million	36.5 million-51.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
33	Nigerian Pidgin, Latin-script profile	49.7 million	44.5 million	34.8 million-49.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
34	Malayalam, Malayalam-script profile	45.9 million	41.1 million	32.1 million-45.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
35	Odia, Odia-script profile	45.1 million	40.4 million	31.6 million-44.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
36	Levantine Arabic, Arabic-script profile	43.7 million	39.1 million	30.6 million-43.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
37	Punjabi, Gurmukhi-script profile	42.9 million	38.5 million	30.1 million-42.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
38	Hausa, Latin-script profile	41.9 million	37.5 million	29.3 million-41.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
39	Polish, Latin-script profile	41.5 million	37.2 million	29.1 million-41.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
40	Xiang Chinese, Simplified-script profile	41.1 million	36.8 million	28.7 million-40.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE
41	Amharic, Ethiopic-script profile	39.2 million	35.1 million	27.4 million-38.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
42	Traditional Written Chinese profile	39.1 million	35.0 million	27.4 million-38.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE
43	Algerian Arabic, Arabic-script profile	39.0 million	35.0 million	27.3 million-38.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
44	Oromo, Latin-script profile	38.3 million	34.3 million	26.8 million-37.9 million	CLDR48_RESOLVED_SCRIPT_PROFILE
45	Sindhi, Arabic-script profile	37.9 million	33.9 million	26.5 million-37.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE

Reach position	Exact edition profile	Population point	Plausibly helped point	Sensitivity range	Population basis
46	Burmese, Myanmar-script profile	37.2 million	33.3 million	26.0 million-36.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
47	Malay, Latin-script profile	36.8 million	33.0 million	25.8 million-36.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
48	Bhojpuri, Devanagari-script profile	34.6 million	31.0 million	24.2 million-34.3 million	CLDR48_RESOLVED_SCRIPT_PROFILE
49	Sundanese, Latin-script profile	33.8 million	30.3 million	23.7 million-33.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
50	Dutch, Latin-script profile	32.9 million	29.4 million	23.0 million-32.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
51	Hakka Chinese, Simplified-script profile	32.6 million	29.2 million	22.8 million-32.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
52	Moroccan Arabic, Arabic-script profile	32.5 million	29.1 million	22.8 million-32.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
53	Yoruba, Latin-script profile	31.8 million	28.5 million	22.2 million-31.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
54	Uzbek, Latin-script profile	31.6 million	28.3 million	22.1 million-31.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
55	Sudanese Arabic, Arabic-script profile	30.8 million	27.6 million	21.5 million-30.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
56	Igbo, Latin-script profile	30.8 million	27.6 million	21.5 million-30.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
57	Saraiki, Arabic-script profile	30.3 million	27.1 million	21.2 million-30.0 million	CLDR48_RESOLVED_SCRIPT_PROFILE
58	Cebuano, Latin-script profile	28.4 million	25.4 million	19.9 million-28.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
59	Awadhi, Devanagari-script profile	27.5 million	24.6 million	19.2 million-27.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
60	Min Nan Chinese, Simplified-script profile	26.9 million	24.1 million	18.8 million-26.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
61	Malagasy, Latin-script profile	26.5 million	23.7 million	18.6 million-26.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
62	Ukrainian, Cyrillic-script profile	24.1 million	21.6 million	16.9 million-23.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
63	Gan Chinese, Simplified-script profile	24.1 million	21.6 million	16.9 million-23.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
64	Bavarian, Latin-script profile	22.8 million	20.4 million	16.0 million-22.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
65	Azerbaijani, Arabic-script profile	22.5 million	20.2 million	15.8 million-22.3 million	CLDR48_RESOLVED_SCRIPT_PROFILE
66	Nepali, Devanagari-script profile	21.7 million	19.5 million	15.2 million-21.5 million	CLDR48_RESOLVED_SCRIPT_PROFILE
67	Maithili, Devanagari-script profile	20.3 million	18.2 million	14.2 million-20.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
68	Romanian, Latin-script profile	20.0 million	18.0 million	14.0 million-19.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
69	Nyanja, Latin-script profile	18.9 million	16.9 million	13.2 million-18.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE

Reach position	Exact edition profile	Population point	Plausibly helped point	Sensitivity range	Population basis
70	Somali, Latin-script profile	18.6 million	16.7 million	13.0 million-18.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
71	Assamese, Bangla-script profile	18.3 million	16.4 million	12.8 million-18.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
72	Madurese, Latin-script profile	17.7 million	15.9 million	12.4 million-17.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
73	Rangpuri, Bangla-script profile	17.0 million	15.2 million	11.9 million-16.9 million	CLDR48_RESOLVED_SCRIPT_PROFILE
74	Haryanvi, Devanagari-script profile	16.9 million	15.1 million	11.8 million-16.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE
75	Magahi, Devanagari-script profile	16.9 million	15.1 million	11.8 million-16.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE
76	Marwari, Devanagari-script profile	16.9 million	15.1 million	11.8 million-16.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE
77	Northeastern Thai, Thai-script profile	16.8 million	15.0 million	11.7 million-16.6 million	CLDR48_RESOLVED_SCRIPT_PROFILE
78	Nigerian Fulfulde, Latin-script profile	15.9 million	14.2 million	11.1 million-15.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE
79	Chhattisgarhi, Devanagari-script profile	15.5 million	13.9 million	10.9 million-15.3 million	CLDR48_RESOLVED_SCRIPT_PROFILE
80	Khmer, Khmer-script profile	15.2 million	13.6 million	10.6 million-15.0 million	CLDR48_RESOLVED_SCRIPT_PROFILE
81	Zulu, Latin-script profile	15.0 million	13.4 million	10.5 million-14.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
82	Sinhala, Sinhala-script profile	14.9 million	13.4 million	10.5 million-14.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
83	Deccan, Arabic-script profile	14.0 million	12.5 million	9.8 million-13.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
84	Shona, Latin-script profile	13.9 million	12.4 million	9.7 million-13.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
85	Akan, Latin-script profile	13.5 million	12.1 million	9.4 million-13.4 million	CLDR48_RESOLVED_SCRIPT_PROFILE
86	Min Nan Chinese, Traditional-script profile	13.5 million	12.1 million	9.4 million-13.3 million	CLDR48_RESOLVED_SCRIPT_PROFILE
87	Swedish, Latin-script profile	13.3 million	11.9 million	9.3 million-13.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
88	Czech, Latin-script profile	13.2 million	11.9 million	9.3 million-13.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
89	Wolof, Latin-script profile	13.2 million	11.8 million	9.2 million-13.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
90	Kazakh, Cyrillic-script profile	13.0 million	11.6 million	9.1 million-12.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
91	Hungarian, Latin-script profile	12.3 million	11.0 million	8.6 million-12.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
92	Greek, Greek-script profile	12.2 million	11.0 million	8.6 million-12.1 million	CLDR48_RESOLVED_SCRIPT_PROFILE
93	Low German, Latin-script profile	12.0 million	10.8 million	8.4 million-11.9 million	CLDR48_RESOLVED_SCRIPT_PROFILE
94	Kinyarwanda, Latin-script profile	12.0 million	10.7 million	8.4 million-11.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE

Reach position	Exact edition profile	Population point	Plausibly helped point	Sensitivity range	Population basis
95	Quechua, Latin-script profile	11.9 million	10.7 million	8.4 million-11.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
96	Iloko, Latin-script profile	11.4 million	10.2 million	7.9 million-11.2 million	CLDR48_RESOLVED_SCRIPT_PROFILE
97	Luba-Lulua, Latin-script profile	11.1 million	9.9 million	7.8 million-11.0 million	CLDR48_RESOLVED_SCRIPT_PROFILE
98	Tigrinya, Ethiopic-script profile	10.9 million	9.8 million	7.7 million-10.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
99	Xhosa, Latin-script profile	10.9 million	9.8 million	7.6 million-10.8 million	CLDR48_RESOLVED_SCRIPT_PROFILE
100	Tunisian Arabic, Arabic-script profile	10.8 million	9.7 million	7.6 million-10.7 million	CLDR48_RESOLVED_SCRIPT_PROFILE

Result 2: the same package divided by standardized token-event workload

This is the first quantitative compute-allocation bridge. It holds the package and nominal QA target fixed, uses a measured language/script workload where a defensible binding exists, and otherwise uses the identical language-blind envelope. Missing tokenization evidence therefore cannot remove a row. The displayed unit is potential readers per one million fresh-equivalent token events.

It is a conditional proxy, not the final efficiency result: it assumes equal incremental educational effect and equal delivery. Fresh-equivalent token events are not physical compute, time, energy, money, or quality.

Common-package proxy position	Exact edition profile	Reach position	Point readers per 1m FECU	Sensitivity range	Workload profile
1	Simplified Standard Written Chinese profile	1	858,078,700	285,579,704-1,815,176,552	zho_Hans
2	Hindi, Devanagari-script profile	2	347,929,450	121,588,218-714,941,435	hin_Deva
3	Spanish, Latin-script profile	3	330,708,224	112,523,621-685,505,384	spa_Latn
4	Arabic, Arabic-script profile	4	215,448,628	65,561,796-486,510,536	common envelope
5	French, Latin-script profile	5	213,167,471	72,947,385-445,370,541	fra_Latn
6	Urdu, Arabic-script profile	6	183,122,705	64,241,656-377,096,747	urd_Arab
7	Portuguese, Latin-script profile	8	167,824,727	56,781,781-352,028,512	por_Latn
8	Indonesian, Latin-script profile	9	165,773,561	40,669,148-346,830,957	ind_Latn
9	Bangla, Bangla-script profile	7	160,790,872	56,660,233-327,339,326	ben_Beng
10	Russian, Cyrillic-script profile	10	126,609,405	43,088,824-262,644,274	rus_Cyrl
11	Swahili, Latin-script profile	11	110,400,876	33,595,385-249,299,286	common envelope
12	Punjabi, Shahmukhi/Perso-Arabic-script profile	12	100,477,258	30,575,593-226,890,489	common envelope
13	German, Latin-script profile	13	92,656,399	31,610,086-191,930,951	deu_Latn

Common-package proxy position	Exact edition profile	Reach position	Point readers per 1m FECU	Sensitivity range	Workload profile
14	Japanese, Japanese-script profile	14	68,675,694	23,998,228-140,625,240	jpn_Jpan
15	Javanese, Latin-script profile	18	61,541,346	21,007,940-128,074,730	jav_Latn
16	Western Panjabi, Arabic-script profile	16	57,435,613	17,477,865-129,696,954	common envelope
17	Vietnamese, Latin-script profile	19	56,859,648	19,670,421-116,635,627	vie_Latn
18	Marathi, Devanagari-script profile	17	54,856,862	19,524,561-110,944,071	mar_Deva
19	Telugu, Telugu-script profile	15	54,715,793	19,426,504-110,731,661	tel_Telu
20	Turkish, Latin-script profile	23	51,706,068	17,708,902-107,688,320	tur_Latn
21	Persian, Arabic-script profile	21	50,739,748	15,440,289-114,576,836	common envelope
22	Wu Chinese, Simplified-script profile	22	48,324,693	14,705,379-109,123,332	common envelope
23	Tamil, Tamil-script profile	20	48,188,313	17,300,503-97,178,539	tam_Taml
24	Korean, Korean-script profile	24	45,091,943	13,721,641-101,823,369	common envelope
25	Egyptian Arabic, Arabic-script profile	27	45,042,843	15,378,427-93,722,097	arz_Arab
26	Italian, Latin-script profile	28	43,669,442	15,129,536-89,714,453	ita_Latn
27	Cantonese, Simplified-script profile	25	41,881,400	12,744,662-94,573,554	common envelope
28	Filipino, Latin-script profile	26	41,633,549	12,669,239-94,013,874	common envelope
29	Gujarati, Gujarati-script profile	29	36,827,395	13,036,851-75,249,249	guj_Gujr
30	Pashto, Arabic-script profile	30	33,024,855	10,049,583-74,574,342	common envelope
31	Thai, Thai-script profile	31	29,876,777	10,801,084-61,115,876	tha_Thai
32	Nigerian Pidgin, Latin-script profile	33	28,277,832	8,605,046-63,854,958	common envelope
33	Levantine Arabic, Arabic-script profile	36	28,193,586	9,626,392-58,924,982	apc_Arab
34	Kannada, Kannada-script profile	32	27,796,911	9,998,822-55,339,835	kan_Knda
35	Odia, Odia-script profile	35	25,647,402	7,804,597-57,915,110	common envelope
36	Hausa, Latin-script profile	38	25,283,771	8,743,942-52,009,619	hau_Latn
37	Traditional Written Chinese profile	42	24,624,002	8,388,300-51,327,781	zho_Hant
38	Malayalam, Malayalam-script profile	34	24,579,283	8,785,283-49,969,023	mal_Mlym

Common-package proxy position	Exact edition profile	Reach position	Point readers per 1m FECU	Sensitivity range	Workload profile
39	Polish, Latin-script profile	39	24,379,279	8,531,559-49,624,459	pol_Latn
40	Xiang Chinese, Simplified-script profile	40	23,356,935	7,107,600-52,742,943	common envelope
41	Algerian Arabic, Arabic-script profile	43	22,198,628	6,755,123-50,127,339	common envelope
42	Dutch, Latin-script profile	50	21,933,157	7,398,669-46,278,700	nld_Latn
43	Oromo, Latin-script profile	44	21,763,944	6,622,847-49,145,766	common envelope
44	Sindhi, Arabic-script profile	45	21,759,491	7,691,771-44,210,075	snd_Arab
45	Sundanese, Latin-script profile	49	21,521,599	7,400,475-44,167,883	sun_Latn
46	Malay, Latin-script profile	47	20,929,961	6,369,063-47,262,527	common envelope
47	Moroccan Arabic, Arabic-script profile	52	20,063,920	6,931,069-41,979,056	ary_Arab
48	Bhojpuri, Devanagari-script profile	48	19,552,816	6,875,905-39,921,220	bho_Deva
49	Punjabi, Gurmukhi-script profile	37	19,393,058	7,214,028-38,398,074	pan_Guru
50	Hakka Chinese, Simplified-script profile	51	18,524,465	5,637,062-41,830,611	common envelope
51	Uzbek, Latin-script profile	54	17,951,469	5,462,697-40,536,711	common envelope
52	Igbo, Latin-script profile	56	17,945,322	6,267,833-37,339,167	ibo_Latn
53	Sudanese Arabic, Arabic-script profile	55	17,509,842	5,328,308-39,539,461	common envelope
54	Saraiki, Arabic-script profile	57	17,224,673	5,241,530-38,895,512	common envelope
55	Cebuano, Latin-script profile	58	16,489,398	5,792,931-33,947,185	ceb_Latn
56	Yoruba, Latin-script profile	53	16,223,657	5,868,853-32,896,594	yor_Latn
57	Awadhi, Devanagari-script profile	59	15,804,968	5,490,259-32,657,733	awa_Deva
58	Min Nan Chinese, Simplified-script profile	60	15,302,819	4,656,703-34,555,721	common envelope
59	Malagasy, Latin-script profile	61	15,076,827	4,587,933-34,045,403	common envelope
60	Burmese, Myanmar-script profile	46	14,876,938	5,746,932-28,802,470	mya_Mymr
61	Gan Chinese, Simplified-script profile	63	13,691,997	4,166,524-30,918,277	common envelope
62	Ukrainian, Cyrillic-script profile	62	13,681,980	4,837,925-27,946,199	ukr_Cyrl

Common-package proxy position	Exact edition profile	Reach position	Point readers per 1m FECU	Sensitivity range	Workload profile
63	Bavarian, Latin-script profile	64	12,979,402	3,949,679-29,309,150	common envelope
64	Azerbaijani, Arabic-script profile	65	12,807,237	3,897,289-28,920,378	common envelope
65	Nepali, Devanagari-script profile	66	12,362,580	3,761,978-27,916,285	common envelope
66	Romanian, Latin-script profile	68	11,796,814	4,138,069-23,954,389	ron_Latn
67	Maithili, Devanagari-script profile	67	11,522,314	4,061,376-23,809,199	mai_Deva
68	Nyanja, Latin-script profile	69	11,166,916	3,896,672-23,170,985	nya_Latn
69	Somali, Latin-script profile	70	10,667,282	3,767,168-21,845,790	som_Latn
70	Amharic, Ethiopic-script profile	41	10,139,854	4,192,238-18,900,993	amh_Ethi
71	Madurese, Latin-script profile	72	10,089,205	3,070,181-22,782,713	common envelope
72	Magahi, Devanagari-script profile	75	9,701,929	3,419,911-19,663,746	mag_Deva
73	Assamese, Bangla-script profile	71	9,698,578	3,498,573-19,312,565	asm_Beng
74	Rangpuri, Bangla-script profile	73	9,683,187	2,946,629-21,865,874	common envelope
75	Haryanvi, Devanagari-script profile	74	9,617,776	2,926,724-21,718,166	common envelope
76	Marwari, Devanagari-script profile	76	9,617,776	2,926,724-21,718,166	common envelope
77	Northeastern Thai, Thai-script profile	77	9,544,676	2,904,479-21,553,099	common envelope
78	Nigerian Fulfulde, Latin-script profile	78	9,427,287	3,267,255-19,652,445	fuv_Latn
79	Chhattisgarhi, Devanagari-script profile	79	8,859,563	3,131,912-18,126,548	hne_Deva
80	Swedish, Latin-script profile	87	8,595,279	2,935,387-17,821,088	swe_Latn
81	Zulu, Latin-script profile	81	8,577,486	3,042,718-17,364,401	zul_Latn
82	Shona, Latin-script profile	84	8,007,978	2,807,045-16,325,172	sna_Latn
83	Deccan, Arabic-script profile	83	7,934,665	2,414,547-17,917,487	common envelope
84	Czech, Latin-script profile	88	7,876,222	2,754,709-16,143,409	ces_Latn
85	Kazakh, Cyrillic-script profile	90	7,743,142	2,709,332-16,243,517	kaz_Cyrl
86	Min Nan Chinese, Traditional-script profile	86	7,663,232	2,331,949-17,304,556	common envelope

Common-package proxy position	Exact edition profile	Reach position	Point readers per 1m FECU	Sensitivity range	Workload profile
87	Wolof, Latin-script profile	89	7,539,675	2,663,154-15,263,071	wol_Latn
88	Akan, Latin-script profile	85	6,968,724	2,450,267-14,448,773	aka_Latn
89	Hungarian, Latin-script profile	91	6,904,214	2,447,869-14,000,203	hun_Latn
90	Kinyarwanda, Latin-script profile	94	6,871,180	2,422,456-14,031,008	kin_Latn
91	Low German, Latin-script profile	93	6,853,344	2,085,497-15,475,727	common envelope
92	Tunisian Arabic, Arabic-script profile	100	6,825,498	2,309,719-14,301,733	aeb_Arab
93	Quechua, Latin-script profile	95	6,792,664	2,067,032-15,338,703	common envelope
94	Sinhala, Sinhala-script profile	82	6,631,827	2,501,608-12,873,484	sin_Sinh
95	Xhosa, Latin-script profile	99	6,406,871	2,245,955-13,115,068	xho_Latn
96	Luba-Lulua, Latin-script profile	97	6,389,488	2,206,449-13,217,364	lua_Latn
97	Iloko, Latin-script profile	96	6,276,195	2,239,045-12,720,226	ilo_Latn
98	Afrikaans, Latin-script profile	105	6,259,618	2,160,727-13,013,521	afr_Latn
99	Greek, Greek-script profile	92	6,236,033	2,268,502-12,345,370	ell_Grek
100	Umbundu, Latin-script profile	101	6,058,396	2,139,308-12,437,394	umb_Latn

Why the two orders can differ

Potential reach is the numerator. The common-package proxy divides it by an estimated end-to-end token-event workload for the same package. Script and tokenizer representation can change the workload estimate, but no claim is made that token counts capture all production difficulty. In the final model, a target with smaller potential reach can lead when a specific package produces a larger additional functional effect, costs less to produce and deliver, or uniquely reaches people not served by another edition.

What remains before a final allocation claim

For each intervention the model still needs a named educational endpoint, an independently usable baseline, a defensible incremental effect, exposure and delivery pathways, accessibility costs, measured physical compute, reusable-work accounting, and beneficiary overlap. The correct final object is portfolio-marginal additional functional access per standardized physical compute, with uncertainty and reversal thresholds, not the nonordinal profile catalogue or this common-package proxy alone.

English academic-to-non-academic register translation remains a coequal but analytically separate intervention because its production process and audience nonoverlap cannot be inferred from ordinary English fluency. Signed-language, accessibility-format, displaced-community, and evidence-supported interlanguage interventions likewise remain possible outrankers outside this spoken/written CLDR table.

Source and nonadditivity cautions

CLDR 48 is an open locale-planning source. Its language-population estimates represent people fluent enough to use a computer or smartphone interface and include L1 and L2 users. Resolving likely scripts improves edition specificity but does not prove literacy in that script. Multilingual people, macrolanguages and constituent languages, and alternative scripts can overlap; these rows cannot be summed into a world total. The direct Indonesian population context measures communicative ability, not independent academic reading. All such distinctions are preserved in the machine-readable evidence.

Appendix A - Full-content academic-register translation

Full-content academic-register translation: evidence and production method

Status: governing research-and-production method, version 1. This is not a proposal to reduce intellectual content and not a test of whether readers deserve access.

Executive conclusion

The programme should treat academic-to-nonacademic English as full-content diaphasic intralingual translation: translation across registers within English, for a specified adult audience, while keeping the entire intellectual object recoverable. The target can be longer and more explicit than the source. It may define, unpack, split, reorder with tracked dependencies, resolve references, and add labeled scaffolding. It may not summarize, abridge, generalize, silently strengthen or weaken, delete, infantilize, or substitute intuition for proof.

The governing premise is a sufficiently plausible access thesis, not a hypothesis that the literature may veto: inherited academic-register familiarity is one institutionally distributed access advantage, and any credible nonzero educational benefit makes a source-faithful register edition eligible. Research determines target profiles, fidelity safeguards, measurement, and revision. It does not turn correlations between schooling, wealth, test performance, language exposure, or attainment into claims of innate capacity, and it cannot use such claims to withhold translation.

What is empirical and what is a governing choice

Several empirical literatures support different links: parent education and socioeconomic position are associated with unequal educational trajectories; institutional rules and academic discourse are often tacit; academic-language knowledge predicts comprehension in several school samples; jargon and document design can causally affect processing or understanding; and making hidden institutional knowledge explicit can improve some outcomes. No identified study tests the entire chain from parental education through academic-register translation to long-run attainment. That gap limits causal claims and determines the evaluation agenda. It does not cancel the production commitment.

The method therefore uses four distinct statements:

- Normative and operational premise: readers are presumed intellectually capable; inherited register fit is not a deservingness test.
- Evidence-supported mechanism: unfamiliar academic forms can add processing, navigation, identity, and institutional burdens.
- Design thesis: a complete, optional, audience-specific register translation can remove part of that burden without removing content.
- Empirical unknown: the size and distribution of learning, persistence, autonomy, or research benefits must be measured by target and task.

Why this is translation rather than simplification

Jakobson's category of intralingual translation supplies the broad foundation. Diaphasic translation names movement between situation-linked registers. A target register is chosen by purpose and audience, but this project adds a stronger invariant than ordinary simplification or popularization: every source proposition and dependency must remain recoverable, and every target assertion must be source-supported or explicitly labeled as an addition.

Ordinary simplification corpora are unsafe as the governing objective. In one major document-level dataset, deletion represented 44% of sampled operations. Factuality studies find omitted and unsupported information that surface metrics miss. The project may borrow splitting, explicitness, anaphora resolution, navigation, and reordering only under claim-level and dependency constraints; deletion is disabled.

There is no single Plain English

English is pluricentric. Academic genres vary by field, institution, country, professional community, educational history, and medium. Word difficulty also varies with first language, proficiency, and reading experience. A single universal Plain English would reproduce the same flattening that the intervention is intended to avoid. Each edition needs a declared adult target profile.

Target profile	Intended users	Design consequences
International adult / ELF-friendly	Readers who use English across language backgrounds	Shorter dependency spans, explicit logical relations, controlled idiom, technical anchors with glosses; no imitation of one metropolitan accent or grammar as a condition of correctness.
Educated non-specialist adult	Readers with general education but not the discipline's institutional register	Define field terms, unpack nominalizations, make agents and relations explicit, add labeled examples, preserve all propositions and evidence.
Profession-adjacent novice	Readers entering a field from work or another discipline	Retain professional terminology with first-use glosses, expose prerequisites, preserve genre and citation structure.
Localized adult English	Readers using a legitimate local or regional English repertoire	Commission locale-specific lexical, pragmatic, orthographic, and example choices without treating metropolitan standards as intrinsically superior.
Academic-convention companion	Readers who want both the content and the prestige register used by institutions	Parallel display of full-content register translation and original academic formulation, with reversible term and phrase mappings.
Accessible-language edition	Readers requiring Easy Language, Easy Read, audio, signed, screen-reader-native, or cognitive-access formats	Separate specification and production path. It may share the content ledger but must not be collapsed into the general adult-register edition.

The accessible-language edition is not a euphemism for the general adult translation. Easy Language, Easy Read, signed, audio, and cognitive-access editions have distinct audiences, standards, and multimodal requirements. They remain fully eligible and can share the same source ledger.

Evidence synthesis

Evidence types are kept separate. Theoretical sources identify mechanisms and audit questions. Qualitative studies show possible institutional processes. Observational studies quantify associations. Experiments support only their tested contrast. Standards guide design but do not prove outcomes.

Intergenerational And Institutional Access

REG-001 - Cataldi, Bennett, and Chen (2018), NCES 2018-421. nationally representative descriptive longitudinal evidence.

- Scope: Three United States longitudinal programmes: ELS:2002 (more than 15,000 high-school sophomores), BPS:04/09 (nearly 17,000 beginning postsecondary students), and B&B:08/12 (more than 17,000 bachelor's recipients).
- Finding: Postsecondary enrolment by 2012 was 72% where parents had never attended college, 84% where parents had some college, and 93% where a parent held at least a bachelor's degree. Six-year degree attainment or continued enrolment in BPS was 55.8%, 63.4%, and 74.5%, respectively.
- Use here: Establishes a large parent-education gradient that an access programme must take seriously.
- Limit: It does not identify academic register as the cause and cannot separate finances, prior schooling, institutional sector, employment, family obligations, discrimination, or other co-varying conditions.

REG-002 - Sirin (2005). meta-analysis of correlational studies.

- Scope: Fifty-eight United States K-12 articles, 101,157 students, 6,871 schools, and 128 districts; the article reports 74 or 75 independent samples in different sections.
- Finding: The student-level random-effects association between socioeconomic status and achievement was r=.27, 95% CI [.23, .30]. Parental education specifically had k=30 and a reported mean association of r=.30.
- Use here: Shows that socially distributed resources and educational attainment are empirically associated.
- Limit: Association is not innate ability and is not a causal estimate of register, schooling, or parent education.

REG-003 - Holmlund, Lindahl, and Plug (2011). causal-evidence review plus Swedish-register comparisons.

- Scope: Conventional, twin, adoption, and compulsory-schooling instrumental-variable designs; main Swedish samples included 336,083 mother-child and 269,648 father-child observations.
- Finding: Twin, adoption, and instrumental-variable estimates were generally smaller and less stable than ordinary regressions.
- Use here: Prevents the framework from treating an intergenerational schooling association as biological or deterministic transmission of capacity.
- Limit: The designs do not isolate academic-register translation and do not imply that parental education is irrelevant.

REG-004 - Ives and Castillo-Montoya (2020). systematic review.

- Scope: Fifty-nine United States higher-education publications concerning first-generation students.
- Finding: Thirty articles primarily framed first-generation learners against conventional performance norms, while only seven used experimental or quasi-experimental methods; a smaller literature identified assets from lived experience.
- Use here: Requires an anti-deficit interpretation and explicit separation of institutional fit from intellectual capability.
- Limit: The literature is heterogeneous and does not estimate the effect of register translation.

REG-005 - Collier and Morgan (2008). qualitative focus-group study.

- Scope: One urban public university, 15 faculty members and 63 students, including six first-generation and two continuing-generation student groups.

- Finding: Both groups wanted clearer expectations; first-generation discussions particularly described unfamiliarity with syllabus use, office hours, citation and writing formats, disciplinary expectations, and lecture terminology.
- Use here: Supports the existence of unequally distributed institutional discourse knowledge.
- Limit: It supplies themes, not prevalence, causal effects, or a representative population estimate.

REG-006 - Yee (2016). two-year ethnography.

- Scope: Thirty-four students at an urban public comprehensive university and more than 800 hours of observation and interviews.
- Finding: Both groups worked hard, but continuing-generation students had a broader repertoire of institutionally rewarded interactive strategies, while first-generation students more often relied on strenuous independent strategies.
- Use here: Treats hidden rules and institutional valuation as access conditions rather than student defects.
- Limit: This is not a register-translation study and does not estimate population prevalence.

Academic Register And Comprehension

REG-007 - Preece (2015). ethnographic case study.

- Scope: A compulsory academic-writing cohort of 93 first-year students at a London university, with repeated interviews of eight focal participants.
- Finding: Participants described academic English as unfamiliar and socially marked and experienced remedial positioning as erasing bilingual and bidialectal resources.
- Use here: Directly motivates optional register translation that respects adult identities and existing repertoires.
- Limit: The study is contextual, not a trial of full-content translation.

REG-008 - Uccelli et al. (2019). small longitudinal observational study.

- Scope: Forty-two English-speaking parent-child dyads observed at 30 months and assessed in Grade 7.
- Finding: Socioeconomic status explained 31% of Grade-7 academic-language variance; parent and child decontextualized talk added variance in sequential models, with child talk remaining significant in the final model.
- Use here: Supports academic-discourse familiarity as learned through differential participation and exposure.
- Limit: The sample is small, observational, and does not establish that parental education causes later register knowledge.

REG-009 - Volodina and Weinert (2020). longitudinal cohort-sequential observational study.

- Scope: N=627 learners in German Grades 2-4, including 361 language-minority learners.
- Finding: Socioeconomic status predicted initial comprehension and growth for academic connectives; these paths became nonsignificant after receptive-language measures entered the model.
- Use here: Treats connectives and explicit logical relations as learned access tools that a translation can preserve and unpack.
- Limit: Statistical mediation is not causal proof and child German-school evidence does not directly estimate adult tertiary effects.

REG-010 - Uccelli et al. (2015). cross-sectional comprehension study.

- Scope: N=218 United States students in Grades 4-6; 65% qualified for subsidized meals and half were current or former English learners.
- Finding: Cross-disciplinary academic-language skills added 12% unique variance to reading comprehension after socioeconomic status, word-reading fluency, and academic vocabulary; the final model reported R-squared=.59.
- Use here: Shows that register knowledge can matter beyond vocabulary and decoding.
- Limit: It is neither adult evidence nor a causal estimate of translation benefit.

Institutional Interventions

REG-011 - Stephens et al. (2012). observational studies plus randomized framing experiments.

- Scope: Four United States studies, including 261 administrators from 60 institutions, a student cohort of 1,424, and randomized experiments with N=88 and N=147.
- Finding: First-generation participants performed worse after an independence-framed welcome message, while no gap appeared after an interdependence frame; first-generation performance improved under the interdependence frame.
- Use here: Demonstrates that institution-created framing can causally affect immediate performance.
- Limit: The outcome is short-term task performance, not document translation or attainment.

REG-012 - Stephens, Hamedani, and Destin (2014). randomized controlled trial.

- Scope: N=168 incoming students at one private United States university.
- Finding: A one-hour difference-education panel made background-linked challenges, strengths, and navigation strategies explicit. The control first-generation/continuing-generation GPA gap was .30 grade points; no significant gap appeared in the intervention condition. The first-generation intervention-control contrast was d=.70.
- Use here: Supports making institutionally implicit knowledge explicit and rejects a fixed-capacity account.
- Limit: The intervention did not isolate language or translate academic content.

Language Interventions And Comprehension

REG-013 - Bullock et al. (2019). online randomized language experiment.

- Scope: N=650 participants reading messages about emerging technologies.
- Finding: Jargon reduced reported processing fluency, which statistically mediated greater resistance, higher perceived risk, and lower support.
- Use here: Treats jargon as a manipulable access burden.
- Limit: The main outcomes were metacognitive and persuasive, not objective educational comprehension.

REG-014 - Santesso et al. (2015). randomized controlled trial.

- Scope: N=143 members of the public across Canada, Norway, Spain, Argentina, and Italy.
- Finding: A redesigned plain-language Cochrane summary yielded correct understanding of benefits, harms, and evidence quality in 53% versus 18% under the existing format; median correct answers were 3 versus 1.
- Use here: Provides objective evidence that a clear-language document package can improve understanding.

- Limit: The intervention combined wording, organization, absolute numbers, and evidence-quality presentation and concerned summaries rather than full courses.

REG-015 - Kim and Kim (2015). randomized controlled trial.

- Scope: N=150 participants comparing standard and simplified cancer-trial consent documents.
- Finding: The redesigned document produced higher objective and subjective understanding.
- Use here: Supports full-document redesign as a plausible comprehension intervention.
- Limit: The intervention mixed plain wording, short sentences, diagrams, pictures, and bullets and was not an academic-course translation.

REG-016 - August et al. (2024). three randomized within-participant online studies.

- Scope: Three studies with N=199, N=191, and N=203 readers of scientific summaries.
- Finding: In the information-preserving study, lower-complexity versions retained 97.7% of manually coded information units. Readers with the lowest topic familiarity reported better ease outcomes, while higher-familiarity readers sometimes skipped more in overly plain versions.
- Use here: Supports audience-specific parallel versions and information-unit accounting.
- Limit: Outcomes were largely subjective and the sources were summaries, not complete courses or proofs.

REG-017 - August et al. (2023), Paper Plain. formative think-aloud and small usability study.

- Scope: Think-aloud N=12 and partial within-participant usability N=24.
- Finding: Question-guided navigation, definitions, and section-level gists beside the original increased perceived ease and confidence; multiple-choice comprehension was not detectably different.
- Use here: Supports a parallel source-plus-translation interface rather than replacement.
- Limit: The sample was small and the intervention was multicomponent.

REG-018 - Stoll et al. (2022). systematic review.

- Scope: 7,714 database records plus 135 additional records; 90 reports included, 33 empirical.
- Finding: Only eight empirical studies directly compared alternative plain-language-summary forms and only six used experiments; heterogeneity prevented firm criteria.
- Use here: Defines the evidence gap and prevents overclaiming certainty.
- Limit: The review concerns brief summaries, not full-content register translation.

Translation Studies And Register Theory

REG-019 - Jakobson (1959). foundational translation theory.

- Scope: Conceptual distinction among intralingual, interlingual, and intersemiotic translation.
- Finding: Intralingual translation is rewording by means of signs in the same language.
- Use here: Establishes that academic-to-adult-register work is translation, not merely editing.
- Limit: The source does not supply a modern production or fidelity standard.

REG-020 - Zethsen (2009). translation-studies synthesis.

- Scope: Descriptive model of intralingual translation using knowledge, time, culture, and space as parameters.
- Finding: Intralingual translation often uses extensive addition, omission, restructuring, and simplification.

- Use here: Supplies a descriptive map of transformations.
- Limit: Ordinary practice permits omission; this project tightens the model by prohibiting loss of source content.

REG-021 - Zethsen and Hill-Madsen (2016). translation-theory synthesis.

- Scope: Conceptual account of source existence, target derivation, and skopos-dependent relevant similarity.
- Finding: Intralingual products can count as translation when a target derives from a source for a communicative purpose.
- Use here: Supports a register-specific skopos while the project adds a stricter completeness invariant.
- Limit: Relevant similarity alone is too permissive for proofs and high-stakes academic claims.

REG-022 - Hill-Madsen (2015; 2024). register-translation analysis and quality-assessment framework.

- Scope: Diaphasic intralingual translation across expert and non-expert registers.
- Finding: Registerial adequacy can be assessed through transparency of meaning, including iconicity, familiar lexis, explicitness, groundedness, and coherence.
- Use here: Supplies the positive target-side criteria used after fidelity passes.
- Limit: A register score cannot compensate for lost content or changed logic.

Plain-Language Standards

REG-023 - ISO 24495-1:2023. international standard.

- Scope: Governing principles and guidelines for plain-language documents, including technical writing and controlled languages.
- Finding: Plain language is audience- and purpose-oriented and may apply to technical communication.
- Use here: Supports findability, understandability, and usability as target-side dimensions.
- Limit: The standard does not by itself prove semantic equivalence or cover all accessibility formats.

REG-024 - CAN-ASC-3.1:2025. national accessibility standard.

- Scope: Audience-centred plain-language standard covering relevance, findability, understanding, and use.
- Finding: Plain language varies by audience, topic, format, and context, and multiple audiences can require separate communications.
- Use here: Rules out a single universal Plain English target.
- Limit: It is a design standard, not an evaluation of full mathematical translation.

World Englishes And Lingua-Franca Use

REG-025 - Kachru (2001). sociolinguistic synthesis.

- Scope: Global histories, functions, norms, and identities of World Englishes.
- Finding: English is pluricentric and used through multiple institutional and local norms rather than one invariant native-speaker standard.
- Use here: Requires locally and functionally specified adult English targets.
- Limit: The framework does not quantify academic-source mismatch for any population.

REG-026 - Jenkins, Cogo, and Dewey (2011). research review.

- Scope: Developments in research on English as a lingua franca.
- Finding: Lingua-franca English is variable, adaptive, and not adequately evaluated only against metropolitan native norms.
- Use here: Separates communicative success from prestige-standard conformity.
- Limit: The review does not identify one ready-made register for educational translation.

REG-027 - Gooding and Tragut (2022). computational and reader-difference study.

- Scope: Lexical complexity across readers with differing first languages, proficiency, and reading experience.
- Finding: Word difficulty is not invariant across readers; first language, proficiency, and exposure affect lexical accessibility.
- Use here: Requires more than one adult English register and audience-specific terminology support.
- Limit: Lexical difficulty alone is not full-document comprehension or mathematical fidelity.

Meaning-Preservation Risk

REG-028 - Sun, Jin, and Wan (2021). document-level simplification corpus and modelling study.

- Scope: D-Wikipedia document pairs and baseline systems.
- Finding: Sentence deletion accounted for 44% of sampled document-level simplification operations; addition, splitting, joining, anaphora resolution, and reordering also occurred.
- Use here: Shows why ordinary simplification objectives are unsafe for a no-content-loss brief.
- Limit: Wikipedia simplification is not academic translation and the operation frequencies are descriptive, not normative.

REG-029 - Sulem, Abend, and Rappoport (2018). metric-evaluation study.

- Scope: Sentence simplification with splitting and human judgments.
- Finding: BLEU had low or no correlation with human grammaticality and meaning-preservation judgments in relevant settings and could correlate negatively with simplicity.
- Use here: Prevents a surface metric from certifying fidelity.
- Limit: The result does not imply that every automatic diagnostic is useless.

REG-030 - Devaraj et al. (2022). factuality evaluation of text simplification.

- Scope: Reference and system simplifications with factual-error analysis.
- Finding: Both references and system outputs contained factual errors that common metrics missed, including deletion and unsupported insertion.
- Use here: Requires claim-level source alignment and support checks.
- Limit: The study does not validate a complete replacement metric.

REG-031 - Agrawal and Carpuat (2024). reading-comprehension evaluation of simplification systems.

- Scope: Question-answerability tests of supervised and other simplification systems.
- Finding: Even the strongest supervised system left at least 14% of comprehension questions unanswerable from the transformed text.
- Use here: Supports zero-tolerance checking of protected content rather than assuming readability implies completeness.

- Limit: Question answerability is an alarm and does not prove all remaining propositions are correct.

REG-032 - Joseph et al. (2024), FactPICO. expert-annotated factuality benchmark.

- Scope: 345 plain-language summaries of randomized-trial abstracts generated by three language models.
- Finding: Existing factuality metrics correlated poorly with expert judgments at the individual-summary level, including correctness of added explanations.
- Use here: Requires structured proposition and addition provenance rather than metric-only release.
- Limit: The benchmark concerns medical abstracts and summaries, not whole academic works.

Mathematical Discourse

REG-033 - Schleppegrell (2007). mathematics-language synthesis.

- Scope: Linguistic challenges of mathematics teaching and learning.
- Finding: Dense noun phrases, relational processes, conjunctions, and implicit logical relations carry mathematical meaning; everyday talk and technical register serve different functions.
- Use here: Makes logical connectives, relations, and technical anchors protected content.
- Limit: The paper does not specify an automated translation pipeline.

REG-034 - Ganesalingam (2013). linguistic and computational monograph.

- Scope: The language and formal interpretation of mathematical texts.
- Finding: Mathematical prose depends on ambiguity resolution, typed objects, notation, and compositional relations between linguistic and symbolic material.
- Use here: Requires notation/type preservation and document-level mathematical parsing.
- Limit: It is not an outcome evaluation of plain-language mathematics.

English-Medium Education And Research

REG-035 - Macaro et al. (2018). systematic review.

- Scope: An in-depth review of 83 higher-education EMI studies within a wider mapped literature.
- Finding: The evidence available then was insufficient to assert that EMI benefited language learning or was clearly detrimental to content learning; stakeholders expressed serious implementation concerns.
- Use here: Prevents automatic credit to English-medium provision as complete educational access.
- Limit: The 2018 evidence base was geographically uneven and has since expanded.

REG-036 - Curle et al. (2026 update of Macaro et al.). updated systematic review.

- Scope: 196 empirical EMI studies published from 2016 through 2023.
- Finding: The update reports positive receptive-language and vocabulary outcomes, mixed productive-language outcomes, no general pattern of worse content attainment, and persistent academic-writing and transition challenges; Asia and Europe remain most studied.
- Use here: Updates the evidence without equating EMI availability with comfortable independent academic access.
- Limit: Heterogeneity and regional imbalance prevent a universal person-level academic-access rate.

REG-037 - Amano et al. (2023). online self-report survey.

- Scope: 908 published environmental scientists from eight nationalities stratified by national income and English-proficiency context.
- Finding: Compared with native-English peers, moderate- and low-English-proficiency nationality groups reported substantially more reading and writing time and more English-related rejection.
- Use here: Makes research-stage reading, writing, and dissemination burdens visible.
- Limit: Nationality categories are not individual proficiency, the data are self-reported, and the sample does not represent all disciplines.

Critical And Anti-Hegemony Evidence

REG-038 - Bourdieu (1977). sociological theory.

- Scope: Linguistic exchanges, social markets, legitimate discourse, and symbolic power.
- Finding: The social value of linguistic forms depends on institutions and speakers' positions rather than intrinsic linguistic merit.
- Use here: Forbids treating academic-register possession as neutral evidence of lower need or greater capability.
- Limit: The theory does not supply a numerical allocation coefficient.

REG-039 - Pennycook (1997). critical EAP analysis.

- Scope: Critique of neutrality and pragmatism in English for Academic Purposes.
- Finding: Presenting academic conventions as neutral can naturalize and reproduce institutional power.
- Use here: Frames academic register as situated and teachable, not the sole legitimate form of thought.
- Limit: This is conceptual critique, not an educational-effect experiment.

REG-040 - Lea and Street (1998). qualitative academic-literacies case study.

- Scope: Twenty-three staff and 47 students at two English universities using interviews, observation, assignments, feedback, and guidance documents.
- Finding: Judgments of good academic writing depended on tacit disciplinary epistemologies, identities, and institutional relations, not only generic skills.
- Use here: Requires discipline-specific translation and separates knowledge from conformity to tacit conventions.
- Limit: The authors did not claim statistical representativeness.

REG-041 - Flores and Rosa (2015). raciolinguistic theoretical synthesis.

- Scope: Educational categories of appropriateness and perceptions of racialized language.
- Finding: Racialized speakers can be perceived as deficient even when approximating supposedly appropriate forms; the listener and institution are part of the judgment.
- Use here: Requires matched-meaning variety audits and prohibits standard-language conformity from serving as semantic QA.
- Limit: The source does not estimate the effect of a translation programme.

REG-042 - Hofmann et al. (2024). matched-guise probing experiments on language models.

- Scope: GPT-2, RoBERTa, T5, GPT-3.5, and GPT-4 tested with matched African American English and Standardized American English prompts.

- Finding: Dialect features alone elicited different stereotypes and hypothetical occupational and criminal decisions; model scaling and human-feedback training did not remove the covert dialect prejudice measured.
- Use here: Requires dialect-counterfactual invariance in AI allocation and translation QA.
- Limit: The constructed decision tasks do not estimate educational translation outcomes.

Anti-hegemony implementation

The analysis treats academic-register familiarity as linguistic capital unequally distributed through institutions and social histories. It does not reward a target because its users already conform to a prestige register, have more money, attend better-funded schools, generate more web data, live in a wealthier state, or use a colonial or globally dominant language. Nor does it punish a target for lacking those advantages.

A register translation must preserve the reader's legitimate linguistic repertoire rather than silently normalize it. Matched-meaning inputs that differ only by racialized, classed, local, or non-metropolitan English features must not produce different judgments of intellectual ability, educational need, translation eligibility, or source fidelity. Surface convention is assessed only against the declared target brief and must remain separate from truth and reasoning.

Alternative access through another language or register counts only when the exact material is technically usable, linguistically intelligible, materially equivalent, voluntarily and politically acceptable, disability-accessible, affordable, discoverable, lawful, safe, and trustworthy. Official status, compulsory exposure, conversational fluency, or a bilingual label cannot establish those conditions.

Protected content

Object	Mandatory invariant
Atomic propositions	Every source claim maps to one or more target locations; every target assertion maps to source support or a labeled explanatory source.
Definitions	Preserve term, category, distinguishing conditions, scope, directionality, and every if-and-only-if relation.
Quantifiers and negation	Preserve all, every, any, no, some, exists, at least, at most, exactly, only, unless, except, and their scope.
Numbers and units	Preserve values, signs, ranges, inequalities, endpoints, precision, uncertainty, denominators, dates, sample sizes, and units.
Terminology	Retain the exact technical anchor on first use, add a plain gloss, and never collapse distinct technical co-hyponyms into one vague word.
Notation and types	Preserve symbols, case, subscripts, superscripts, primes, binding, domains, codomains, object types, and equation references.
Theorems and proofs	Preserve hypotheses, constructions, witnesses, cited lemmas, cases, inference types, equality chains, contradiction assumptions, and conclusions.
Cross-references	Resolve every section, figure, table, equation, theorem, note, and citation link after restructuring.
Coreference and cohesion	Resolve ambiguous pronouns and ellipsis; preserve rhetorical relations such as cause, contrast, concession, condition, and conclusion.
Epistemic and deontic force	Preserve may, suggests, is consistent with, proves, should, must, confidence, limitations, and conjectural status.
Prerequisites and assumptions	Inventory and introduce global and local assumptions before use without moving them outside their logical scope.

Object	Mandatory invariant
Additions	Label each gloss, example, intuition, background note, or translator explanation and record its source.
Multimodal relations	Keep formulas, diagrams, captions, tables, axes, accessibility descriptions, and surrounding prose mutually consistent.

Production workflow

1. Freeze the source identity, version, date, byte count, and SHA-256.
2. Assign stable identifiers to sections, paragraphs, equations, theorems, figures, tables, notes, and citations.
3. Build an atomic content ledger covering claims, definitions, assumptions, quantifiers, numbers, exceptions, examples, evidence status, and attribution.
4. Build a dependency graph for prerequisites, definitions, theorems, proof steps, cross-references, symbols, and coreference chains.
5. Define one adult target-register profile per deliverable: audience, purpose, field knowledge, English repertoire, locality, medium, and accessibility requirements.
6. State the positive translation purpose and the no-loss invariants.
7. Create a termbase with exact term, plain gloss, first-use treatment, later short form, prohibited flattenings, and notation.
8. Translate coherent sections with the whole document graph available; do not process isolated sentences without context.
9. Prefer expansion and explicitness: unpack nominalizations, restore agents and omitted subjects, split overloaded sentences, and make logical relations visible.
10. Retain every technical anchor on first use; examples and intuitions are additions beside, not replacements for, the abstract claim or proof.
11. Reorder only with a source-target map and a dependency-order check.
12. Maintain bidirectional alignment from every source ledger item to target text and from every target assertion to source support.
13. Run a full fidelity pass over every protected item; sampling is not allowed for definitions, claims, formulas, proofs, numbers, citations, or epistemic markers.
14. Run a separate register-quality pass for familiar lexicon, explicitness, groundedness, coherence, navigation, genre fit, locale fit, and audience fit.
15. Run matched-meaning dialect and register counterfactual tests so surface variety cannot change capability or priority judgments.
16. Run structural checks for numbers, units, math spans, symbols, cross-references, citation keys, terminology, quantifiers, negation, conditionals, and modal markers.
17. Use readability, similarity, NLI, generated QA, and language-model critique as alarms only; none certifies semantic equivalence.
18. Build and render phone, desktop, print, screen-reader, low-bandwidth, and offline forms where applicable.
19. Publish a reversible version with the source hash, target brief, termbase, content ledger, alignment, test results, model provenance, and correction channel.

20. Measure use and learning where possible and revise; missing human evidence never blocks a completed, mechanically audited edition.

Release and QA rule

A critical fidelity error changes or loses a definition, proposition, quantifier, condition, exception, number, unit, mathematical symbol, proof dependency, cross-reference target, attribution, epistemic or deontic force, or prerequisite. Critical errors allowed in a released edition: zero. Major errors that could cause the target reader to form a materially different model of the source must also be resolved. Minor register or style issues remain revisable and may not be traded against accuracy.

Readability, sentence length, lexical frequency, SARI, BLEU, semantic similarity, natural-language inference, generated questions, and model critique are diagnostic alarms. None is a semantic certificate. Deterministic checks reconcile claims, numbers, units, formulas, symbols, links, citation keys, terms, quantifiers, negation, conditions, and modality. A second model pass may challenge the translation, but the auditable content ledger and exact source-target alignment remain the core evidence.

Human evidence can improve later revisions but is not a publication hold. A completed edition proceeds through deterministic and model-audit gates and remains versioned and reversible. Use studies, classroom studies, or community feedback measure outcomes and guide later profiles; they do not decide whether the audience was worthy of receiving the first edition.

Research design for measuring benefit

The production premise is fixed; the following outcomes remain research questions:

- immediate proposition-level comprehension, including definitions, conditionality, and uncertainty;
- time and effort needed to reach the same answer or proof understanding;
- independent navigation, prerequisite recovery, exercise completion, and error correction;
- delayed retention and transfer to new problems;
- persistence through a course or research text;
- ability to move between the translated and conventional academic registers when the reader chooses;
- differences across adult target registers, fields, educational stages, disability modes, devices, and prior familiarity;
- whether a parallel source-plus-translation interface outperforms either version alone.

Measure the actual learning task, not conformity to one accent, orthography, or metropolitan grammar. Publish null and adverse effects, but use them to revise target design, not to infer deficient readers. Because an optional edition leaves the original available, heterogeneous preferences support a portfolio rather than one compulsory register.

Immediate Open Logic implementation brief

The first bounded implementation can translate one complete Open Logic module into one declared international adult/ELF-friendly register while preserving the original beside it. Freeze the source; create stable claim, definition, formula, example, exercise, and reference IDs; build the termbase and dependency graph; produce the full-content translation; and run all protected-item checks. The pilot should compare at least two adult target profiles rather than declare one global Plain English.

Commissioning language:

> Produce a full-content diaphasic intralingual translation for the specified adult English register. Preserve every proposition, definition, quantifier and its scope, condition, exception, number and unit, term distinction, symbol and type, proof step and dependency, cross-reference, attribution, epistemic and deontic status, and prerequisite. Do not summarize, abridge, generalize, or delete. You may expand, define, split, reorder with tracked dependencies, resolve anaphora, add labeled scaffolding, and lower unnecessary formality. Retain each technical term and notation at first use with a plain gloss. Maintain bidirectional source-target claim alignment. Treat readability metrics as diagnostics only. Release only with zero critical accuracy errors.

Bottom line

The access intervention is neither a claim that academic language is useless nor a claim that every reader wants the same register. It separates intellectual content from one inherited institutional form, keeps the conventional form available, and offers complete parallel routes. The literature makes the mechanism and the design obligation plausible; the project adopts production as the standing decision and uses research to make the editions more faithful, more varied, and more useful.

Appendix B - Anti-hegemony model audit

Anti-hegemony audit for educational-access reach models

Status: controlling method and deterministic failure specification. This document governs the hybrid reach comparison and later package comparisons. It is not a rhetorical caveat and it does not reduce any community's eligibility.

Governing position

The analysis presumes learner capability. Familiarity with the register that schools, universities, publishers or professional institutions reward is an unequally distributed access condition, not an intelligence measure. Schooling, income, test performance, dominant-language exposure, formal literacy and academic-register familiarity are entangled with earlier access to education and resources. None may be used as a proxy for intrinsic capacity or as a reason to withhold a potentially useful edition.

The register-translation programme is justified by a sufficiently plausible access thesis: academic-register distance is one removable institutional barrier, and a source-faithful translation that produces a credible nonzero benefit is worth making. Research improves target selection, design, semantic fidelity and evaluation. It is not a gate that can veto production by adopting a deficit explanation.

Access-potential model

For language/register community `l`, intersectional stratum `s`, and exact content/register/task `d`, define:

```
N_ld = sum_s(P_ls x q_lsd)
```

`P_ls` is the plausible population of potential users. It includes people who are not currently enrolled, people whose earlier education limited literacy, displaced and diaspora populations, signing communities, and people requiring non-print formats. `q_lsd` is the source-user mismatch rate: the share who cannot reach the intended learning objective from the actual academic-English source independently and comfortably enough for the task, without bespoke human mediation. It is not a property of a language or a person. It changes with domain, difficulty, genre, register, modality, supports and circumstances.

Here, “independently” permits ordinary assistive technology and culturally normal collaborative learning. It does not mean solitary print reading.

When `q_lsd` is not measured comparably, the published hybrid comparison uses the disclosed common imputation and wide sensitivity range. Research replaces that imputation only when population, register threshold, task, stage and material endpoint match. Missingness therefore widens uncertainty; it never lowers the access-potential population.

When another language or edition counts as coverage

An alternative `a` can reduce distinct reach only for people facing the academic-source mismatch and only when the following joint gates pass:

Gate	Question
Technical access	Can the person obtain and use it on the relevant device, bandwidth, encoding, font and offline conditions, without DRM blocking the use?

Gate	Question
Linguistic access	Is the exact variety, register, script and modality intelligible for this task?
Material equivalence	Is it correct, complete enough, current, and matched to the same domain and educational level?
Voluntary acceptability	Is use socially and politically acceptable rather than merely compulsory, colonial, stigmatized or geopolitically coercive?
Disability access	Do captions, sign, screen-reader semantics, audio, visual, cognitive and motor-access requirements pass for the relevant stratum?
Practical availability	Is it affordable, discoverable and lawfully obtainable?
Safety and trust	Can it be used without unreasonable surveillance, identity exposure or other safety cost?

Only a defensible lower bound on their joint satisfaction can be deducted. Missing evidence for any indispensable gate means zero coverage deduction. The model must not multiply marginal gate rates as though they were independent. With marginal lower bounds `g_1 ... g_K`, the conservative Frechet lower bound is `max(0, sum(g_k) - (K - 1))`; it will often correctly be zero.

For verified alternative coverage `o_lsd`, residual eligible reach is:

```
R_ld = sum_s(P_ls x q_lsd x (1 - o_lsd))
```

Delivery usability for the proposed new edition, `u_lsd`, remains separate:

```
R_initial_ld = sum_s(P_ls x q_lsd x (1 - o_lsd) x u_lsd)
```

A low `u_lsd` is a design problem to repair through offline, phone-readable, audio, signed, printable or assistive formats. It is not evidence of lower underlying need.

Mechanisms that would reproduce privilege

Exclusion mechanism	Prohibited shortcut	Required protection
Elite linguistic capital	University, urban or online-test respondents stand for everyone; ordinary English or years of schooling imply academic autonomy.	Use task-matched evidence, retain unsampled strata, and impute missing groups without a downward penalty.
Data abundance	Confidence or source count multiplies reach; complete-case filtering drops poorly documented communities.	Confidence changes interval width and research priority only.
National wealth	GDP, enrolment, internet access or predicted uptake lowers need.	Keep underlying need separate from production and delivery feasibility.
Colonial-language exposure	Official status, compulsory schooling or historical exposure proves proficiency or voluntary preference.	Require task competence and voluntary, safe, community-acceptable use.
Standard-language dominance	A macrolanguage, state standard or one orthography automatically covers local varieties, scripts, mixed codes or oral communities.	Use exact variety/register/script/modality gates; preserve every unresolved community.
Corpus and model abundance	Better NLP support or lower production cost makes a language more deserving.	Cost is operational information, never a need or eligibility multiplier.
Literacy norms	Only current readers enter the denominator.	Preserve people excluded from literacy and design audio, sign, visual and supported routes.
Able-bodied assumptions	An online PDF or uncaptioned video is called accessible.	Retain disability-specific strata and format gates.

Exclusion mechanism	Prohibited shortcut	Required protection
Geopolitical coercion	Forced dominant-language use or a census label proves acceptable coverage.	Compulsion is non-evidence; record affiliation, safety and political acceptability separately.
Aggregate averaging	High-proficiency elites suppress rural, classed, gendered, caste, disability, age or displaced tails.	Stratify where lawful and test aggregate results for masking and Simpson reversals.
Population thresholds	Small communities disappear from a headcount order.	Publish raw headcount separately from protected inclusion, rights, vitality and prestige-domain lanes.
Material-count proxies	One title, one machine translation or a large catalogue proves coverage.	Require semantic, curricular, level, licensing and delivery equivalence for the same task.
“Bilingual” labels	Knowing another language proves access in every domain and register.	Count only an actually suitable alternative that passes every joint gate.
Prestige-English testing	Accent, grammar or conformity to British/American norms stands in for comprehension.	Test comprehension of the actual source; World Englishes are legitimate target and user registers.
English-first-languag e shortcut	English identity sets academic-source mismatch to zero.	Apply the same task/register test to English users and retain multiple adult English targets.
Deficit framing	The user is described as deficient rather than the source as inaccessible.	Describe removable source-user mismatch and assume intellectual capability.

Critical-theory crosswalk

The implementation must explicitly engage several partly overlapping bodies of work rather than use “inclusion” as a generic label:

- Linguistic capital and social reproduction: institutions reward language practices unequally distributed through family and schooling; the model must not treat possession of the rewarded register as neutral evidence of lower need.
- Critical EAP and EMI: access to academic English can be useful while the institutions and norms that make it compulsory reproduce unequal power. A translation portfolio expands choices; it does not demand assimilation to a prestige register.
- Raciolinguistics: judgements of “appropriate,” “professional,” or “academic” language can reside in a powerful listener or institution rather than in objective communicative failure. Target design and QA must not encode white, elite or metropolitan listening norms as semantic correctness.
- Linguistic imperialism and decolonial multilingual education: colonial and global prestige languages cannot be credited as neutral substitutes merely because they are official or widespread. Language choice, local epistemologies, material control and the right to continue a local language in high-prestige domains remain visible.
- World Englishes and English as a lingua franca: there is no one universal non-academic English. Audience-specific adult registers may differ while preserving the same intellectual content.
- Disability and signed-language justice: written-language counts do not measure access for signing, blind, low-vision, print-disabled, DeafBlind, cognitive-access or motor-access communities.

These lenses do not convert structural disadvantage into an opaque bonus score. They define denominators, prevent invalid coverage deductions, require separate strata, and identify design obligations.

Evidence supporting the audit design

The following sources support different parts of the audit. Theoretical work identifies mechanisms and questions; qualitative cases show possible processes; observational studies quantify associations or reported burdens; and experiments support only their tested contrasts. None supplies a universal causal coefficient for the allocation model.

Source	Design and bounded finding	Operational use here
Bourdieu (1977), The economics of linguistic exchanges	Sociological theory: linguistic value is produced in social markets and institutions help establish a legitimate discourse.	Do not mistake prestige-register conformity for intrinsic linguistic or intellectual quality.
Pennycook (1997), Vulgar pragmatism, critical pragmatism, and EAP00019-7)	Critical analysis of EAP's claimed neutrality.	Present academic conventions as situated choices and power relations, not a natural benchmark that every target must imitate.
Lea and Street (1998), Student writing in higher education	Case study at two English universities: 23 staff and 47 students, using interviews, observation, assignments, feedback and guidance documents. The authors did not claim statistical representativeness.	Use discipline-aware register analysis and separate content fidelity from tacit institutional writing conventions.
Macaro et al. (2018), Systematic review of English-medium instruction in higher education	Mapped 285 empirical studies and reviewed 83 in depth. At that point, evidence did not settle broad claims that EMI improves English or harms/benefits content; regions were unevenly studied.	Give English-medium provision no assumed return premium and treat geographical evidence gaps as uncertainty, not lower need.
Flores and Rosa (2015), Undoing appropriateness	Critical synthesis: racialized speakers may be heard as deficient even when approximating supposedly appropriate forms.	Audit the model and evaluator with matched-meaning variety tests; standard-language matching alone cannot certify fairness.
Flores (2020), From academic language to language architecture	Conceptual and ethnographic application argues that “academic language” can operate as a deficit-producing raciolinguistic ideology.	Describe multiple ways of constructing disciplinary meaning rather than remediating supposedly deficient speakers.
Hofmann et al. (2024), Dialect prejudice predicts AI decisions about people's character, employability, and criminality	Matched-guise probing across GPT-2, RoBERTa, T5, GPT-3.5 and GPT-4 found different stereotyped and consequential outputs for African American English versus Standardized American English while propositional cues were held comparable.	Require dialect-counterfactual invariance and prohibit register or dialect from becoming a proxy for merit, readiness, risk or value.
Phillipson (1992), Linguistic imperialism	Historical and theoretical account of structural and cultural inequalities sustaining English dominance.	Treat official status, global centrality and market size as potentially endogenous to power, never as positive need weights.
Veronelli (2015), The coloniality of language	Historical-conceptual analysis of conquest, racialization and denial of colonized people as communicative agents.	Attribute uncertainty to system coverage and method rather than deficient speakers; retain pragmatic, cultural and epistemic expression.
Sah and Li (2024), Translanguaging or unequal languaging?	Four-month ethnography in one Nepalese EMI school: English-Nepali translanguaging coexisted with exclusion and stigmatization of Bhojpuri.	A national prestige language plus English cannot discharge exact home-language obligations.
Laitin, Ramachandran and Walter (2019), The legacy of colonial language policies and their impact on student learning	Twelve Kom-medium primary schools were matched with twelve English-medium schools. The programme also changed teacher training and textbook supply, so language was not isolated; early advantages weakened after return to English-only schooling.	Investigate sustained local-language routes and never transplant the bundled programme's effect as a universal language coefficient.

Source	Design and bounded finding	Operational use here
Amano et al. (2023), The manifold costs of being a non-native English speaker in science	Online self-report survey of 908 published environmental scientists across eight nationalities found substantial extra reported time and career burdens associated with working in English. Nationality categories are not individual proficiency measures.	Retain research-stage reading, writing and dissemination burdens and intersectional strata without generalizing the sample to all fields.
Blasi, Anastasopoulos and Neubig (2022), Systematic inequalities in language technology performance	Global analysis across several NLP tasks found strong concentration in a small set of languages and showed that approximate GDP predicted NLP publication volume better than speaker population. The relation is associational.	Present tool and research abundance as an outcome needing audit, not evidence that already favored languages deserve more access compute.
Kreutzer et al. (2022), Quality at a glance	Manual audit of 205 language-specific web corpora found at least 15 with no usable text and many with less than half acceptable sentences.	Raw crawl volume is not readiness or coverage; scarcity triggers provenance, language-ID and curation work rather than exclusion.
Nekoto et al. (2020), Participatory research for low-resourced machine translation	Participatory case study produced datasets and benchmarks for more than 30 African languages and enabled contributors without formal NLP training to make scientific contributions.	Treat institutional and geographic exclusion as part of low-resourcedness and budget for speaker/community governance, curation and evaluation.
UNESCO GEM (2025), The right multilingual education policies can unlock learning and inclusion	Global policy synthesis supports sustained home-language instruction and identifies material and implementation shortages.	Do not treat a brief transition to a dominant language as complete coverage; preserve local-language materials and implementation routes.

The synthesis used by this model is an inference across these sources: observed demand, corpus size, academic prestige and standardized-language visibility are partly produced by earlier institutional investment and exclusion. They are not neutral measurements of educational need.

Allocation order without an opaque equity score

If a decision must be sequenced, use a transparent Pareto or lexicographic view: (1) minimum coverage floor breaches; (2) worst-group source-user mismatch; (3) stakes of the educational task; (4) directly documented burden; (5) underserved users and locally expressed priorities; and only then (6) verified marginal production cost. Publish unweighted headcounts beside any equity or rights-based lane. GDP, revenue, citation volume, academic prestige, web share, existing product use and colonial-language status receive no positive benefit weight.

Observed demand is censored by availability, price, schooling and historical exclusion. Data scarcity is therefore a repair signal: it routes work to provenance, language identification, curation and evaluation instead of lowering the target. A minoritized language also requires knowledge-in and knowledge-out support; one-way extraction into a prestige language is not reciprocal access.

Required deterministic tests

1. Missingness monotonicity: deleting evidence cannot lower inclusive planning reach or remove a row.
2. Barrier monotonicity: increasing P or q cannot lower reach.
3. Joint-gate veto: any false or unknown indispensable alternative gate makes the coverage deduction zero.
4. Disaggregation safety: splitting a stratum cannot lower the sum unless new verified coverage is discovered.

5. Partition invariance: splitting or merging identical populations without changing facts preserves aggregate reach.
6. Label invariance: changing only “language,” “dialect,” “official,” “L1” or “bilingual” labels has no numerical effect.
7. Wealth counterfactual: communities differing only in GDP, connectivity, enrolment, test participation or data density have identical underlying need.
8. Colonial-status counterfactual: official or compulsory prestige-language status cannot lower mismatch or create coverage without task evidence.
9. World-Englishes counterfactual: nonconformity to metropolitan accent or grammar cannot create mismatch if the actual source is understood; ordinary fluency cannot erase mismatch if the academic source is not understood.
10. Tail audit: recompute lawful class, rurality, gender, age, disability, displacement, caste/ethnicity and educational strata and flag aggregate masking.
11. Source deletion: removing any source cannot make a community disappear.
12. Delivery separation: worsening bandwidth or predicted uptake affects realizable delivery, not underlying need.
13. Small-community retention: no minimum-size or cost-per-person threshold can remove eligibility.
14. No capability inference: attainment, schooling and register familiarity do not appear in a formula as innate intellectual-capacity parameters.

Hard failures

The model fails if a missing record becomes zero; if evidence confidence, wealth, enrolment, literacy, connectivity, official status or NLP support multiplies need downward; if an alternative creates coverage while a required gate is unknown; if a standard variety erases a distinct variety, script, modality or political community; if English first-language or bilingual status forces mismatch to zero; if a point estimate is presented without provenance and sensitivity; or if potential access is described as actual uptake or learning.

The central rule is simple: absence of evidence may widen uncertainty and make research more urgent, but it may never be converted into evidence of access.

Appendix C - Intervention-effect evidence

Intervention-effect evidence for educational-access compute

Working evidence register-not a universal multiplier or an eligibility test.

Governing thesis

Academic-register familiarity is an institutionally distributed access advantage. Source-faithful language and adult-register editions with any credible nonzero educational benefit remain eligible. Evidence estimates effects and improves audience, package, format, sequence, fidelity and scale; it never functions as a permission test, innate-capacity screen or language-level withholding rule.

A null or adverse result for one document, audience or delivery design redirects the intervention. It does not establish low reader capacity and does not remove a language, register or community from eligibility.

What the evidence currently says

The closest same-content translation evidence is an eight-module Swedish/English programming-course RCT and a short professional-reading RCT among Scandinavian physicians. Adult register-translation experiments in health, legal and scientific material often improve immediate comprehension, but the size varies sharply by source and null or adverse controls exist. Large foundational effects mostly belong to language-of-instruction or structured-pedagogy bundles. Free/open access chiefly changes acquisition; offline delivery changes practical access; worked examples, retrieval practice, explanatory feedback and self-regulation change learning conditional on use.

No located controlled study measures a complete proposition-preserving mathematics curriculum translated into another language or adult register. The scenarios below are therefore disclosed planning intervals, not a new meta-analysis.

Closest direct and short-text anchors

ID	Study	Stages	Comparison	Exact result	Transfer boundary
EFF-001	Leroy et al.	A|U|R	Translated general-register passages versus originals.	Correct comprehension 63% versus 52% (+11 percentage points, p<.001); learning gain +18% versus +9% (p=.003); free recall unchanged; perceived difficulty 2.3 versus 3.2/5.	Not a full-course, mathematics, long-run attainment or universal register effect.
EFF-002	Martínez, Mollica, and Gibson	A|V|U	Equivalent-meaning ordinary English versus legalese.	Comprehension 73.5% versus 67.7% (+5.8 percentage points); register coefficient remained significant after below-chance items were excluded.	Not a complete-statute, complete-contract, academic-course or global effect.
EFF-003	Martínez, Mollica, and Gibson	A|V|U	Ordinary-language versus legalese contract registers.	Both groups comprehended plain versions better (beta=.354, SE=.088, p<.001) and recalled more (beta=.360, SE=.121, p=.003); register-by-legal-training interactions were nonsignificant.	Not evidence that operative legal text can be freely altered or a long-form education effect.

ID	Study	Stages	Comparison	Exact result	Transfer boundary
EFF-004	Feinberg et al.	A|V|U	Original clinical-consent wording followed by a plain-language rewrite.	Mean comprehension 10.53/15 versus 9.15/15 (+1.38 items, +9.2 points of the test maximum), t(191)=9.36, p<.001, d=.68; 52.6% improved, 32.6% unchanged, 15.1% decreased.	Original-always-first order prevents treating the full gain as translation-only.
EFF-005	Beskow et al.	A|V|U	Concise form versus traditional form communicating the same required information.	Knowledge 17.6/21 versus 17.5/21 (noninferior); median reading time 5 versus 10 minutes, p<.0001; targeted review and retest improved all strata.	Not every source sentence was preserved; not evidence of superior long-run learning.
EFF-006	Sayfi et al.	A|V|U	Plain-language recommendations versus originals.	WHO comprehension +19.8 percentage points (95% CI 14.7-24.9, p<.001); CDC +3.9 points (95% CI -0.7 to 8.3, p=.096); accessibility and satisfaction each +1.2/7.	The WHO effect cannot be applied to the CDC text, a textbook or another audience.
EFF-009	August et al. (Paper Plain)	U|R	Original PDF versus original-plus-definitions, question navigation, and localized gists.	Perceived difficulty 2.00 versus 4.00/5 and perceived understanding 3.50 versus 2.00; actual comprehension 3.67/7 versus 3.50/7, F(4,20)=1.38, p=.267, with reported noninferiority.	Small college-educated sample; curated summaries; no measured knowledge gain.
EFF-030	Piesche et al.	S	English/German bilingual physics versus German-only physics.	Smaller science gains immediately, d=-.21, and after six weeks, d=-.23.	One topic and teacher; only partially transferable to self-study translation.
EFF-031	Bälter et al.	V|U	Otherwise identical translated self-study course.	All assigned/analyzed: 7.24 versus 4.18 correct of 42, Cliff's delta .15; course starters: 16.9 versus 14.3, delta .085; dropout 57% versus 71% (14-point difference), phi=.20.	Highly educated self-selected Swedish sample, high attrition and excluded post-randomization switchers limit transport.
EFF-032	Gulbrandsen et al.	R|V	Mother tongue versus English and paper versus screen.	Median recall 4/13 facts (IQR 3-6) in mother tongue versus 3/13 (2-4) in English, p=.01; paper versus screen was null.	Small convenience sample, short exposure, one medical article and closely related languages.

Planning intervals

Scenario	Intervention	Endpoint	Range	Evidence status
SCN-REG-ADVERSE	well-audited full-content adult register translation	immediate objective comprehension probability	-3.0 to 0.0 absolute_percentage_points	scenario_not_meta_analysis
SCN-REG-NEAR-NULL	well-audited full-content adult register translation	immediate objective comprehension probability	0.0 to 3.0 absolute_percentage_points	scenario_not_meta_analysis
SCN-REG-CENTRAL	well-audited full-content adult register translation	immediate objective comprehension probability	3.0 to 10.0 absolute_percentage_points	scenario_not_meta_analysis
SCN-REG-HIGH-MISMATCH	well-audited full-content adult register translation	immediate objective comprehension probability	10.0 to 20.0 absolute_percentage_points	scenario_not_meta_analysis
SCN-SELF-STUDY-TRANSLATED	otherwise identical translated self-study course among starters	course-test performance	0.0 to 0.085 Cliffs_delta	direct_anchor_bounded_transport
SCN-SELF-STUDY-PERSISTENCE	otherwise identical translated self-study course	dropout reduction	0.0 to 14.0 absolute_percentage_points	direct_anchor_bounded_transport
SCN-FOUNDATIONAL-STRUCTURED-L1	structured early-grade L1 literacy package	sentence reading or comprehension	0.1 to 0.3 standard_deviation	synthesis_based_package_scenario
SCN-OPEN-REPLACES-PAID	free/open book replaces comparable paid book among enrolled learners	course learning	-0.05 to 0.1 standard_deviation	meta_analysis_informed_scenario

Scenario	Intervention	Endpoint	Range	Evidence status
`SCN-PASSIVE-PUBLICATION`	file merely published without active dissemination or usage evidence	average learning effect per eligible person offered the file	0.0 to 0.02 standard_deviation	`conservative_scenario`
`SCN-STATIC-ACTIVE-USE`	level-appropriate static package with worked examples, retrieval, answers and explanatory feedback among active users	learning change over roughly a term or year	0.05 to 0.25 standard_deviation	`component_synthesis_informed_scenario`
`SCN-ADAPTIVE-ROLLOUT`	level-adaptive system with repeated explanatory feedback	rollout intent-to-treat learning change	0.12 to 0.24 standard_deviation	`meta_analysis_informed_scenario`
`SCN-OFFLINE-SUBSTITUTION`	offline digitization substituting for conventional instruction without major pedagogical redesign	knowledge change	-0.1 to 0.2 standard_deviation	`systematic_review_informed_scenario`
`SCN-SMS-ONLY`	SMS or phone content without instruction	rollout intent-to-treat learning change	0.0 to 0.05 standard_deviation	`direct_anchor_bounded_transport`
`SCN-SMS-GUIDED`	SMS plus brief structured tutoring or guided practice	rollout intent-to-treat learning change	0.03 to 0.21 standard_deviation	`direct_anchor_bounded_transport`
`SCN-AUTONOMOUS-LIBRARY`	curated offline digital library with devices and observable voluntary use	secondary-school learning change	0.05 to 0.15 standard_deviation	`direct_anchor_bounded_transport`
`SCN-SELF-REGULATION`	planning, goal setting, progress monitoring, retrieval schedules and study protocols	incremental performance among users	0.1 to 0.3 standard_deviation	`meta_analysis_informed_scenario`
`SCN-PRESCHOOL-UNQUANTIFIED`	preschool language-accessible package	age-appropriate learning and dual-language development	unmeasured-not zero	`unmeasured_not_zero`
`SCN-RESEARCH-RECALL`	translated professional or research reading	immediate factual recall after ten minutes	0.0 to 1.0 additional_facts_out_of_13	`direct_anchor_bounded_transport`

Calculation rule

For passive or lightly supported releases, model E[learning change | eligible] = p(reached) × p(acquired | reached) × p(usable | acquired) × p(active | usable) × active-user effect. Do not apply those multipliers to an intent-to-treat estimate from a comparable rollout because uptake and nonuse are already included.

Report newly usable access, sustained active use or completed units, and standardized learning change separately. Do not add effects measured in incompatible units or from overlapping components. Do not multiply a language population directly by a comprehension uplift without an exposure/use model.

Design implications

- Preserve every proposition, definition, condition, negation, quantifier, exception, equation, theorem status, example, exercise, answer, citation and cross-reference.
- Pair stable technical terms with inline accessible glosses; do not replace the advanced vocabulary needed for later transfer.
- Include correct worked examples, retrieval practice, immediate answers with explanations, prerequisite diagnostics, mastery checks, study plans, restart instructions and download-once navigation.
- Keep original and translated registers available as user-selectable parallel routes; develop more than one adult and World English target profile.
- Treat readability, fluency and downloads as diagnostics-not proof of semantic fidelity, active use or learning.
- Use proposition-level source-target reconciliation, protected-token and formula maps, and independent model/mechanical checks. No human-dependent release gate is introduced.

Appendix D - Conditional global portfolios

Conditional global portfolios for educational-access compute

Working result, not an ordinal world-language ranking. The current evidence does not identify a unique global winner. This document nevertheless gives an actionable first allocation for a funder who must act now, while publishing exactly what could displace it.

How to use the result

The recommended first action is a ten-program measurement-and-production tranche. Each program uses the same 100,000-source-token benchmark protocol for comparable workflow measurement while retaining the different target-specific first package listed below. The work records actual target-specific production, QA, accessibility and delivery costs and evaluates the resulting functional-access tasks. Subsequent sequencing and scale follow the measured intervals. The ten programs are slots in one feasible diversified hedge, not ranks 1-10 or a unique solution.

The accompanying 100-profile portfolio is the broader research and production queue. It is also not ordered. All 10,525 entities outside that communication set remain possible outrankers in the universal register. Missing evidence cannot exclude them.

`P` = preschool/early childhood; `F` = foundational/primary; `S` = secondary; `V` = vocational/technical/practical; `U` = university/taught tertiary; `R` = postgraduate/research.

The ten-program first tranche

Bahasa Indonesia secondary catch-up and autonomous STEM progression

- Scope: Bahasa Indonesia in Latin orthography; first-language and additional-language users kept separate; Javanese, Sundanese, signed and other regional-language routes are not treated as covered.
- Stages: `F|S|V|U|R` - foundational/primary; secondary; vocational/technical/practical; university/taught tertiary; postgraduate/research.
- First package: Secondary mathematics/science prerequisite repair with exercises and complete solutions, offline/mobile delivery, followed by statistics, computing, finance/business and open tertiary/research bridges.
- Why it is in this hedge: BPS reports 248,501,794 age-5+ users by its conversational-ability definition, while the source-bound Indonesian secondary cohort shows a large below-Level-2 context. Neither figure is treated as exact edition uptake or effect.
- Evidence and scaling rule: The Indonesian programme remains eligible. A source-and-task audit selects the stage, subject, register and format with the largest credible additional access; strong existing supply redirects the next package rather than removing Indonesian learners from consideration.
- Named possible outrankers: Javanese, Sundanese, Bangla, Filipino and named Philippine languages, Telugu, Tamil, Marathi.

Bangla/Bengali Bangladesh-India parallel foundational and STEM package

- Scope: Bangladesh Bangla and India Bengali outputs with terminology and curriculum overlays; Sylheti and Chittagonian remain separate profiles rather than automatic beneficiaries.
- Stages: F|S|V|U|R - foundational/primary; secondary; vocational/technical/practical; university/taught tertiary; postgraduate/research.
- First package: Foundational numeracy and reading-supported mathematics, then secondary science/statistics and practical finance/health modules, with one reusable source core and two explicit national interfaces.
- Why it is in this hedge: The country-disjoint parallel-package population context is 264,588,675-264,605,657 people, while Bangladesh has a high learning-poverty context. The aggregate is not copied into either output's uptake estimate.
- Evidence and scaling rule: Bangladesh Bangla and India Bengali remain eligible as distinct outputs. Supply audits choose the most useful common task and determine how much source preparation can be shared without flattening national terminology or separately named varieties.
- Named possible outrankers: Sylheti, Chittagonian, Hindi-Urdu package, Telugu, Hausa, Kiswahili.

Hindi-Urdu Devanagari/Nastaliq parallel educational package

- Scope: Separate Standard Hindi/Devanagari and Urdu/Perso-Arabic-Nastaliq outputs plus an experimentally shared mathematical register; other Hindi-group mother tongues and Punjabi varieties remain separate.
- Stages: F|S|V|U|R - foundational/primary; secondary; vocational/technical/practical; university/taught tertiary; postgraduate/research.
- First package: Parallel foundational/secondary mathematics and science, practical finance/business and tertiary logic/statistics, measuring what terminology and formal structure can be reused across scripts.
- Why it is in this hedge: The enumerated L1 lower-bound package context is 395,205,166 country-and-census-category-disjoint people. Pakistan's country context also shows high learning and literacy gaps; those rates are not multiplied into the language counts.
- Evidence and scaling rule: Retain the native Hindi and Urdu outputs in every scenario. Treat a shared-register layer as an additional experiment whose scale follows task-specific comprehension and production-reuse evidence; it never substitutes for either native output by name alone.
- Named possible outrankers: Western/Pakistani Punjabi, Eastern Punjabi, Telugu, Marathi, Tamil, Bangla, Pashto, Sindhi, Saraiki.

Simplified Standard Written Chinese targeted tertiary, research and accessibility package

- Scope: PRC Standard Written Chinese in normative simplified characters; Putonghua ability, written-character use, formal enrolment and other Sinitic languages are distinct measures.
- Stages: S|V|U|R - secondary; vocational/technical/practical; university/taught tertiary; postgraduate/research.
- First package: Audit first for missing open, downloadable and accessible advanced mathematics/science/computing; translate high-value research and tertiary gaps, with MathML/HTML, reflow and offline formats rather than duplicating adequate school supply.
- Why it is in this hedge: A conditional adult standardized-character-use bound exceeds 1.05 billion and formal enrolment is 280,404,300. The opportunity scale is enormous even though existing supply may make target-specific additionality smaller.

- Evidence and scaling rule: The Simplified Written Chinese programme remains eligible. Independent supply audits identify the exact stage, subject, register and accessibility gaps where a new edition adds most; existing provision redirects the package and never turns industrialization into evidence of low need.
- Named possible outrankers: Vietnamese, Japanese research access, Korean research access, Thai, Burmese, Traditional Chinese regional profiles.

Philippine multilingual foundational and secondary package

- Scope: Filipino and Cebuano plus separately named Hiligaynon, Ilocano, Bikol, Waray, Kapampangan and Filipino Sign Language interfaces; Filipino exposure is not Tagalog L1 or nationwide comprehension of one edition.
- Stages: P|F|S|V - preschool/early childhood; foundational/primary; secondary; vocational/technical/practical.
- First package: Early numeracy and reading-supported mathematics, secondary science, disaster/health education and practical financial capability in downloadable mobile formats, sharing diagrams and exercises while localizing prose.
- Why it is in this hedge: The Philippines country sentinel is 116,786,962 people and carries a very high learning-poverty context. The Filipino exposure ceiling of 112,729,484 is not used as a language-specific need count.
- Evidence and scaling rule: Retain every named Philippine interface as a candidate. Exact home-, school- and task-language evidence determines whether to expand, split or repartition the package rather than forcing separate language communities through Filipino alone.
- Named possible outrankers: Other Philippine languages, Burmese, Central Khmer, Lao, Indonesian regional languages, named Pacific languages.

Ethiopian multilingual foundational, science and practical-learning package

- Scope: Amharic, West-Central Oromo, Tigrinya and Somali interfaces are separate; Ethiopia's population and country indicators are not assigned to any one language.
- Stages: P|F|S|V|U - preschool/early childhood; foundational/primary; secondary; vocational/technical/practical; university/taught tertiary.
- First package: Foundational literacy/numeracy, secondary science and locally adaptable agriculture, health, technical and small-enterprise modules designed for offline/low-bandwidth use.
- Why it is in this hedge: The Ethiopia sentinel covers 135,472,051 people and shows severe country-level learning, literacy and connectivity gaps. Exact language audiences remain a first-order measurement requirement, not a reason to omit the context.
- Evidence and scaling rule: Begin with bounded multilingual production while resolving language-of-instruction, literacy, script, region, stage and supply cells. Scale multiple interfaces wherever one national-language output would miss learners; missing measurements do not postpone eligibility.
- Named possible outrankers: Other Ethiopian languages, Somali displaced learners, Sudanese language portfolios, South Sudanese portfolios, DRC multilingual package.

DRC multilingual and displacement-continuity package

- Scope: Lingala, Kivu/Congo Swahili, French, Tshiluba, Kikongo varieties and other named languages plus host-language interfaces; no national or refugee total is assigned to one language.
- Stages: P|F|S|V|U - preschool/early childhood; foundational/primary; secondary; vocational/technical/practical; university/taught tertiary.

- First package: Resumable foundational and secondary mathematics/science, public-health and practical livelihood modules, with school-continuity and credential-navigation paths for displaced learners.
- Why it is in this hedge: The DRC sentinel is 112,832,473 people with extremely severe country-level learning and connectivity contexts. The current displacement row lacks a publication-ready language denominator and remains visible rather than scored down.
- Evidence and scaling rule: Begin with a bounded origin-host continuity package and keep origin, host, stage and language as separate cells. Evidence refines its scale and composition; offline, print and accessible routes remain part of the design wherever connectivity would otherwise prevent use.
- Named possible outrankers: Named DRC languages not yet profiled, Sudan displacement, South Sudan displacement, Somali displacement, Hausa, Kiswahili national profiles.

Hausa Boko/Ajami cross-border foundational and practical package

- Scope: Hausa country profiles with Boko and Ajami literacy routes kept separate; Nigerian and cross-border totals are not added.
- Stages: P|F|S|V|U - preschool/early childhood; foundational/primary; secondary; vocational/technical/practical; university/taught tertiary.
- First package: Foundational numeracy/literacy, secondary science and practical health, agriculture, financial capability and enterprise modules in low-bandwidth text/audio and both evidenced literacy interfaces.
- Why it is in this hedge: The source register retains a global lower bound of 100 million mixed L1/L2 users and an 80 million Nigeria lower bound. Nigeria's country context shows major literacy and connectivity gaps without supplying a Hausa-specific rate.
- Evidence and scaling rule: Measure Boko/Ajami literacy and national supply separately; preserve a parallel interface where shared content lowers production cost without excluding either script community.
- Named possible outrankers: Nigerian Pidgin, Yoruba, Igbo, Central Kanuri, Nigerian Fulfulde, other West African language profiles.

Kiswahili multistandard cross-border STEM and practical-learning package

- Scope: Tanzania, Kenya, Uganda, Rwanda, Burundi and Kivu/Congo Swahili standards and L1/L2 audiences kept explicit; a global lower bound is not inherited by one edition.
- Stages: F|S|V|U - foundational/primary; secondary; vocational/technical/practical; university/taught tertiary.
- First package: Secondary mathematics/science, statistics/computing and practical health, agriculture, finance and management modules with national terminology overlays and offline/mobile delivery.
- Why it is in this hedge: The source register retains a cross-border lower bound of 200 million mixed L1/L2 users. Multiple education systems may allow substantial reuse, but usable academic register and national supply must be measured.
- Evidence and scaling rule: Retain the national and variety-specific Kiswahili outputs as candidates. Audience and production evidence determines the useful balance between a common core and national overlays without treating all Swahili varieties as identical.
- Named possible outrankers: Hausa, Amharic, Oromo, Somali, Kinyarwanda, Kirundi, named DRC languages.

Modern Standard Arabic with explicit vernacular and displacement interfaces

- Scope: MSA as a learned written educational register plus separately specified Egyptian, Palestinian, Syrian/North Levantine, Sudanese and other vernacular interfaces; no vernacular is replaced by the Arabic umbrella.
- Stages: `F|S|V|U|R` - foundational/primary; secondary; vocational/technical/practical; university/taught tertiary; postgraduate/research.
- First package: Reading-supported mathematics/science bridges into MSA, practical health/finance/management modules and advanced open tertiary/research sources, with host/origin-language continuity for displaced learners.
- Why it is in this hedge: The Arab World education-exposure context is 503,356,133 and the separate umbrella estimate is 450 million speakers. Their relation to exact written comfort, vernacular use and unmet supply remains unmeasured.
- Evidence and scaling rule: Treat MSA, vernacular and host-language interfaces as distinct eligible routes. Task-level evidence determines their relative scale and combination; MSA exposure never erases a vernacular route by assumption.
- Named possible outrankers: Dari/Pashto displaced learners, Kurdish varieties, Somali, Tigrinya, Haitian Creole, Ukrainian displacement continuity.

The 100-profile portfolio

The table is grouped by analytical lane. Member numbers identify records only; there is no ordinal rank.

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-001	Indonesian	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-ID-SECONDARY-TO-TERTIARY
P100-002	Hindi	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-HI-UR-PARALLEL
P100-003	Bangla	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-BN-BD-IN-PARALLEL
P100-004	Standard Written Chinese, Simplified	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-ZH-HANS-TARGETED
P100-005	Urdu	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-HI-UR-PARALLEL
P100-006	Telugu	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-007	Tamil	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-008	Marathi	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-009	Vietnamese	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-010	Hausa	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-HAUSA-MULTISCRIPT
P100-011	Standard Kiswahili	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-DRC-MULTILINGUAL-DISPLACEMENT, T10-SWAHILI-MULTISTANDARD
P100-012	Modern Standard Arabic	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-ARABIC-REGISTER-INTERFACES

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-013	Brazilian Portuguese	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-014	Spanish, Latin American educational profiles	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-015	English, underserved learner profiles	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_NO_CURRENT_ROUTE	-
P100-016	French, underserved learner profiles	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_NO_CURRENT_ROUTE	-
P100-017	Turkish	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_NO_CURRENT_ROUTE	-
P100-018	Iranian Persian	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-019	Thai	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-020	Burmese	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-021	Filipino	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	T10-PH-MULTILINGUAL
P100-022	Standard Malay, Malaysia	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-023	Gujarati	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-024	Kannada	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-025	Odia	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-026	Malayalam	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-027	Eastern Punjabi	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	F|S|V|U	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_NO_CURRENT_ROUTE	-
P100-028	Russian	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-029	Japanese	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-030	Korean, South Korean standard	HIGH_REACH_AND_MAJOR_EDUCATIONAL_LANGUAGE	U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_NO_CURRENT_ROUTE	-
P100-031	Amharic	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	T10-ETH-MULTILINGUAL
P100-032	West-Central Oromo educational profile	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	T10-ETH-MULTILINGUAL
P100-033	Somali	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	T10-ETH-MULTILINGUAL
P100-034	Yoruba	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-035	Igbo	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-036	Kinyarwanda	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-037	Kirundi	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-038	Lingala	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	T10-DRC-MULTILINGUAL-DISPLACEMENT
P100-039	Central Kanuri	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-040	Wolof	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-041	Bambara	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-042	Pulaar	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-043	Nigerian Fulfulde	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-044	Zulu	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-045	Northern Sotho / Sepedi standard	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-046	Haitian Creole	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-047	Nepali	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-048	Sinhala	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-049	Northern Pashto	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-050	Dari	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-051	Sindhi, Pakistan standard	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-052	Saraiki	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-053	Bhojpuri	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-054	Maithili	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-055	Chhattisgarhi	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-056	Haryanvi	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-057	Santali	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-058	Sylheti	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-059	Chittagonian	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-060	Javanese	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-061	Sundanese	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-062	Cebuano	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	T10-PH-MULTILINGUAL
P100-063	Central Khmer	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-064	Lao	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-065	Northern Uzbek / Uzbekistan standard	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-066	Kazakh	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	S|V|U|R	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-067	Tajik	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	S|V|U	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_AND_ROUTES_PRESENT	-
P100-068	Uyghur	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-069	Central Kurdish / Sorani	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-070	Northern Kurdish / Kurmanji	REGIONAL_DEPTH_AND_LANGUAGE_TRANSITION	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-071	Ukrainian	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V|U|R	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-072	Polish	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-073	Romanian	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-074	Bulgarian	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-075	Belarusian	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-076	Serbian standard	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	S|V|U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-077	Tigrinya	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	T10-ETH-MULTILINGUAL
P100-078	Kabyle	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V|U	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-079	Central Atlas Tamazight	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-080	Tachelhit	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-081	Ayacucho Quechua	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-082	Cusco Quechua	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-083	Paraguayan Guarani	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-084	Central Nahuatl	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-085	K'iche'	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-086	Tok Pisin	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-087	Bislama	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	F|S|V	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-088	Kalaallisut / West Greenlandic	RESEARCH_ACCESS_COMPARATOR_AND_REGIONAL_DEPTH	S|V|U|R	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-089	American Sign Language	NAMED_SIGN_LANGUAGE	F|S|V|U|R	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-090	Brazilian Sign Language / Libras	NAMED_SIGN_LANGUAGE	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-091	Indian Sign Language	NAMED_SIGN_LANGUAGE	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-092	Kenyan Sign Language	NAMED_SIGN_LANGUAGE	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-093	South African Sign Language	NAMED_SIGN_LANGUAGE	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-094	Interslavic mathematical register	INTERLANGUAGE_AND_SHARED_REGISTER	U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-095	Hindi-Urdu shared mathematical register	INTERLANGUAGE_AND_SHARED_REGISTER	S|V|U	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_NO_CURRENT_ROUTE	T10-HI-UR-PARALLEL
P100-096	Malay-Indonesian shared technical profile	INTERLANGUAGE_AND_SHARED_REGISTER	S|V|U	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-

Member	Profile	Lane	Stages	Current evidence band	Ten-program link
P100-097	Persian-Dari-Tajik shared technical profile	INTERLANGUAGE_AND_SHARED_REGISTER	S|V|U|R	SOURCE_BOUND_POPULATION_OR_PACKAGE_CONTEXT_NO_CURRENT_ROUTE	-
P100-098	Oghuz-focused Inter-Turkic design hypothesis	INTERLANGUAGE_AND_SHARED_REGISTER	U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-
P100-099	N'Ko literary Manding educational profile	INTERLANGUAGE_AND_SHARED_REGISTER	F|S|V|U	ROUTES_PRESENT_WITHOUT_HIGH_REACH_POPULATION_CELL	-
P100-100	Romance mathematical intercomprehension profile	INTERLANGUAGE_AND_SHARED_REGISTER	U|R	PROSPECTIVE_PROFILE_NO_CURRENT_ROUTE_OR_HIGH_REACH_CELL_NOT_ZERO_NEED	-

Mandatory overlays

Every applicable language package must consider all 17 accessibility modes. These include semantic responsive HTML, reflowable EPUB, synchronized text/audio, captions/transcripts, named sign-language video, plain-language/Easy Read, braille, tactile graphics, MathML and low-bandwidth/offline modes. They are not added as independent beneficiary populations.

The 10 current displacement portfolios remain separate origin/host/language/stage overlays. Status-cohort totals are not language counts. Ukrainian school continuity, Afghan, Palestinian, Syrian, Rohingya, Sudanese, Somali, South Sudanese, DRC and Venezuelan routes remain visible even when exact language distributions are missing.

The 10 bridge designs remain experiments, including Interslavic, Hindi-Urdu, Malay-Indonesian, Persian-Dari-Tajik, Oghuz, Romance and Germanic alternatives. Mathematics may lower prose burden for prepared readers, but every design still requires task-specific comprehension, script, prerequisite and overlap bounds.

Partial-order result

Among the 100 candidate profiles there are 4,950 unordered pairs. None currently has a demonstrated global dominance relation on compatible exact audience, incremental effect, physical compute and overlap intervals. This is not equality: it is an explicit statement of what the evidence has not yet resolved.

The first-tranche recommendation is therefore a risk-managed allocation under uncertainty. It protects enormous potential reach, severe underserved contexts, cross-border reuse, advanced research access and displacement while producing the measurements needed for a less ambiguous next decision.

Appendix E - Normalized conditional model

Normalized conditional-portfolio model

Machine-checkable research model-not a world-language rank or optimizer result.

Governing thesis

Inherited familiarity with academic language is an institutionally distributed access advantage. A complete, source-faithful translation into an audience-specified adult register remains eligible whenever any credible nonzero educational benefit is plausible. Evidence determines design, effect, cost and scale; it never functions as an innate-capacity screen, permission test or language-level withholding rule.

The thesis is not scored. Effect evidence determines implementation, expected yield and scale; it cannot make a community ineligible.

What is normalized

- 83 source-typed population frames.
- 1,069 single-stage intervention records: 1,022 source routes and 47 first-tranche program-stage records.
- 4,284 typed parameters with `direct`, `inference`, `scenario`, or `unmeasured` evidence mode.
- 217 endpoint records; comparisons require identical endpoint, stage, horizon and population frame.
- 75 overlap sets, none assigned invented intersections or union totals.
- 36 preserved matched-endpoint threshold comparisons and zero robust global dominance edges.
- 10,625 possible-outranker entities from the complete synthesis universe.

What it can support now

1. Reproduce the existing matched-endpoint break-even inequalities without calling either side a winner.
2. Apply named effect scenarios to compatible endpoints while keeping scenario values out of measured fields.
3. Compare a common 100,000-source-token workflow benchmark across targets without mistaking it for the target-specific educational package or physical compute.
4. Express four disclosed portfolio-governance templates without claiming that their quotas are empirical need weights.
5. Preserve every unselected entity as an eligible possible outranker.

What it cannot support yet

It cannot produce a measured global efficiency ranking. Route-specific effects and standardized physical compute remain unmeasured; exact edition audiences and overlap are mostly unresolved; cross-stage benefits have no agreed scalar weights. The ten-program tranche is therefore one replaceable feasible hedge, and the Top 100 is a nonordinal communication set.

Portfolio fronts

Front	Members	Interpretation
Conditional first tranche	10 programs	One expert-constructed, nonunique measurement-and-production hedge; no ordinal positions.
Top 100 communication set	100 profiles	A broad research/production roster; member numbers are identifiers, not ranks.

All 10,625 entities in the source universe remain possible outrankers. Selection from the displayed portfolios never changes eligibility.

Validation

All 36 constructor checks pass. See `qa/CONDITIONAL_PORTFOLIO_MODEL_QA_v1.json` for exact checks and hashes.

Publication status

WORK IN PROGRESS - ONGOING - NON-EXHAUSTIVE - NOT THE FINAL PRODUCT

The PDF is the human-readable entry point. The editable document, full HTML reader, Markdown snapshot, portfolio CSVs, manifest, and source archive preserve the current evidence and calculations. Missing or unfinished research never makes a community ineligible.